

Global Temperature and Global Mortality*

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Abstract

This paper documents large indirect mortality impacts of climate change. Exploiting natural global temperature variability, we find that 1°C of warming increases annual mortality by 2.5 deaths per 1,000 individuals, compared to 0.1 deaths per 1,000 due to the associated local heat exposure. Using external estimates of the impact of global income on mortality, we show that the macroeconomic effects of global temperature changes explain two thirds of the gap between these estimates. We use this evidence to estimate damage functions in a neoclassical growth model that links mortality and economic activity and accounts for adaptation to mortality impacts. Mortality contributes \$400 per ton of CO₂ to the Social Cost of Carbon, or one quarter of total climate damages. These results suggest that economic adaptation has a double dividend by also reducing climate-related mortality.

JEL classification: I15, Q51, Q54, Q58, E01, O44

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1 Introduction

Mortality is one of the most salient consequences of climate change. The associated loss of life and reduced life expectancy can dramatically disrupt the fates of families and communities. A changing climate might affect mortality through a myriad of channels: directly through local heat exposure—the focus of the vast majority of existing studies—but also through other extreme weather realizations, or even indirectly through associated economic contractions. Do the mortality consequences of climate change primarily depend on localized heat exposure, or are wider impacts quantitatively relevant?

In this paper, we construct a comprehensive estimate of the impact of temperature on mortality by leveraging time-series variation in global mean temperature. Our estimates imply that a permanent 1°C increase in global temperature leads to 2.5 additional deaths per 1,000 individuals annually, compared to 0.1 deaths per 1,000 when mediated through local heat exposure only. We show that our estimate based on global temperature is primarily mediated by the income losses caused by global temperature fluctuations: combining external estimates of global output losses associated with global temperature fluctuations and external estimates linking income and mortality accounts for two thirds of our baseline effect. We then use our reduced-form evidence to estimate structural damage functions in a simple model that links mortality and economic losses and accounts for adaptation to mortality impacts. Mortality contributes over \$400 per ton of CO₂ to the Social Cost of Carbon (SCC), representing one quarter of the total SCC, instead of \$40 and under one fifth of overall losses when based on local heat only.

In the first part of the paper, we develop our empirical approach. We construct a new climate-mortality dataset with comprehensive coverage across the globe. We combine worldwide annual country-level mortality data from the Human Mortality Database and World Health Organization, complemented with direct sources for India and China. This dataset spans 1960–2019 and covers over 100 countries and 70% of the world’s population. We link these data with Gross Domestic Product (GDP) information from the Penn World Tables, global temperature data from Berkeley Earth, and gridded extreme weather indicators from the Inter-Sectoral Impact Model Intercomparison Project.

The joint trending behavior of temperature and mortality complicates the identification of causal impacts. To isolate exogenous variation, we construct global temperature *shocks* as innovations to the temperature process that are orthogonal to long-run trends, following Hamilton (2018). Importantly, the geoscience literature attributes much of this

medium-run variation in global temperature to internal climate phenomena such as the El Niño cycle and to external forcings such as solar variability and volcanic eruptions—sources of variation that are plausibly exogenous to contemporaneous economic and health conditions.

We estimate the dynamic causal effects of global temperature shocks on mortality using panel local projections. A global temperature shock scaled to 1°C leads to a gradual increase in the crude death rate that peaks at 1 death per 1,000 after 12 years, with a 95% confidence interval of (0.54, 1.53). The effect is statistically significant over medium- to long-run horizons and gradually mean-reverts after 20 years. Peak effects are five times larger among individuals aged 65 and older.

Temperature *shocks* are a useful identification device, but the fundamental determinant of mortality is the *level* of global temperature. Because the temperature shock serves as our instrument, we account for the partially persistent effect of global temperature shocks on the level of global temperature, which remains elevated near 0.5°C for several years.

We convert our estimates to the effect of a permanent increase in global temperature by taking the ratio of cumulative impulse responses of mortality and temperature. This calculation implies that a permanent 1°C rise in global temperature increases annual mortality by 2.5 deaths per 1,000 people in the long run. Our estimate relies on similar identifying assumptions to existing work that focuses on local heat: the relationship between temperature and mortality extends beyond the range of shocks observed in sample and also applies to anticipated permanent changes. We relax these assumptions in our structural model.

The time-series nature of our analysis and the limited number of annual observations warrant particular care in interpreting our results. We consider three main identification concerns, which we address in a series of robustness exercises. First, omitted variables could bias our estimates if temperature shocks coincide with other adverse global health shocks. Our results remain stable across different sets of controls, including commodity price indices, regional or global trends, the age structure of countries, major wars and epidemics, and indicators for global recessions. The estimates do not appear to be driven by any particularly influential years and remain largely unchanged under alternative definitions of temperature shocks.

Second, reverse causality may bias our estimates if developments in contemporaneous

economic activity affect emissions and thereby temperature. While this channel is comparatively more relevant for output responses than for mortality outcomes, we nonetheless adjust our mortality estimates for the feedback between output, associated changes in carbon dioxide, methane, and sulfur dioxide emissions, and temperature. Our results remain virtually unchanged.

Third, we verify that our estimates are likely to be externally valid. They remain stable in shorter samples (1980–2019 and 1960–2000) and across alternative country samples—excluding China and India or expanding coverage to include African countries—and after removing specific sources of temperature variation such as El Niño events or volcanic eruptions. While no set of sensitivity checks is ever fully exhaustive, the stability of our results across specifications addressing omitted variables, reverse causality and external validity suggests that our specification may credibly identify the causal effect of global temperature on worldwide mortality.

In the second part of the paper, we ask what mechanisms can account for the mortality effects of global temperature changes. We consider three potential candidates. First, we examine changes in local heat and cold exposure associated with changes in global temperature. To maximize comparability, we estimate mortality responses to local heat shocks—innovations in the fraction of days when country-average temperature exceeds historical thresholds—using the same empirical specification and data but replacing global temperature shocks with local heat shocks. We find that 1°C of warming increases mortality by approximately 0.1 deaths per 1,000 when local heat is the only mediator. If anything, including local cold exposure attenuates our results. These effects are consistent with estimates in Carleton et al. (2022), but are one to two orders of magnitude below our global temperature estimate.

Second, we examine whether other extreme events that correlate with global temperature but that are not captured by local heat may directly influence mortality. We show that accounting for the additional direct effect of droughts, extreme wind and precipitation on mortality raises the overall contribution but still falls short of closing the gap.

Third, we consider the possibility that global temperature changes cause income losses which, in turn, increase mortality. This mechanism is consistent with a long tradition in development and health economics (Preston, 1975; Cutler, Deaton, and Lleras-Muney, 2006). We find strong evidence supporting this income channel by constructing an indirect estimate of the impact of global temperature on mortality that isolates the role of

income.

To construct our indirect estimate, we start from the GDP impact of global temperature fluctuations estimated in Bilal and Känzig (2026): a permanent 1°C increase in global temperature reduces world GDP per capita by 22% in the long run. We then combine this response of GDP to global temperature with the response of mortality to global GDP estimated in Bilal, Känzig, and Lisiecki (2026). They exploit innovations in global GDP instrumented with monetary policy surprises to assess the impact of global income on mortality.

We find that global GDP losses mediate 66% of the aggregate mortality response to global temperature shocks after combining the estimate of global temperature shocks on GDP with the estimate of GDP on mortality. These findings suggest that the primary reason why global temperature fluctuations increase mortality is because of the induced macroeconomic contractions and the associated worsening of health outcomes rather than direct environmental exposure.

We corroborate this mechanism by exploring heterogeneity in mortality responses along several dimensions. Lower-income regions and countries with low levels of public healthcare spending exhibit larger mortality responses. Cardiovascular disease and HIV-related deaths drive the aggregate results, while respiratory, tropical, and heat- or cold-related mortality contribute substantially less. These patterns are consistent with mortality operating primarily through economic disruptions and access to healthcare rather than direct heat exposure.

In the final part of the paper, we develop a structural model to evaluate the welfare costs of climate change incorporating both mortality and economic damages, and taking into account adaptation to mortality impacts. We use a version of the neoclassical growth model with finite lifespans in the spirit of Blanchard and Katz (1992) and similar to Bilal, Känzig, and Lisiecki (2026), where households face stochastic mortality and firms produce using capital and labor. In addition to investing, households can spend their income on final consumption or on health, which lowers their mortality risk according to a health production function. This health production function captures a broad range of mechanisms such as air conditioning, nutrition, or healthcare. Consistent with our empirical results, mortality depends on aggregate output in addition to private health spending. Excess temperature relative to baseline affects mortality rates in addition to aggregate output, and affects total factor productivity, through separate damage functions.

We estimate these damage functions by matching the estimated impulse response functions of mortality and output to temperature shocks. We obtain estimates of the effect of global and local GDP on mortality from Bilal, Känzig, and Lisiecki (2026), which match the corresponding empirical impulse response functions in the same model. Thus, we discipline the importance of adaptation to mortality impacts with the empirical relationship between country-level GDP and mortality. We calibrate the value of life at the end of the lifespan using an estimate of the Value of Statistical Life (VSL) for the United States. Our baseline estimation conservatively adjusts this VSL downward accounting for the gap between world income per capita and United States income per capita. We also conservatively assume that all mortality losses are borne by individuals aged 65 years or older, implying a further downward age adjustment to the VSL.

We conduct four main counterfactuals, all of which are relative to a reference scenario that features 2% economic growth per year. Our first counterfactual is a gradual increase in global mean temperature that starts in 2026 and reaches 3°C above pre-industrial levels by 2100, so a 2°C increase above current levels. The estimated model indicates that mortality rises by 6 excess deaths per 1,000 by the end of the century, close to the value implied by our reduced-form calculations despite adaptation and equilibrium adjustments. Mortality impacts alone from 2°C warming imply a present-value welfare loss of approximately 14%. Combining mortality with economic losses, total welfare losses exceed 40%.

Our second counterfactual evaluates the cost of carbon associated with the warming trajectory caused by the release of one ton of carbon in 2026. We find a Social Cost of Carbon (SCC) of approximately \$1,640 per ton of CO₂, with a 95% confidence interval of (\$750, \$2,520) under damage functions based on global temperature. Mortality contributes \$400 per ton to this total, amounting to one quarter of the SCC. The mortality component of the SCC includes both indirect losses through GDP and direct losses, each contributing approximately half. By contrast, damage functions calibrated to estimates based on local heat imply a smaller total SCC of \$250 per ton. Mortality then accounts for roughly \$40 per ton of this total, or less than one fifth, with most of this component due to indirect effects. Thus, accounting for global temperature damages raises both the total SCC and the mortality share of climate damages.

Our third counterfactual assesses mortality-based damages associated with individual regions following the release of one ton of carbon anywhere in the world: Domestic Costs of Carbon (DCCs). We re-estimate our model region by region, and find substantial

heterogeneity: populous or wealthy countries face higher domestic mortality costs. For instance, China faces a DCC of just over \$40 per ton due to its large population, while the United States faces a DCC of just over \$70 per ton due to its high income levels. As a continent, Africa's DCC is under \$30 due to its low income per capita.

Our fourth counterfactual assesses the role of adaptation. We consider two margins of adaptation: the ability to change private health spending, and benefits from economic growth due to health spillovers. Perhaps surprisingly, when we shut down adaptation through private health spending, mortality, welfare and the SCC remain largely unchanged. On the one hand, private health spending increases in response to climate change, but the strength of private adaptation is moderate because the empirical relationship between country-level income and mortality is modest (Preston, 1975; Cutler, Deaton, and Lleras-Muney, 2006). On the other hand, households invest more when prevented from spending on health, leading to higher aggregate GDP and thus lower mortality due to spillovers in the health production function. Quantitatively, these two forces offset each other.

However, we find that aggregate economic growth leads to larger benefits of adaptation. When we shut down economic growth in our reference scenario, climate change causes about 9 excess deaths per 1,000 by the end of the century instead of 6 excess deaths per 1,000. The welfare costs of climate-induced mortality grow from 14% to 21%, and the mortality SCC increases from \$400 to more than \$500 per ton. Consistent with empirical evidence, economic growth lowers the mortality rate through its dependence on aggregate GDP in addition to private health spending. This spillover elasticity is larger than the private adaptation elasticity. Without this spillover, a proportional change in mortality due to climate change implies larger effects on the level of the mortality rate, implying larger losses.¹

Of course, an important determinant of the mortality costs of climate change is the VSL. Adjusting the VSL for regional income per capita and for age is quantitatively significant. Without the income adjustment, the worldwide mortality SCC would be just under \$1,400. The ranking of DCCs between regions would also change. China's and India's DCCs would approach \$300 per ton, and Africa's would exceed \$300 per ton. Adjusting the VSL for age is similarly important. Assuming instead that all individuals are equally exposed to climate-related mortality, the mortality SCC is just under \$1,100 per

¹While the spillovers between growth and mortality affect the level of economic activity and mortality, they leave all our (relative) counterfactuals in response to climate change virtually unchanged.

ton.

Our results have two implications for policy and economic adaptation—which is distinct from adaptation to mortality impacts. First, domestic costs suggest that unilateral decarbonization policies may be cost-effective for populous or wealthy economies even without international coordination and on the basis of mortality alone. Second, the mechanism we identify—macroeconomic disruptions leading to heightened mortality—suggests that policies that adapt the economy to climate change have a double dividend: they also reduce the mortality consequences of climate change. Thus, society’s willingness to pay for economic adaptation is larger than on the basis of economic losses alone.

Related literature. Our findings contribute to several literatures. First, we contribute to the important strand of work that evaluates the health and mortality impacts of climate change. Deschênes and Greenstone (2011), Barreca et al. (2016), Burgess et al. (2017), Hsiang et al. (2017), and Carleton et al. (2022) focus on contemporaneous local heat exposure as the main mediator of climate-related mortality. Recent work further examines the role of air pollution caused by wildfires (Burke et al., 2021; Burke et al., 2022) and air pollution more broadly (Aguilar-Gomez et al., 2022). We show that global temperature fluctuations appear to have a larger impact on mortality than implied by local heat and air pollution, largely mediated by associated economic losses.

Second, we relate to work estimating temperature impacts on economic activity (Dell, Jones, and Olken, 2012; Burke, Hsiang, and Miguel, 2015; Newell, Prest, and Sexton, 2021; Nath, Ramey, and Klenow, 2024). Our empirical approach and our analysis of the income channel builds on Bilal and Känzig (2026): we directly estimate the impact of global temperature fluctuations on mortality, and we separately leverage their estimates of the impact of global temperature fluctuations on output.

Third, our analysis linking income losses to mortality caused by climate change connects to the broader literature on development and health (Preston, 1975; Pritchett and Summers, 1996; Cutler, Deaton, and Lleras-Muney, 2006; Cutler, Huang, and Lleras-Muney, 2016; Ma et al., 2022). The estimated effects of global GDP on mortality in Bilal, Känzig, and Lisiecki (2026) that we use are qualitatively in line with the well-known negative relationship between economic development and mortality. Quantitatively, their estimates are based on world output and are larger than estimates linking local income and mortality in the cross-section or panel of countries due to global longevity spillovers.²

²In the United States, mortality has been found to be procyclical, falling when local unemployment

The remainder of the paper proceeds as follows. Section 2 describes the data and construction of temperature shocks. Section 3 presents the baseline estimates of global temperature impacts on mortality. Section 4 compares global temperature to local heat and explores mechanisms. Section 5 develops the structural model and estimates damage functions. Section 6 studies the mortality, economic and welfare consequences of climate change. Section 7 concludes.

2 Data and Empirical Specification

Our goal is to estimate the impact of climate change on mortality. Climate change represents shifts in a broad range of phenomena relative to pre-industrial times: heat, cold, precipitation, wind, ocean currents and many others. The scalar measure used by the Intergovernmental Panel on Climate Change (IPCC) to summarize this collection of phenomena is global mean temperature. This definition motivates our focus on the relationship between global mean temperature and mortality.

2.1 Mortality Data

To study the effects of global temperature on mortality, we assemble a comprehensive dataset of country-level mortality outcomes. Annual mortality data are less widely available than measures of economic activity. We compile information for Europe, North America, and a subset of countries in Asia and South America using the Human Mortality Database (HMD) and the World Health Organization (WHO). We complement these data with information for India and China for 1960–2019 from national statistics. We measure mortality using the crude death rate. We use deaths per 1,000 individuals as our units.

Because our analysis only requires country-level mortality data, we are able to expand coverage relative to Carleton et al. (2022). Our baseline sample coverage rises from 60 countries in 1960, to over 100 countries in the second half of our sample. These countries represent an approximately constant share of world population over the period 1960–2019 that fluctuates between 60 and 80%. Appendix Figure A.1 illustrates the evolution

rises (Ruhm, 2000; Stevens et al., 2015; Finkelstein et al., 2025), although this relationship appears to have weakened over time (Ruhm, 2015). Bilal, Känzig, and Lisiecki (2026) show that these local business-cycle relationships can differ substantially from the relationship between global economic activity and mortality, with global income expansions producing sizable mortality reductions through international spillovers. Our results on the income channel may also help rationalize recent findings by Young and Hsiang (2024) who show that tropical cyclones have delayed mortality effects that appear to be unrelated to initial accidental deaths.

of country and population coverage over our sample period.

Despite this broad coverage, reliable annual mortality data remain unavailable for many countries, particularly in Africa, where civil registration systems are often incomplete or absent. The United Nations provides comprehensive mortality estimates through the World Population Prospects, constructed by combining censuses, household surveys, and partial registration systems, and by applying demographic models to interpolate missing observations and ensure internal consistency. While well suited for measuring long-run trends and cross-country differences, these estimates smooth high-frequency variation and are therefore less appropriate for our empirical design.

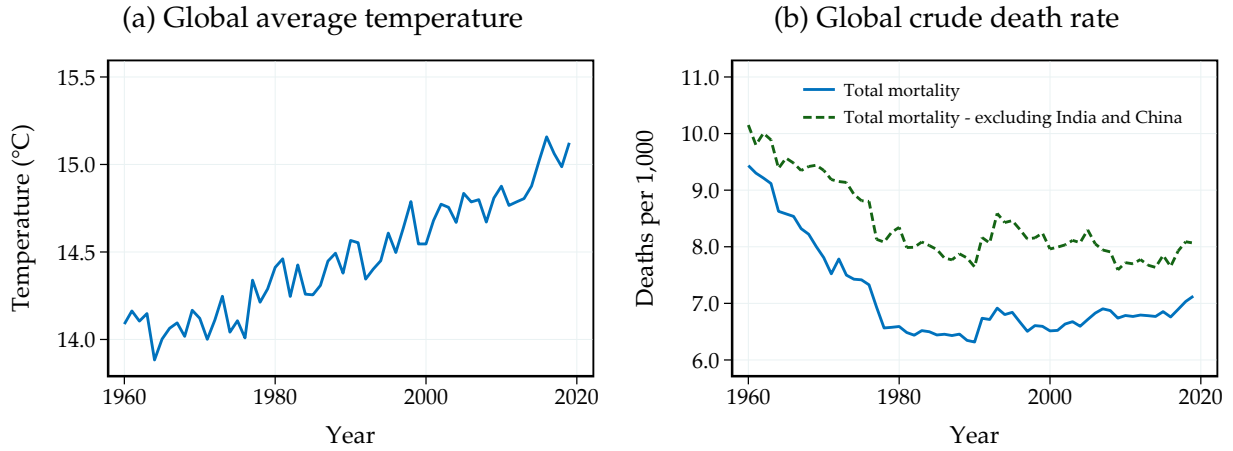
To nonetheless provide suggestive evidence for regions not covered in the baseline sample, we use the United Nations data to examine the mortality implications of global temperature changes in a broader set of countries, including Africa, in Section 4.3.

2.2 Economic and Climate Data

Our economic and climate data build on the dataset constructed in Bilal and Känzig (2026). We draw GDP and population data from the Penn World Tables, which provide comprehensive coverage for our sample of countries over the period 1960–2019. We obtain a global commodity price index from Bloomberg. We take world recession dates from the World Bank (Kose, Sugawara, and Terrones, 2020). We obtain data on armed conflicts from the Uppsala Conflict Data Program. We compile information on major epidemics and pandemics from publicly available historical sources.

We construct global mean temperature data from the Berkeley Earth Surface Temperature Database, supplemented with direct measures from the National Oceanic and Atmospheric Administration (NOAA). We construct country-level indicators of exposure to extreme weather events using the Inter-Sectoral Impact Model Intercomparison Project (ISIMIP) observed climate dataset. This source provides global, daily reanalysis data on temperature, wind speed, and precipitation from 1901 to 2019 at a $0.5^\circ \times 0.5^\circ$ resolution. For extreme heat, cold, wind, precipitation, and droughts, we construct exposure indices as in Bilal and Känzig (2026) by recording, for each country and year, the fraction of days in which country-wide average weather exceeds or falls below fixed percentiles of its 1950–1980 daily weather distribution. We provide more details in Appendix A.

Figure 1: Global Mean Temperature and Mortality Since 1960



Notes: Panel (a): evolution of global average temperature (including sea surface) from Berkeley Earth since 1960. Panel (b): evolution of global crude death rate (deaths per 1,000 population) computed from HMD, WHO, and national statistics weighted by country population.

2.3 Global Temperature Shocks

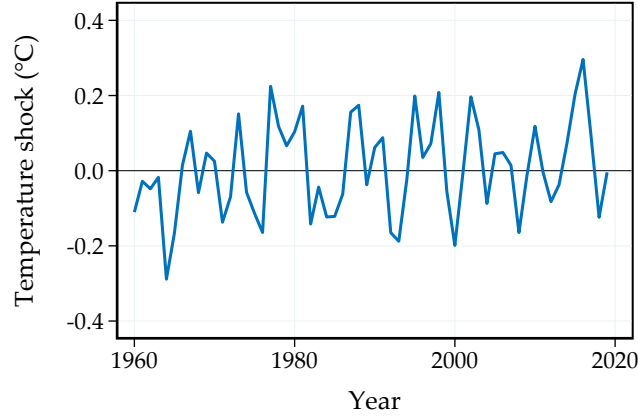
Our empirical analysis exploits fluctuations in global mean temperature. Figure 1 displays the evolution of global mean temperature and global mortality since 1960. Global mean temperature remains relatively stable around 14°C until the mid-1970s. From the 1970s onward, global mean temperature begins to steadily rise. At the same time, the global crude death rate declines from approximately 9 deaths per 1,000 in 1960 to around 7 deaths per 1,000 by 2019.³

The joint trending behavior of temperature and mortality creates a well-understood challenge, as simple correlations will deliver a spurious link between the variables (Granger and Newbold, 1974). Therefore, we follow the approach in Bilal and Känzig (2026) and focus on *temperature shocks* for identification. We define such shocks as possibly persistent deviations from the long-run trend in global mean temperature.

The geoscience literature suggests that two types of phenomena cause global mean temperature to fluctuate around its trend independently from human activity: external causes such as solar cycles and volcanic eruptions (National Oceanic and Atmospheric Administration, 2009; National Oceanic and Atmospheric Administration, 2005), and internal climate variability such as the El Niño-La Niña cycle (Kaufmann, Kauppi, and

³The flattening of the crude death rate beginning around the 1980s largely reflects changes in population composition, in particular population aging. Mortality rates within age groups continued to decline throughout the entire sample period.

Figure 2: Global Temperature Shocks



Notes: Global temperature shocks for 1960–2019, computed using the Hamilton (2018) filter with $h = 2$ and $p = 2$. Shocks represent deviations from the long-run temperature trend driven by solar cycles, volcanic eruptions, and internal climate variability such as El Niño events.

Stock, 2006; National Oceanic and Atmospheric Administration, 2023).

We isolate the trend and innovations of the temperature process with the method proposed by Hamilton (2018) and applied in Bilal and Känzig (2026). We project global mean temperature on its lags as of period $t - h$ and construct the temperature shock as the corresponding innovation:

$$\widehat{T}_t^{\text{shock}} = T_t - (\hat{\alpha} + \hat{\beta}_1 T_{t-h} + \dots + \hat{\beta}_{p+1} T_{t-h-p}), \quad (1)$$

where $\hat{\beta}_i$ denote the coefficient estimates of the regression of temperature on its lag $j = (h+i-1)$ and $\hat{\alpha}$ is the estimated intercept. This exercise amounts to isolating shocks that persist typically for h periods.

Motivated by the observation that phenomena such as El Niño or solar cycles can last for several years, we select a horizon of $h = 2$. We set the number of lags to $p = 2$. Varying these values leaves our results virtually unchanged. Results are also robust to using alternative detrending methods such as the Hodrick-Prescott filter, as we show in Appendix C.2.

Figure 2 shows the resulting global temperature shocks over our sample. They fluctuate around zero, are usually less than 0.2°C , and are weakly autocorrelated because we allow for relatively persistent deviations from the long-run temperature trend. In our empirical specification, we therefore control for lagged temperature shocks to avoid bias from serial correlation.

2.4 Empirical Specification

The mortality effects of temperature shocks may take time to materialize. Therefore, we estimate the dynamic effects of temperature shocks up to 20 years following the initial shock. We do so with panel local projections (Jordà, 2005; Jordà and Taylor, 2025), which allow us to estimate medium-run effects directly without extrapolating from short-term impacts.⁴

We estimate the following series of panel regressions, one for each horizon $h = 0, \dots, 20$:

$$m_{i,t+h} - m_{i,t-1} = \alpha_{i,h} + \theta_h T_t^{\text{shock}} + \mathbf{z}_t' \boldsymbol{\beta}_h + \mathbf{z}_{i,t}' \boldsymbol{\gamma}_h + \varepsilon_{i,t+h}, \quad (2)$$

where $m_{i,t}$ is the mortality rate (deaths per 1,000 individuals) for country i in year t , T_t^{shock} is the global temperature shock constructed in Section 2.3, and θ_h is the dynamic causal effect of interest at horizon h . The sequence $\{\theta_h\}_{h=0}^{20}$ is the impulse response function to the global temperature shock. Because we use the cumulative change in mortality as the dependent variable, we estimate a possibly persistent level effect on the mortality rate. The country fixed effects $\alpha_{i,h}$ absorb time-invariant differences in mortality levels across countries, such as baseline health system quality or demographic composition. The vector $\mathbf{z}_t$ contains global controls and $\mathbf{z}_{i,t}$ includes country-specific controls. The error term $\varepsilon_{i,t+h}$ captures residual variation in mortality outcomes.

Our specification differs from standard panel studies of local temperature or heat in one critical respect: the identifying variation in global temperature shocks T_t^{shock} is common across all countries. Consequently, we cannot include year fixed effects, as these would absorb all variation in global temperature. Instead, we control for global economic shocks that may correlate with both temperature and mortality. Specifically, $\mathbf{z}_t$ includes lags of the global temperature shock, lags of world GDP growth, commodity price inflation, and global recession indicators.⁵

At the country level, $\mathbf{z}_{i,t}$ includes lags of the country's own mortality rate, local GDP growth, and regional linear trends. We include two lags of the shock and four lags of all other control variables. We weight regressions by country population in period $t-1$ to

⁴We use local projections rather than vector autoregression or distributed lag models because they tend to be more robust at longer horizons (Montiel Olea et al., 2024). Local projections directly estimate the effects of interest at each horizon rather than extrapolating from the first few autocovariances, and they allow for more flexible control specifications.

⁵We include a dummy for global recession dates as well as its lags to allow for persistent effects of past global economic contractions on mortality.

reflect that larger countries contribute more to global mortality outcomes.

Because global temperature variation is common across countries, the error terms $\varepsilon_{i,t+h}$ are potentially serially and cross-sectionally correlated. We therefore compute standard errors using the approach of Driscoll and Kraay (1998), which is robust to both serial and cross-sectional dependence of arbitrary form.⁶

3 Global Temperature and Mortality

We now present our main empirical results on the impact of global temperature shocks on mortality. We first report our baseline estimates before discussing three identification concerns: omitted variable bias, reverse causality, and external validity.

3.1 Baseline Estimates

Figure 3 displays the impulse responses of global temperature and world mortality to a global temperature shock scaled to 1°C. The left panel shows that, by construction, global temperature increases by 1°C on impact. The effect is highly persistent: after 10 years, global temperature remains elevated by nearly 0.5°C, before gradually decaying over time.

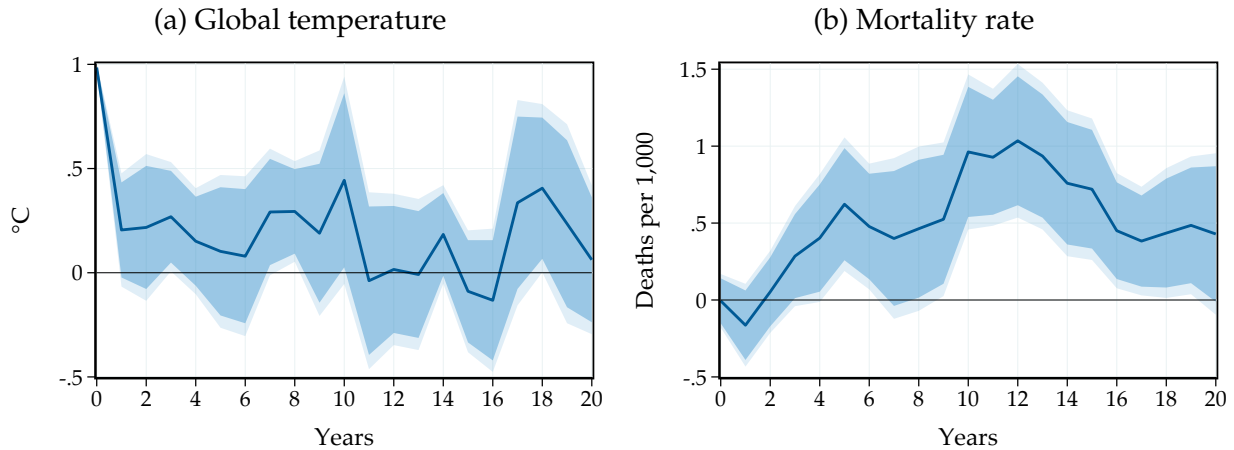
The right panel shows the mortality response. Mortality responds with a meaningful lag, and the effect builds up over time. The peak effect occurs 12 years after the shock, when mortality is elevated by 1 death per 1,000. The effect remains statistically significant at the 5% level in years 5 and 6, and then again in years 9 to 19. The point estimates eventually revert back to zero over time.

The mortality response is not uniform by age. Appendix Figure B.1 indicates that individuals aged 65 years or more exhibit much stronger responses, peaking near 5 excess deaths per 1,000 five years after the global temperature shocks, and account for the majority of the total mortality impact.

The gradual increase in mortality represents the accumulating effects of persistently elevated temperature over time: the fundamental determinant of mortality is the level of temperature. The temperature shock only serves as an instrument. Hence, the effects are larger than implied by a fully transitory 1°C temperature increase, but are lower than the effect of a permanent 1°C increase.

⁶Our baseline specification does not account for estimation uncertainty in the global temperature shock. In Appendix B.4, we show that inference is very similar when accounting for this uncertainty either using a Delta-method correction or by estimating the responses in a one-step procedure.

Figure 3: The Effect of Global Temperature Shocks on Temperature and Mortality



Notes: Impulse responses of global mean temperature and mortality rate (deaths per 1,000 individuals) to a 1°C global temperature shock. The mortality response is estimated using panel local projections in equation (2). Sample: 1960–2019. Controls: two lags of global temperature shock, four lags of world GDP growth, global recession indicators, commodity price inflation, local mortality rate, local GDP growth and regional linear trends. The temperature response is estimated from the global time series, following Bilal and Känzig (2026). Shaded areas: 90% and 95% confidence bands based on Driscoll-Kraay standard errors.

We assess the role of temperature persistence by constructing a counterfactual mortality path that corresponds to a fully transitory temperature change using the method of Sims (1986). Appendix B.2 shows that accumulated temperature persistence accounts for a substantial portion of the peak effect: under a purely transitory shock, the peak mortality effect would be 0.6 deaths per 1,000.

To quantify the long-run impact of a permanent temperature increase, we calculate the ratio of the cumulative impulse response of mortality to the cumulative impulse response of temperature. This ratio corresponds to the long-run effect of a sustained 1°C rise in global temperature. We find that a permanent 1°C increase in global temperature leads to a long-run increase of approximately 2.5 deaths per 1,000 population. For comparison, this increase is approximately twice the current global mortality burden attributable to all forms of air pollution combined.

Scaling our estimates to a permanent 1°C increase relies on two assumptions that are common in the literature on climate impacts. First, the relationship between temperature and mortality must hold beyond the range of shocks observed in our sample. A 1°C shock is large relative to historical variation: in practice, we observe shocks of about 0.1–0.3°C. Reassuringly, Appendix B.3 indicates that there is little evidence of nonlinearities within the range of shocks we observe. Of course, future climate change may involve adaptation, tipping points or other nonlinearities that have not yet materialized (Dietz et al., 2021b),

which could make actual impacts smaller or larger than our linear exercise suggests.

The second assumption is that permanent changes are well approximated by the cumulative effects of partially transitory shocks. This condition is more difficult to verify empirically, and is also subject to the same caveats about adaptation and nonlinear impacts. We relax both assumptions in our structural model in Section 5.

3.2 Identification Challenges and External Validity

While we estimate effects using country-level mortality data, our identifying variation comes from global temperature shocks common to all countries. This time-series variation warrants additional care in interpreting our results. We discuss the key identification concerns in turn.

Omitted Variable Bias. The first identification concern is that global temperature shocks may coincide with other global or regional shocks affecting mortality. For instance, if global or regional mortality shocks just so happen to be correlated with global temperature variation over time, we would mistakenly attribute these effects to temperature. This concern is particularly relevant given our relatively short annual sample.

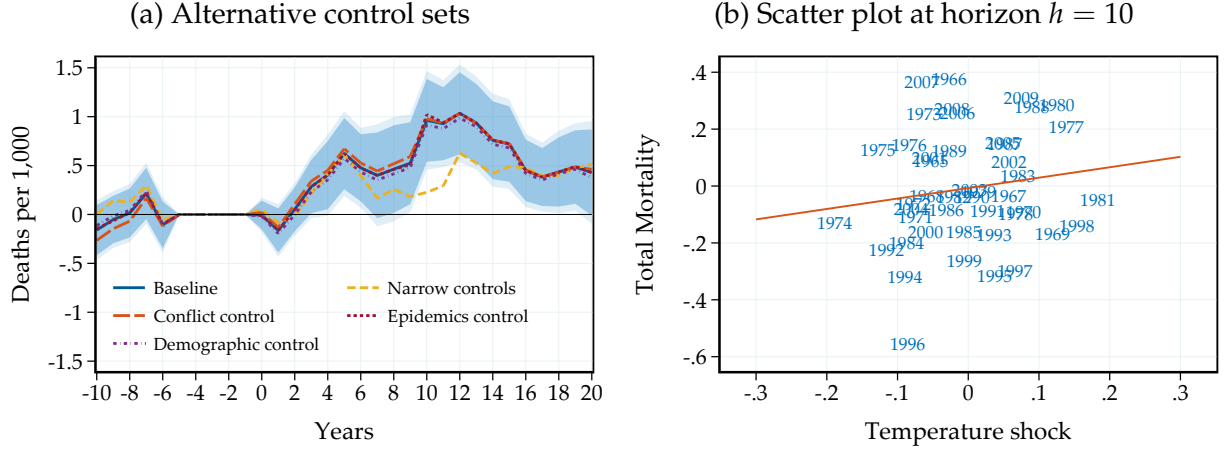
To address this concern, we already include a rich set of global and country-level controls in our baseline specification. Figure 4(a) shows that the results are robust to alternative control sets. First, the results hold when we restrict attention to a narrower set of global controls, excluding commodity price inflation and the global recession indicator. Second, we show that results are robust when we additionally control for demographics, armed conflicts, or epidemics. Across all these specifications, estimates remain similar and largely overlap with the baseline confidence intervals.

We also examine the evolution of mortality prior to the shock to assess the presence of pre-trends. In particular, the estimated effects may capture pre-existing trends rather than a causal response to the shock. Reassuringly, we find no evidence of economically meaningful or statistically significant pre-trends across specifications. Estimates are close to zero throughout the pre-treatment period.⁷

We further assess whether our results are driven by spuriously correlated mortality shocks by examining the influence of individual years. For each year, Panel (b) plots the 10-year-ahead change in mortality against the contemporaneous temperature shock, after

⁷Note that the coefficients for horizons $h = -(p+1), \dots, -1$ are mechanically equal to zero due to the inclusion of lagged controls in the local projection specification.

Figure 4: Sensitivity to Control Variables and Specification Choices



Notes: Panel (a): impulse responses under alternative control specifications. Baseline: four lags of mortality, global and local GDP growth, commodity price inflation, and global recession dummy, two lags of temperature shocks, and subregion trends. Narrow controls: excludes commodity price inflation and global recession dummy. Conflict control: number of conflicts a country is involved in a given year. Epidemics control: adds indicator for pandemic and epidemic outbreaks. Demographic control: population share 65 years and over. Panel (b): mortality change from year -1 to year $+10$ against temperature shock for each country-year observation, with regression line from baseline specification. Shaded areas: 90% and 95% confidence bands.

residualizing both variables with respect to the baseline controls. The positive relationship does not appear to be driven by a small number of influential years or outliers.

We conduct two additional exercises in Appendix C. First, Appendix Figure C.1 reports a jackknife exercise in which we sequentially set each temperature shock to zero and re-estimate the impulse responses. The resulting estimates remain stable, indicating that our findings are not driven by any single shock. Second, Appendix Figure C.2 verifies that our results are robust to alternative methods of constructing temperature shocks, including HP filtering and different choices of the Hamilton filter parameters (h, p). Overall, these results suggest that our estimates are unlikely to be driven by mortality shocks spuriously correlated with temperature shocks nor by the influence of specific countries or specification choices.

Reverse causality. The second identification concern is reverse causality. Global temperature shocks reduce economic activity (Bilal and Känzig, 2026), and thus greenhouse gas emissions. Reductions in emissions alter the future evolution of temperature, thereby potentially biasing our estimates. To address this concern, we follow Bilal and Känzig (2026) and adjust the estimated impulse response function to remove the component attributable to anthropogenic emissions.

We separately estimate the impact of global temperature on GDP per capita. We use

these estimates and standard emissions elasticities to obtain the dynamic response of emissions to global temperature shocks. We then use a simple climate model to predict the temperature contribution from observed CO₂, CH₄, and SO₂ emissions over our sample. Finally, we adjust our estimates by removing the greenhouse gas feedback between global temperature shocks and subsequent temperature in a bivariate analysis.

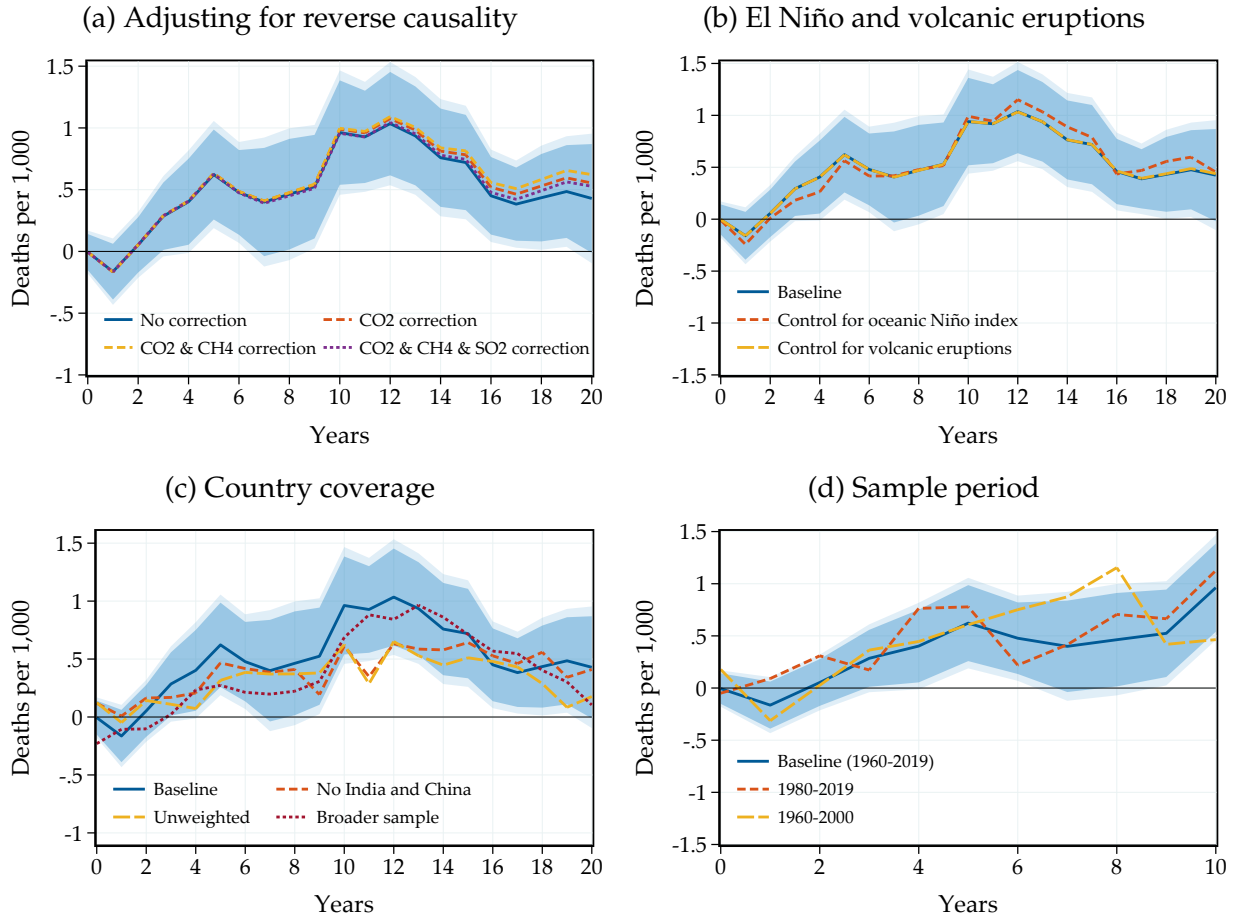
Carbon dioxide and methane are greenhouse gases and therefore raise global temperatures. To the extent that economic activity increases emissions of these gases, reverse causality would attenuate our estimated temperature effects, biasing them downward rather than upward. By contrast, sulfur dioxide is a major anthropogenic pollutant with a cooling effect, implying an offsetting bias in the opposite direction.

Figure 5(a) shows that the adjustments move our estimates in precisely these expected directions. Quantitatively, however, the corrections are very small: the estimated mortality responses are virtually unchanged over the first 15 years. At longer horizons, the adjustments become somewhat more noticeable, but remain modest. Moreover, the opposing effects of the different gases largely offset one another, leaving the impulse response that accounts for all three gases very close to the baseline throughout. Overall, reverse causality operating through the effects of global temperature on economic activity and, in turn, emissions appears quantitatively negligible for our estimates.

External Validity. The third concern is external validity. Do our estimates vary across different types of temperature shocks, geographic coverage, or time periods? For example, if the mortality–temperature relationship has changed over time due to adaptation, estimates based on earlier decades may not generalize to more recent or future climate conditions. Similarly, geographic coverage may matter: if countries that are highly exposed or vulnerable to climate change are underrepresented in our baseline sample—such as many African economies—our estimates may not capture the full extent of temperature-related mortality effects. Finally, if the relationship depends on the underlying source of temperature variation, our average estimates may not be broadly applicable.

We begin by examining whether the type of temperature variation matters. Figure 5(b) compares our baseline estimates to specifications that control for major sources of global temperature fluctuations, including El Niño events and volcanic eruptions. The impulse response functions are virtually identical across specifications. This suggests that the estimated mortality effects reflect temperature changes broadly, rather than factors

Figure 5: Reverse Causality and External Validity



Notes: Impulse response of mortality to a 1°C global temperature shock. Panel (a): impact of temperature shocks after removing predicted temperature contribution from CO₂, CH₄, and SO₂ emissions using a simple climate model. Panel (b): impulse responses when netting out temperature variation driven by El Niño or volcanic eruptions, compared with the baseline. Panel (c): impulse responses based on different samples of countries and weighting schemes. Broader sample includes mortality data from UNWPP. Panel (d): impulse responses estimated over different sample periods. Full sample: 1960–2019. Early period: 1960–2000. Late period: 1980–2019. Shaded areas: 90% and 95% confidence bands.

specific to particular climate phenomena.

Next, we assess the role of country composition. Panel (c) shows that our results are robust to alternative country samples. We begin by expanding the sample using UNWPP mortality data to include a broader set of countries, particularly in Africa. While this entails a trade-off between external validity and data quality, the resulting estimates are close to the baseline. We then examine whether the results are driven by large countries. Excluding India and China—two countries that extend our sample relative to earlier work and receive considerable weight in the baseline—produces comparable estimates, with slightly lower peak effects of around 0.6 deaths per 1,000. Consistent with this result,

estimating unweighted local projections produces very similar results. Taken together, these findings indicate that our estimates are robust to broader geographic coverage and are not driven by large countries or weighting choices.

Finally, Panel (d) examines stability over time by comparing the full sample (1960–2019) to subsamples covering the first four decades (1960–2000) and the most recent four decades (1980–2019). Given our limited sample, we necessarily lose precision when splitting the sample and restricting the analysis to 40 years. For this reason, we report the first ten years of the impulse response function rather than twenty years as in the baseline estimation. We find that the impulse response functions remain close across samples.

Additional sensitivity. Appendix C.3 shows that our results are robust to a range of alternative specification choices. In particular, the estimated mortality response is similar when varying the number of lags of mortality changes, temperature shocks, and global economic controls, as well as under alternative treatments of time trends. We also obtain similar results when specifying mortality in logs and estimating the response as an elasticity rather than in levels of the mortality rate.

4 Understanding the Mortality Effects of Global Warming

What drives the mortality effects of global temperature changes? We answer this question in three steps. First, we ask whether they reflect classic effects operating through extreme heat and cold. We estimate mortality responses to local extreme heat in our sample with a comparable empirical specification and find that they are smaller than global temperature impacts. Second, we show that additional extreme events that correlate with global temperature fluctuations cannot account for the difference either. Third, we show that income reductions associated with global temperature shocks appear to account for the majority of the associated mortality impact.

4.1 Mortality Effects of Local Heat and Cold

Do the effects of global temperature on mortality reflect the well-known effects of exposure to local heat and cold? We answer this question by estimating the effects of local temperature extremes on mortality. We use the following panel local projection specification:

$$m_{i,t+h} - m_{i,t-1} = \alpha_{i,h} + \theta_h^X X_{i,t} + \sum_{\ell=1}^L \rho_{h,\ell}^X X_{i,t-\ell} + \mathbf{z}_t' \boldsymbol{\beta}_h + \mathbf{z}_{i,t}' \boldsymbol{\gamma}_h + \varepsilon_{i,t+h}, \quad (3)$$

where $X_{i,t}$ denotes exposure to a local temperature extreme in country i and year t , and $\mathbf{z}_t$ and $\mathbf{z}_{i,t}$ collect the remaining global and country-specific controls.

We consider both extreme heat and extreme cold, $X \in \{H, C\}$. To construct these measures, we first compute average daily temperature at the national level. Extreme heat exposure is defined as the fraction of days in a given country-year for which daily temperature exceeds a country-specific threshold. We choose the threshold such that average extreme heat exposure equals 0.05 over the historical period 1950–1980. Extreme cold exposure is defined analogously as the fraction of days below a country-specific threshold, which we calibrate such that average exposure equals 0.25 over the same historical period.

For comparability with existing work, we include $X_{i,t}$ directly in the local projection and condition flexibly on its recent history rather than first filtering the extreme-temperature measures and using the resulting innovations as shocks. In particular, we include a rich set of lags chosen to mirror the conditioning information in our Hamilton-filter approach. This allows the coefficient θ_h^X to capture the response to an innovation in local temperature extremes, conditional on their recent realizations.⁸ The remaining controls are the same as in our global temperature specification in (2).

The specification shares several features with the global temperature local projections in (2). In both cases, we trace the cumulative response of mortality over horizon h while controlling flexibly for the history of the climatic variable and other predetermined covariates. The key differences concern the source of identifying variation. Whereas the analysis based on global temperature exploits innovations in aggregate global temperature, the specification in (3) uses cross-country and time-series variation in exposure to local temperature extremes. For comparability with our analysis based on global temperature, the baseline specification in (3) does not include year fixed effects, so that common variation in exposure to extreme temperatures remains part of the identifying variation.

In Appendix D.1.1, we assess the sensitivity of the local extreme responses to alternative controls for common shocks and potential confounders. We re-estimate (3) either including year fixed effects or controlling directly for global temperature shocks. The resulting responses are broadly comparable to the baseline and, if anything, somewhat attenuated.

⁸We also estimate heat and cold exposure jointly. By controlling for lagged realizations of the different temperature extremes, this one-step approach allows us to isolate innovations in each extreme conditional on the other extremes and their recent histories.

Direct vs. indirect effects. Temperature extremes can affect mortality not only through direct physiological exposure, but also indirectly through their effects on economic activity and income. Indeed, Bilal and Känzig (2026) document meaningful GDP effects of temperature extremes. Our baseline mortality responses may therefore reflect both channels. To isolate the component of the mortality response that does not operate through income, we additionally estimate responses conditional on the path of income following the temperature extreme. Following the logic of Holm, Paul, and Tischbirek (2021), we estimate:

$$m_{i,t+h} - m_{i,t-1} = \alpha_{i,h} + \tilde{\theta}_h^X X_{i,t} + \sum_{\ell=1}^L \tilde{\rho}_{h,\ell}^X X_{i,t-\ell} + \sum_{s=0}^h \delta_{h,s} \Delta_s y_{t+s} + \mathbf{z}'_t \tilde{\beta}_h + \mathbf{z}'_{i,t} \tilde{\gamma}_h + \varepsilon_{i,t+h}, \quad (4)$$

where $\Delta_s y_{t+s} = y_{t+s} - y_{t-1}$ denotes the change in global GDP from $t - 1$ to $t + s$. Including these income changes separately at each intermediate horizon allows us to flexibly control for the path of income following the temperature extreme.

The coefficient $\tilde{\theta}_h^X$ therefore measures the mortality response to a local temperature extreme holding global income fixed over the response horizon. We interpret this conditional response as capturing the component of the mortality effect that does not operate through contemporaneous and subsequent changes in income. Comparing $\tilde{\theta}_h^X$ with the baseline response θ_h^X provides a way to assess the quantitative importance of indirect mortality effects of extremes through income.

Appendix Figure D.1 visualizes exposure to both extreme heat and extreme cold for selected countries. While local temperature extremes are related to fluctuations in global mean temperature, they also exhibit substantial country-specific variation. Countries may therefore experience unusually high exposure to heat or cold even in years with relatively modest global temperature fluctuations. This reflects the fact that local extremes are shaped both by changes in global temperature and by regional weather dynamics.

Heat and cold impacts. Figure 6 presents the mortality effects of local extreme heat and cold. To facilitate comparison with the effects of global temperature, we scale the local heat and cold shocks to match the peak changes in extreme-temperature exposure following a 1°C global temperature shock, reported in Appendix Figure D.2. Our extreme heat index has a baseline frequency of 0.05 over 1950–1980, while a 1°C global temperature shock raises the fraction of extreme heat days by about 0.19 at its peak. Thus, the local heat shock corresponds to an almost five-fold increase in the frequency of extreme

heat relative to its historical frequency. Our extreme cold index has a baseline frequency of 0.25 over the same period, and a 1°C global temperature shock predicts a peak decline in the fraction of extreme cold days of about 0.11. The corresponding local cold shock therefore represents a roughly 44% decline in the frequency of extreme cold relative to its historical frequency.

Panel (a) shows that greater exposure to extreme heat raises mortality. A rise in extreme heat increases contemporaneous mortality by about 0.1 excess deaths per 1,000, with the effect statistically significant at the 5% level. The response remains elevated for several years, pointing to effects that extend beyond the immediate consequences of heat exposure. This persistence appears to operate largely through the economic effects of heat: once we keep income fixed over the impulse horizon, the longer-run mortality response is markedly attenuated and no longer statistically significant.

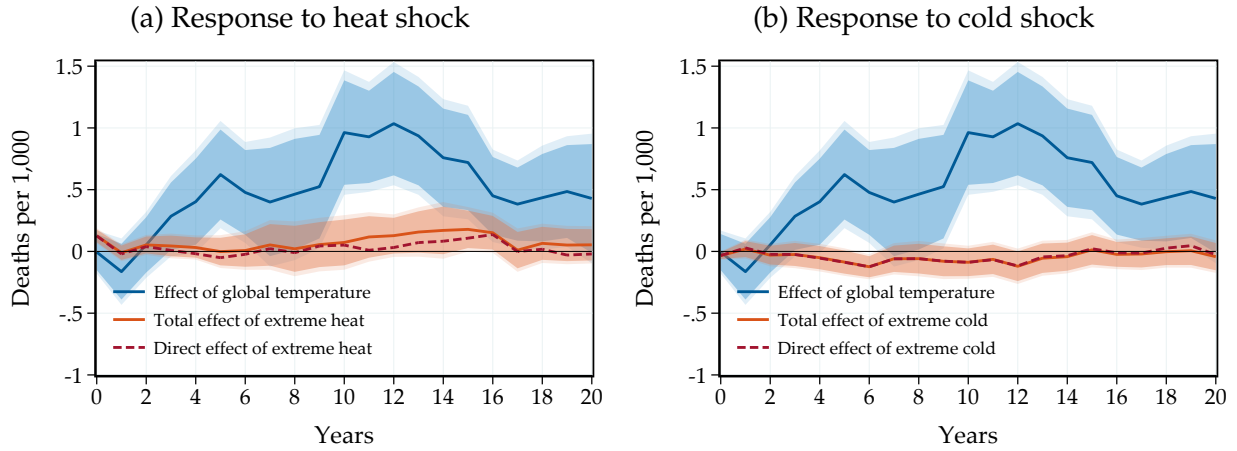
Panel (b) shows the mortality response to a decline in extreme cold of the magnitude induced by a 1°C global temperature shock. Mortality falls by close to 0.1 excess deaths per 1,000 at its peak. While greater exposure to extreme cold raises mortality, a rise in global temperature reduces the incidence of extreme cold. Changes in cold exposure therefore partially offset the adverse mortality effects of exposure to heat.

Taken together, these results suggest that the direct mortality effect of extreme heat is roughly an order of magnitude smaller than the overall mortality response to a global temperature shock. Moreover, exposure to cold operates in the opposite direction.

Accounting for persistent changes in extremes. Is the difference between global temperature and local heat due to specification choices? As discussed above, we scale the underlying heat and cold shocks to match the peak response of extreme local temperature to global temperature shocks. Therefore, the difference in mortality responses cannot be directly attributed to different instantaneous magnitudes of the underlying treatments. However, the timing of the underlying treatments may differ: the dynamic response of local temperature extremes to a local extreme-temperature shock need not coincide with their response to a global temperature shock. Indeed, Appendix Figure D.2 shows that exposure to local temperature extremes responds persistently following a global temperature shock.

To assess the importance of these differences in treatment dynamics, Appendix Figure D.4 imposes the same persistence of extreme-temperature exposure using the method of Sims (1986). We convolve the mortality response to local extreme-temperature shocks

Figure 6: Mortality Impacts of Local Heat and Cold



Notes: Impulse responses of mortality to local extreme heat and cold shocks. To facilitate comparison with the effects of global temperature, we scale the local heat and cold shocks to match the peak changes in extreme-temperature exposure following a 1°C global temperature shock, reported in Appendix Figure D.2. Panel (a) reports the response to extreme heat, and Panel (b) the response to extreme cold. Solid lines show total effects. Dashed lines show effects holding global income fixed over the response horizon. Shaded areas denote 90% and 95% confidence bands based on Driscoll-Kraay standard errors.

with the response of local extremes to a global temperature shock, thereby recovering the mortality effect implied by local extremes under the magnitude and persistence induced by global temperature fluctuations. This adjustment increases the contribution of local extremes, especially for heat. However, the implied mortality response remains well below the aggregate effect of global temperature shocks, with much of the additional effect arising at longer horizons, and is estimated imprecisely.

Is the difference between global temperature and local heat due to how we construct local heat exposure? Existing work often uses local high-degree-day exposure—regional exposure to days with temperature above a certain threshold—rather than the country-level fraction of extreme heat days that we use (Deschênes and Greenstone, 2011; Carleton et al., 2022). To compare our estimates to existing work, we convert them into the mortality effect of a permanent 1°C temperature increase operating solely through local extreme heat. To do so, we take the ratio of the cumulative impulse response of mortality to local heat, and local heat to global temperature.

We estimate that a permanent 1°C warming operating solely through local extreme heat would increase mortality by 0.1 deaths per 1,000 per year when considering only the contemporaneous effect of heat, and by 0.3 deaths per 1,000 when accounting for the full dynamic response. These magnitudes are consistent with previous work: Carleton et al. (2022) likewise estimate an increase of about 0.1 deaths per 1,000 from 1°C warming

when focusing on contemporaneous heat exposure.

Even allowing for delayed effects, the mortality impact attributable to local extreme heat remains roughly an order of magnitude smaller than our estimated global-temperature effect of 2.5 deaths per 1,000. Together with the offsetting effects of reduced extreme-cold exposure, these estimates imply that local heat and cold exposure alone cannot account for the mortality response to global temperature.

4.2 Broader Weather Extremes and Mortality

Global temperature shocks may affect mortality not only through local heat and cold but also through other extreme weather events. The geoscience literature indicates that global temperature—in particular ocean temperature—determines the frequency and intensity of multiple climatic phenomena including droughts, extreme precipitation, and extreme wind (Seneviratne et al., 2016; Wartenburger et al., 2017; Seneviratne et al., 2021; Domeisen et al., 2023).

We next assess whether accounting for the broader effects of global temperature on local weather extremes can explain the gap between the mortality effects of local heat and global temperature. Following Bilal and Känzig (2026), we construct exposure indices for five extreme weather events: extreme heat (national temperature above the 95th percentile, as in Section 4.1) and extreme cold (national temperature below the 25th percentile, as in Section 4.1), as well as drought (national precipitation below the 25th percentile), extreme precipitation (national precipitation above the 99th percentile), and extreme wind (national average wind speed above the 99th percentile).

Appendix Figure D.2 displays the impulse responses of these extreme event indices to global temperature shocks. In addition to the extreme heat and cold impacts discussed above, we find that global temperature shocks also increase the frequency of droughts, extreme precipitation, and extreme wind, with the latter two imprecisely estimated—consistent with the geoscience literature.

We quantify the mortality contribution of these broader weather extremes in three steps. First, we estimate the direct mortality effect of each extreme by regressing mortality on the extreme-event indices, controlling for the same set of covariates as in our baseline specification (3). We estimate the effects jointly to account for the correlation across extreme events. Appendix Figure D.3 reports the resulting responses. The mortality responses to extreme heat and cold remain broadly similar when conditioning on the other extremes. Drought and extreme wind also raise mortality persistently, while

extreme precipitation tends to lower mortality.

Second, we construct the indirect mortality effect of global temperature operating through extreme events. We interact the impulse response of each extreme event to global temperature shocks from Figure D.2 with the mortality impact of that extreme event from Figure D.3. This interaction gives us the mortality effect that would occur if global temperature affected mortality solely through its impact on each specific extreme event.

Third, we aggregate across all extreme events, accounting for their dynamic correlation with global temperature shocks. Specifically, we compute: $\theta_h^{\text{extremes}} = \sum_X \sum_{s=0}^h \phi_s^X \psi_{h-s}^X$, where ϕ_s^X is the response of extreme event X to a global temperature shock at horizon s , and ψ_{h-s}^X is the mortality impact of a transitory increase in extreme event X at horizon $h - s$. The sum over X includes extreme heat, extreme cold, droughts, extreme precipitation, and extreme wind.

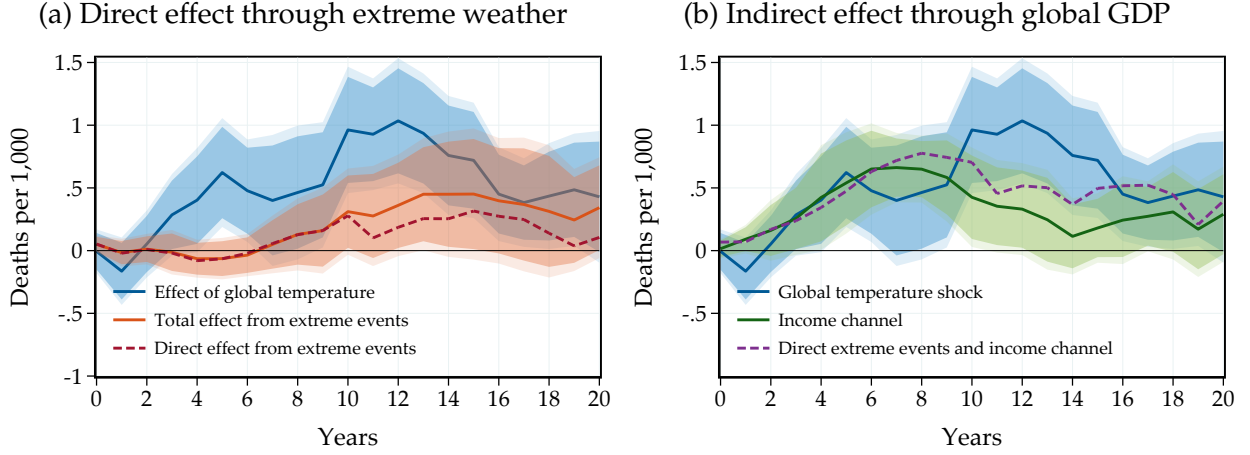
We find that extreme events can account for approximately 0.1–0.4 deaths per 1,000 at longer horizons, as shown in Figure 7(a). The solid red line reports the implied mortality response obtained by aggregating the effects of individual extreme events. When we condition on income to isolate the direct effects of extreme-weather exposure, the implied response drops to less than one quarter of the cumulative mortality effect of global temperature, as shown by the dashed red line. While this contribution is economically meaningful, it falls short of accounting for the sizable mortality response to global temperature shocks shown in blue.

4.3 The Income Channel

Why, then, does global temperature affect mortality more than local heat and other extreme weather events? In this section, we ask whether macroeconomic disruptions caused by global temperature fluctuations increase mortality due to a link between mortality and income. First, we assess the role of income directly by quantifying its contribution, combining separate estimates of the effect of global temperature on income and of the effect of income on mortality. Second, we present corroborating evidence consistent with income playing an important role in the transmission of global temperature shocks to mortality.

GDP losses and mortality. We start by investigating how global temperature affects mortality through its impact on economic output and income. Bilal and Känzig (2026) document that global temperature shocks cause large and persistent declines in global GDP per capita: a permanent increase in global mean temperature of 1°C leads to a 22%

Figure 7: Direct Exposure to Extremes and Indirect Income Effects



Notes: Solid blue lines: direct estimate θ_h of the effect of global temperature on mortality. Panel (a): solid red line: indirect estimate $\theta_h^{\text{extremes}}$ obtained by aggregating the five extreme events; dashed dark red line: the same aggregation holding global income fixed over the response horizon. Panel (b): solid green line: indirect estimate $\theta_h^{\text{indirect}}$ of the effect of global temperature on mortality, mediated by global GDP per capita; dashed purple line: sum of the income channel and the direct effect of extreme events. Shaded areas: 90 and 95% confidence bands based on Driscoll-Kraay standard errors.

reduction in global GDP. If income losses in turn increase mortality—as suggested by the development and health literature (Preston, 1975; Cutler, Deaton, and Lleras-Muney, 2006)—then this income channel may help rationalize the effect of global temperature on mortality.

To assess this mechanism quantitatively, we construct an indirect measure of the effect of global temperature on mortality through income. We interact two components. The first component is the impact of global temperature shocks on global GDP per capita. Appendix Figure D.9 reproduces the analysis from Bilal and Känzig (2026) in our sample of countries, with comparable controls to our mortality specification and up to a 20 year horizon. We obtain similar results.

The second component is the effect of GDP on mortality. We address two challenges in constructing this component. The first challenge is to choose the relevant measure of GDP. Is country-level GDP or worldwide GDP relevant for mortality across countries? A large literature going back to Preston (1975) suggests that mortality reductions over time arise partly because of country-level income growth. However, an important part of mortality reductions appear to take place independently from country-level income changes, but may instead be associated with income changes at the economic frontier, and thus might be more tightly linked to worldwide income changes. For this reason and for consistency with the estimates in Bilal and Känzig (2026), we focus on global GDP as

our main mediating variable.

The second challenge is to obtain an estimate of the effect of global GDP on mortality. We use the estimates from Bilal, Känzig, and Lisiecki (2026) who leverage global GDP fluctuations in the time series. They show that global GDP shocks—unconditionally and when instrumented with U.S. monetary policy surprises to address confounding forces such as epidemics or geopolitical events—lead to larger long-run mortality elasticities than local GDP shocks. Their estimates imply that a 100 log point increase in global GDP per capita reduces mortality by approximately 6 deaths per 1,000 in the long run. For reference, we reproduce their estimates in Appendix Figure D.10.

With these estimates at hand, we construct the indirect mortality response to a global temperature shock by convolving the GDP response to temperature with the mortality response to GDP:

$$\theta_h^{\text{indirect}} = \sum_{s=0}^h \phi_s^{\text{GDP}} \psi_{h-s}^{\text{mort|GDP}}. \quad (5)$$

Here, ϕ_s^{GDP} is the response of world GDP per capita to a global temperature shock at horizon s from Figure D.9. The coefficients $\psi_{h-s}^{\text{mort|GDP}}$ are not estimated directly: they represent the mortality impact of a transitory GDP shock at horizon $h - s$ as implied by our GDP shock or monetary policy IV estimates.

Figure 7(b) presents the results. The indirect estimate tracks the direct temperature–mortality response closely. Using the full dynamic convolution, global GDP losses account for 66% of the cumulative mortality impact of global temperature shocks.

We also assess the additional contribution of extreme weather channels—including heat, cold, droughts, precipitation, and wind—beyond income-mediated effects that are partly captured in the global GDP contraction that follows a global temperature shock. These direct extreme weather impacts account for 23% of the total mortality response.

Overall, these results indicate that global temperature shocks have large mortality effects primarily because they generate persistent global income losses. Estimates based solely on exposure to extreme heat understate the overall mortality consequences of warming because they omit mortality impacts mediated by global income.

Heterogeneity by cause of death and country. We corroborate the interpretation that global temperature shocks lead to an increase in mortality because of associated income losses with two additional sets of results. First, we show that deaths following global temperature shocks are primarily cardiovascular- and HIV-related, not heat- and cold-related.

Second, we show that countries with a lower fraction of public healthcare spending display larger mortality responses.

We first describe which causes of death are responsible for the estimated response of mortality to global temperature. For this purpose, we use data from the Institute for Health Metrics and Evaluation (IHME) covering 31 years and 198 countries. Of course, time-series variation is necessarily more challenging to exploit in this shorter sample. These results should thus be interpreted with more caution.

Figure 8(a) starts by reproducing our estimates in this dataset. Reassuringly, the estimates in this different dataset are very similar to our baseline in Figure 3. We limit our analysis to a horizon of 10 years due to the shorter sample.

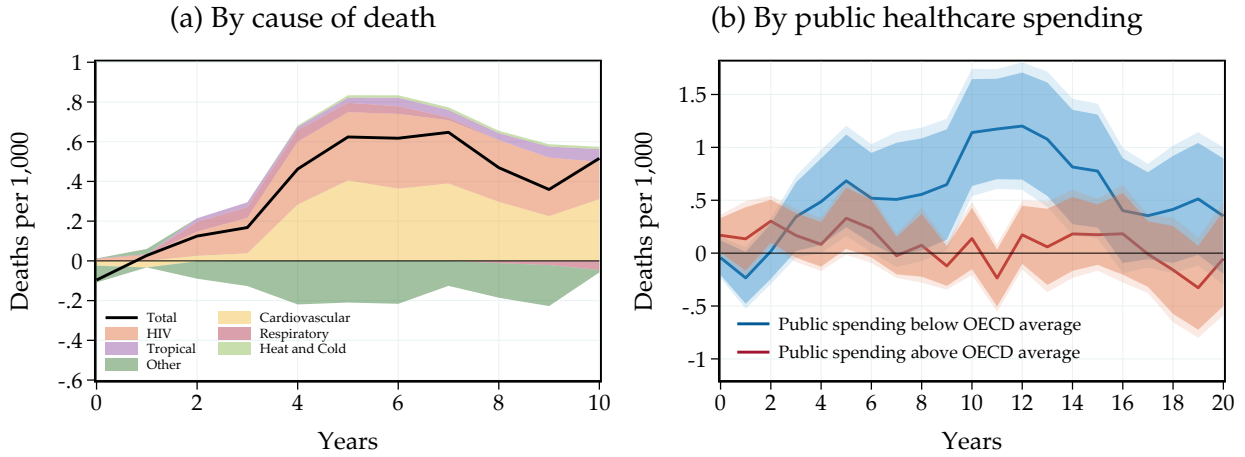
The figure then disaggregates the rise in the crude death rate by cause of death. Cardiovascular-related causes are the leading cause of death following global temperature shocks. As for the overall impulse response function, these deaths peak several years after the shock. The magnitude exceeds the estimated impact of heat on mortality from Figure 6 by an order of magnitude, suggesting that most of these cardiovascular excess deaths are not due to heat exposure.

HIV-related deaths are the second most important cause of death after global temperature shocks. As for cardiovascular deaths, they peak several years after the shock occurs. It is not obvious how to directly link HIV deaths to extreme weather exposure. By contrast, HIV treatments are expensive and can be tightly linked to worldwide income. For instance, the United States President's Emergency Plan for AIDS Relief (PEPFAR) has historically funded antiretroviral therapy for two-thirds of all antiretroviral patients worldwide, who themselves represent three quarters of all infected patients.

By contrast, heat- and cold-related deaths represent a small fraction of overall deaths. This result is consistent with those in Figure 6, showing that direct effects of extreme heat exposure are small relative to the overall effects of global temperature shocks.

Next, we study which countries display stronger mortality responses to global temperature shocks. Many countries in Europe and Oceania have near-universal healthcare systems that offer cheap access for individuals, and may thus be less dependent on the global economy. By contrast, the United States' healthcare system is more reliant on employer-based private insurance and a sizable fraction of the population is not insured, thereby relying more directly on individual income. Similarly, in lower-income countries, public healthcare coverage tends to be more limited and out-of-pocket costs are more important.

Figure 8: Mortality Impacts By Cause of Death and By Public Healthcare Spending



Notes: Panel (a): Mortality impacts of global temperature shocks by cause of death, deaths per 1,000. Data: Health Metrics and Evaluation, covering 31 years and 198 countries. Black line: total mortality response. Colored areas: contributions by cause of death. Panel (b): Mortality impacts of global temperature shocks by public spending as a fraction of total healthcare spending, deaths per 1,000. Solid lines: point estimates. Shaded areas: 90 and 95% confidence bands.

We first estimate heterogeneous effects by broad regions of the world. Appendix Figure D.7 reveals that Europe and Oceania exhibit weak and largely statistically insignificant mortality responses to global temperature shocks. North America, Asia and Latin America display mortality responses that are all close to the world average and peak between 1 and 2 excess deaths per 1,000. Despite data limitations, Africa exhibits the largest mortality response of all regions, which peaks above 2 excess deaths per 1,000.

Second, we directly test whether countries with higher shares of public healthcare spending exhibit different responses to global temperature shocks. We classify countries as high public spending in healthcare if their fraction of public spending in total healthcare expenditure exceeds the OECD average value of 72%. Figure 8(b) shows that countries with a share of public healthcare spending above the OECD average exhibit no significant response of mortality to global temperature shocks. By contrast, countries with public healthcare spending below the OECD average exhibit larger and statistically significant mortality responses to global temperature shocks.

That the mortality impacts from global temperature shocks transmit primarily through causes of death and countries where healthcare is income-sensitive is qualitatively consistent with a transmission channel through worldwide income.

5 A Model of Climate Change, Mortality and the Economy

To study the long-run economic and health consequences of climate change, we develop a model of climate change, mortality, and the economy. Our framework builds on the standard neoclassical growth model that underpins integrated assessment models in the spirit of Nordhaus (1992). We enrich this framework by introducing three elements following Bilal, Känzig, and Lisiecki (2026). First, we include mortality risk. Second, we allow for private health expenditures that reduce mortality and allow individuals to adapt to climate change. Third, we allow for health spillovers that are tied to aggregate output. In addition, we allow for damage functions that relate productivity and mortality to temperature. We discipline the model using our reduced-form estimates of the impact of global temperature shocks on output and mortality, which we use to structurally identify temperature damage functions.

5.1 Setup

Preferences. Time is continuous and runs forever. There is a unit continuum of infinitely-lived identical households who populate the world economy. Households have constant relative risk aversion flow preferences over a consumption good which we use as the numeraire: $U(C) = \frac{C^{1-\gamma}-1}{1-\gamma}$. Labor supply is exogenous and set to $L_t = 1$. The pure rate of time preference of households is ρ .

Households face stochastic lifespans τ as in Blanchard (1985). When they die, they receive a terminal value reduction $\bar{V}$ and give rise to a single offspring with respect to which they are fully altruistic. Households thus have the following dynamic preferences:⁹

$$V_0(K_0) = \mathbb{E}_0 \left[\int_0^\tau e^{-\rho t} U(C_t) dt + e^{-\rho \tau} (V(K_\tau) - \bar{V}) \right]. \quad (6)$$

Lifespans end stochastically following a Poisson process with time-varying intensity M_t .

Final good production. Final good firms produce the final consumption good with a Cobb-Douglas production function in capital K_t and labor L_t with time-dependent Total Factor Productivity (TFP) Z_t : $Y_t = Z_t K_t^\alpha L_t^{1-\alpha}$. They hire labor at wage w_t and rent capital

⁹We use this dynastic formulation rather than annuities to focus on efficiency properties of investment due to healthcare improvements and avoid other sources of dynamic inefficiency. The consequence is that disutility from death $\bar{V}$ must be introduced as a one-time cost rather than as a flow shift in the utility function as in Hall and Jones (2007) and Jones and Klenow (2016), in which case it would become neutral.

from households at net interest rate r_t in competitive factor markets. Capital depreciates at rate δ , which is constant over time and covered by firms. The path of productivity Z_t is perfectly foreseen.

Health and mortality. Households can spend final goods on health. We denote their health spending by H_t . Healthcare purchases reduce mortality. Mortality reductions also occur externally to healthcare expenditures due to income growth and to exogenous factors X_t . The mortality rate M_t follows:

$$M_t = \bar{M} X_t H_t^{-\mu} Y_t^{-\nu}. \quad (7)$$

The parameter μ governs the effect of healthcare expenditures on mortality, while ν governs the effect of aggregate output on mortality. Consistent with our results on the global income channel in Section 4.3, we interpret aggregate output Y_t in equation (7) as the model counterpart to global GDP, while all other variables are understood to be defined at the country level. We omit country subscripts whenever unambiguous.

Household budget constraint. Firms make zero profits given constant returns to scale, so we omit profits in the budget constraint of the household, which writes: $C_t + H_t + \dot{K}_t = w_t + r_t K_t$. Households are endowed with an initial capital stock K_0 .

Climate change and damages. We model climate change as changes in productivity Z_t and the mortality shifter X_t over time, relative to their baseline values Z_0 and X_0 . This representation follows existing work (Nordhaus, 1992; Moore and Diaz, 2015; Carleton et al., 2022) and parsimoniously but comprehensively captures the combined economic and health consequences of excess global temperature documented in the data.

We take the path of global mean temperature T_t relative to a reference level T_0 as given, and denote by $\hat{T}_t \equiv T_t - T_0$ the path of excess temperature. Global mean temperature affects productivity and mortality through structural damage functions $\{\zeta_s, \xi_s\}_{s \geq 0}$:

$$Z_t = Z_0 \exp \left(\int_0^t \zeta_s \hat{T}_{t-s} ds \right) \quad (8)$$

$$X_t = X_0 \exp \left(\int_0^t \xi_s \hat{T}_{t-s} ds \right). \quad (9)$$

The structural damage functions $\{\zeta_s, \xi_s\}$ govern the persistence of the effects of transitory global temperature shocks on productivity and mortality. When temperature is

constant at $T_t \equiv \bar{T}$, the economy converges to a steady state with corresponding productivity and mortality levels determined by the cumulative damage functions, $\bar{Z} = Z_0 \exp((\bar{T} - T_0) \int_0^\infty \zeta_s ds)$ and $\bar{X} = X_0 \exp((\bar{T} - T_0) \int_0^\infty \zeta_s ds)$. The cumulative damage functions $\int_0^\infty \zeta_s ds$ and $\int_0^\infty \xi_s ds$ thus govern the long-run effects of permanent warming.

We abstract from feedback between economic activity, emissions, and temperature dynamics, as our focus is on climate damages.

Competitive equilibrium. A competitive equilibrium is a collection of sequences $\{C_t, K_t, r_t, w_t, M_t, Z_t, X_t\}_{t=0}^\infty$ such that households and firms optimize given prices, mortality and productivity evolve according to the temperature damage functions, and factor markets clear: $K_t = K_t^D$ and $1 = L_t^D$.¹⁰

Adaptation to mortality impacts. Our framework allows for adaptation to mortality impacts of climate change. As we show in Appendix E.1, demand for final good consumption and health implies that:

$$H_t = \bar{H} X_t^{\frac{1}{1+\mu}} C_t^{\frac{\gamma}{1+\mu}} Y_t^{-\frac{\nu}{1+\mu}} \quad M_t = \tilde{M} X_t^{\frac{1}{1+\mu}} C_t^{-\frac{\mu\gamma}{1+\mu}} Y_t^{-\frac{\nu}{1+\mu}}. \quad (10)$$

where $\bar{H}$ and $\tilde{M}$ are constants that depend on parameters and are given in Appendix E.1.

Health demand is increasing in the mortality shifter X_t : when climate change raises mortality, households demand more health to offset some of this excess mortality risk. Higher consumption also raises health demand: when the marginal utility of consumption is low, it is more advantageous to spend on health to lower mortality. By contrast, higher output lowers health spending: when aggregate income spillovers already lower mortality, households substitute from health toward final good consumption.

The equilibrium level of mortality reflects adaptation. Higher mortality impacts from climate change increase mortality, but by less than without taking adaptation into account. The elasticity of mortality M_t to climate damages X_t is $\frac{1}{1+\mu}$ after adaptation instead of 1 before adaptation. Therefore, the elasticity of mortality to health spending is a key determinant of the strength of adaptation in the framework.

Similarly, higher aggregate output Y_t continues to lower mortality even after adaptation, but with a lower elasticity $\frac{\nu}{1+\mu}$ than before adaptation. Finally, higher consumption

¹⁰We show in Appendix E that the decentralized equilibrium is efficient for $\nu = 0$ and inefficient for $\nu \neq 0$. Households do not internalize the health spillovers of income growth. We highlight this property with the Euler equation: $\gamma_{C_t}^C = \alpha Z_t K_t^{\alpha-1} - \rho - \delta + \mathbf{I}^{\text{SP}} \alpha \nu \frac{M_t}{K_t u'(C_t)} \bar{V}$, where $\mathbf{I}^{\text{SP}} = 1$ in the efficient allocation, and zero in the decentralized equilibrium.

implies lower mortality as households spend more on health.

Social Cost of Carbon. In our framework, we define the Social Cost of Carbon (SCC) as the one-time dollar amount $\mathcal{C}$ that households would pay at time 0 to make them indifferent between a world with an additional ton of CO₂ emitted at time 0, and a world starting in steady state, without emissions, but having paid $\mathcal{C}$.

Given that we do not model emissions directly, we must map a one-time CO₂ pulse into a temperature path in order to calculate the SCC. We follow Folini et al. (2024) and use the temperature response of global mean temperature to a CO₂ pulse from Dietz et al. (2021a) and Joos et al. (2013).

We denote by $\{\hat{T}_t^{\text{SCC}}\}_{t \geq 0}$ the path of excess warming implied by a one-time pulse of one ton of CO₂ emitted at time 0. The median response in Dietz et al. (2021a) indicates that after a 1 gigaton pulse of carbon, temperature rises steadily and eventually stabilizes at 0.002°C after 15 years.¹¹ Our welfare numbers do not depend on the temperature response to a CO₂ pulse, but instead on a particular warming scenario.

We then construct productivity and mortality paths $\{Z_t^{\text{SCC}}, X_t^{\text{SCC}}\}_{t \geq 0}$ according to equations (8)-(9) in which we use the temperature path $\{\hat{T}_t^{\text{SCC}}\}_{t \geq 0}$ rather than a global warming scenario. The model delivers a path of value functions $\{V_t^{\text{SCC}}(K)\}_{t \geq 0}$, equilibrium capital stocks $\{K_t^{\text{SCC}}\}_{t \geq 0}$ with initial condition $K_0 = K^{\text{ss}}$, leading to a path of realized values $\{V_t^{\text{SCC}}(K_t^{\text{SCC}})\}_{t \geq 0}$, in response to this CO₂ pulse-induced warming. Our definition requires that the SCC $\mathcal{C}$ be given implicitly by: $V_0(K^{\text{ss}} - \mathcal{C}) = V_0^{\text{SCC}}(K^{\text{ss}})$. Since a single ton of carbon implies marginal losses, we can use the first-order formula:

$$\mathcal{C} = \frac{V_0(K_0) - V_0^{\text{SCC}}(K_0)}{u'(C_0)}. \quad (11)$$

Equation (11) and our specification of preferences in equation (6) ensure that the SCC $\mathcal{C}$ is additively separable into welfare losses from consumption and welfare losses from mortality. We exploit this property in our quantitative decompositions.

5.2 Estimation

We start by setting all parameters unrelated to adaptation or damages to standard values from the literature. Specifically, we set risk aversion to $\gamma = 1$, the capital share to $\alpha = 0.33$,

¹¹For brevity, we refer to the ‘best fit CMIP5 ensemble’ response in Dietz et al. (2021a) as ‘median response’.

and the annual capital depreciation rate to $\delta = 0.08$. Our choice of annual rate of time preference is $\rho = 0.02$, following Rennert et al. (2022), consistent with a 2% annual interest rate in steady state.

Estimating adaption parameters. Next, we estimate the elasticity of mortality to health spending μ and aggregate output ν as in Bilal, Känzig, and Lisiecki (2026). We identify these elasticities by targeting the empirical impulse response functions of mortality to country-level GDP shocks and to global GDP shocks that we obtain from Bilal, Känzig, and Lisiecki (2026).

To do so, we interpret our model as representing a given country i , except for aggregate output Y_t in the health production function, which we interpret as global GDP. For any given country i in year t , equilibrium mortality (10) implies that:

$$\log M_{it} = a_i - \frac{\nu}{1 + \mu} \log Y_t - \frac{\mu\gamma}{1 + \mu} \log C_{it} + \frac{1}{1 + \mu} \log X_{it}, \quad (12)$$

where the a_i denote country fixed effects. When we include time fixed effects, they absorb the effect of spillovers from global GDP to mortality, implying that the estimation is block-recursive and we can estimate μ first. We leverage these observations for estimation.

We assume that idiosyncratic country-level productivity shocks Z_{it} generate variation in country-level consumption C_{it} and country-level output Y_{it} . We assume that these productivity shocks are uncorrelated with the residual mortality shocks in $\log X_{it}$. After log-linearization, we invert the model to recover the sequence of idiosyncratic shocks Z_{it} that exactly rationalizes the empirical impulse response functions of GDP to a country-level GDP shock (defined similarly to temperature shocks). We then estimate μ so that the corresponding empirical and model impulse response functions of mortality to the same GDP shock coincide. We obtain $\mu = 0.12$, and report the model fit in Appendix Figure F.1.

We then turn to the estimation of ν by unpacking the time fixed effect. We assume that fluctuations in global GDP are driven by common productivity shocks Z_t , which thus also affect country-level consumption C_{it} , and estimate ν so that the empirical and model impulse response functions coincide. We obtain $\nu = 0.24$, and report the model fit in Appendix Figure F.2.

Our choice of empirical targets implies that the strength of private adaptation is governed by the empirical relationship between country-level income per capita and mor-

tality. If mortality falls steeply when a country becomes richer—holding aggregate GDP constant—then private adaptation is potent. If, by contrast, mortality falls moderately when a country becomes richer, private adaptation is weaker. The literature relating country-level income and mortality has consistently found that this relationship is moderate at most (Preston, 1975; Cutler, Deaton, and Lleras-Muney, 2006), which is also the case in the impulse response functions in Bilal, Känzig, and Lisiecki (2026). The resulting estimate of $\mu = 0.12$ thus implies that the strength of private adaptation is moderate.

By contrast, adaptation through aggregate economic growth is informed by the empirical relationship between global income per capita and mortality. As shown in Bilal, Känzig, and Lisiecki (2026), this empirical relationship is approximately four times larger than the corresponding relationship between country-level income per capita and mortality. The resulting estimate of $\nu = 0.24$ thus implies that the strength of adaptation through aggregate economic growth is likely stronger.

With these elasticities in hand, we estimate the shifter of the health production function $\bar{M}$ and the disutility of end of life $\bar{V}$. We set $\bar{M}$ to target an average global life expectancy of 70 years in 2019. We set the disutility of end of life using the identity: $VSL = \frac{\bar{V}}{u'(C)}$, where VSL denotes an empirical value of statistical life estimate. We start from an estimate of 7 million USD. Consistent with our identity under $\gamma = 1$, we further divide the VSL by United States per capita consumption. This transformation corresponds to our “income adjustment.” Finally, we adjust it proportionally to the number of remaining life-years for an individual older than 65 relative to an average adult. This transformation corresponds to our “age adjustment,” which conservatively assumes that all mortality effects of climate change are borne by individuals older than 65.

Estimating damage functions. Our next step is to estimate the structural damage functions $\{\zeta_s, \xi_s\}_{s \geq 0}$ by matching the reduced-form impulse response functions of output and mortality to global temperature shocks documented in Section 3.1.

Our estimation accounts for the fact that global temperature shocks affect output and therefore mortality. The mortality damage function ξ thus only captures residual reasons why mortality responds to global temperature, above and beyond output. Define $y_t = \log Y_t$ and $m_t = \log M_t$, and denote deviations from steady state with hats. We obtain the following identification result:

Proposition 1. (*Model inversion*)

There exist $\mathcal{K}_{t,s}^z, \mathcal{K}_{t,s}^m, \mathcal{M}_{t,s}^z, \mathcal{M}_{t,s}^m$ given in Appendix E.4.2, that only depend on steady-state ob-

jects and are independent from $\{\zeta_s, \tilde{\zeta}_s\}_{s \geq 0}$, such that, to a first order in $\{\hat{T}_t\}_{t \geq 0}$:

$$\begin{aligned}\hat{y}_t &= \hat{z}_t + \alpha \int_0^\infty \mathcal{K}_{t,s}^z \hat{z}_s ds + \alpha \int_0^\infty \mathcal{K}_{t,s}^m \hat{x}_s ds \\ \hat{m}_t &= \frac{\hat{x}_t - \nu \hat{z}_t}{1 + \mu} + \int_0^\infty \mathcal{M}_{t,s}^z \hat{z}_s ds + \int_0^\infty \mathcal{M}_{t,s}^m \hat{x}_s ds.\end{aligned}$$

Proof. See Appendix E.4.2. □

Proposition 1 delivers an identification result. Given the observed output and mortality responses $\hat{y}_t, \hat{m}_t$, we can recover the underlying sequence of productivity and mortality shifter shocks $\hat{z}_t, \hat{x}_t$. The first component in each equation of Proposition 1 corresponds to the direct effect of productivity and mortality shocks on output and mortality, respectively. The second component in each equation corresponds to the equilibrium response of capital, output and consumption, which ultimately feed back into mortality. It is an integral over all times because investment is forward-looking and capital accumulates over time. This second component involves the sequence-space Jacobians $\mathcal{K}_{t,s}^z, \mathcal{K}_{t,s}^m, \mathcal{M}_{t,s}^z, \mathcal{M}_{t,s}^m$ (Auclert et al., 2021; Bilal and Goyal, 2023). In the context of our model, these Jacobians admit closed-form expressions as a function of parameters and steady-state objects. When the corresponding mapping is invertible, productivity and mortality shocks are identified. Proposition 1 allows us to obtain the sequence of productivity and mortality shocks $\hat{z}_t, \hat{x}_t$ that correspond to any sequence of temperature shocks $\hat{T}_t$.

We use Proposition 1 to estimate $\zeta_s, \tilde{\zeta}_s$. We consider the response of output to an observed temperature shock from Bilal and Känzig (2026). We consider the response of mortality to an observed temperature shock from Figure 3, which corresponds to the underlying temperature path $\hat{T}_t$ in the same figure. Proposition 1 delivers the corresponding sequence of shocks $\hat{z}_t, \hat{x}_t$. We then identify $\zeta_t, \tilde{\zeta}_t$ as the innovations to these sequences as per equations (8)-(9): $\hat{z}_t = \int_0^t \zeta_s \hat{T}_{t-s} ds$ and $\hat{x}_t = \int_0^t \tilde{\zeta}_s \hat{T}_{t-s} ds$.

This approach is consistent with rational expectations. After a temperature shock, agents anticipate that temperature remains persistently elevated, as observed in the data. Thus, the identified damage functions are disciplined by impulse responses to shocks that are themselves persistent, rather than purely transitory. Our estimation takes into account the internal persistence of realized temperature.

In practice, we face two challenges. First, the empirical impulse responses are observed only over a finite horizon, whereas the inversion requires the entire path. Second, unconstrained estimation could imply large but imprecise long-run effects of transitory

temperature shocks. We address both issues by imposing smooth parametric forms for the structural damage functions. Specifically, we constrain $x_s = A_x(e^{-B_x s} - e^{-C_x s})$ for $x \in \{\zeta, \xi\}$. We estimate the parameters by nonlinear least squares, matching model-implied and empirical impulse responses over the observed horizon.

Figure 9 presents the estimated damage functions. The first panel reproduces the underlying temperature path from Figure 3. The second panel shows the mortality and output responses to the temperature shock—in the data and based on our model. The estimated model closely fits the empirical responses of mortality, despite its limited degrees of freedom. Of course, the model fit relies on our constrained functional form: if we did not constrain the damage function, the fit would be one-to-one.

The third panel depicts the estimated mortality damage function ζ_s . It peaks at 0.17 deaths per 1,000 and exhibits substantial persistence up to 20 years after the shock. We compute confidence bands with the Delta method accounting for the variance-covariance matrix between the empirical mortality and GDP impulse response functions.

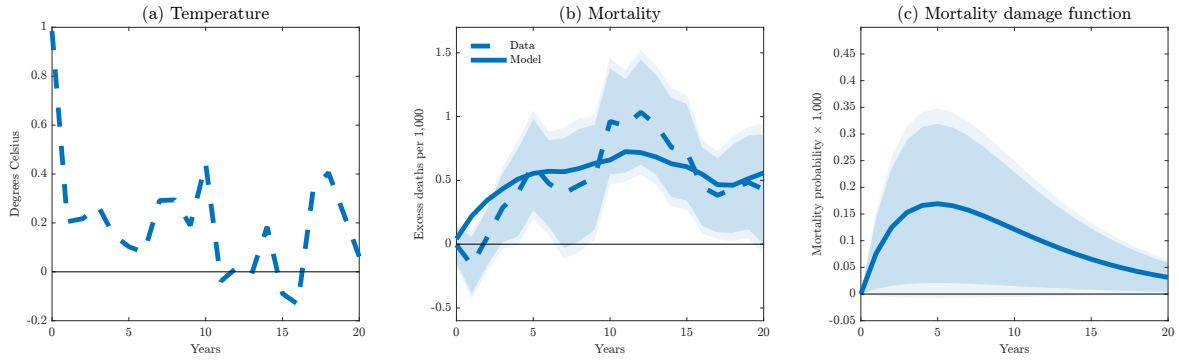
Why is the estimated mortality damage function borderline significant, while the mortality impulse response function in panel (b) is strongly significant? A large fraction of the mortality response in panel (b) is already accounted for by indirect effects operating through aggregate output. Therefore, the mortality damage function in panel (c) only relies on the gap between the mortality impulse response function in panel (b) and, given the health production function (7), the mortality path predicted from the empirical output response in Bilal and Känzig (2026) and reproduced in Appendix Figure F.3. This gap is closer to zero than the response in panel (b), and therefore the mortality damage function that controls direct effects is only borderline significant.

How do the mortality effects of global temperature shocks compare to those associated with local heat shocks? Given that the empirical responses for local heat shocks are closer to zero as shown in Figure 6, such shocks likely also imply smaller damages.

To answer this question quantitatively, we re-estimate our damage functions targeting the mortality and output impulse responses to local heat shocks.¹² For comparability with the literature, we target the initial mortality response to extreme heat only and exclude its lagged responses, which, in any case, are imprecisely estimated. Doing so yields a mortality damage function that peaks at approximately 0.11 deaths per 1,000, is transitory, and is not statistically different from zero at the 5% level (see Appendix Figure F.4).

¹²To remain consistent with previous work, we use the output response to average annual temperature for output, and the mortality response to local extreme heat.

Figure 9: Mortality Damage Function Estimates



Notes: Estimated mortality damage functions obtained by matching model-implied impulse responses to the empirical responses to global temperature shocks. Dashed lines: empirical responses. Solid lines: model fit. Dark and light shaded areas: 90 and 95% confidence bands, computed using the Delta method and accounting for the variance-covariance matrix between empirical GDP and mortality impulse response functions.

6 Climate Change, Mortality, and Welfare

With the estimated damage functions at hand, we are now in a position to evaluate the effects of climate change on mortality, the economy, and welfare.

6.1 Climate Change in the 21st Century

We start from a reference scenario that includes 2% annual exogenous productivity growth, consistent with historical evidence. To evaluate the consequences of climate change, we specify an exogenous path for global mean temperature. The baseline year $t = 0$ corresponds to 2026. The world subsequently warms by 2°C above current levels by 2100—and thus approximately 3°C above pre-industrial levels—after which temperature asymptotes to 2.2°C . This scenario is broadly consistent with IPCC business-as-usual pathways implying 3 to 4°C of warming by 2100 (Hausfather and Peters, 2020; Lee et al., 2023).

We evaluate the transitional dynamics given the specified warming path under three alternative damage specifications. First, we use the structural damage functions estimated from global temperature shocks. Second, we construct counterfactuals using the damage functions estimated from local heat and cold shocks within our framework. Third, we incorporate mortality damages implied by Carleton et al. (2022), mapped into our model environment. This comparison isolates how different empirical identification strategies translate into long-run mortality, welfare, and SCC implications under the same temperature trajectory.

In each case, we compare allocations and welfare in an economy that warms according to $\hat{T}_t$ to an economy under $\hat{T}_t = 0$ that still experiences 2% productivity growth. Welfare losses are reported as equivalent percent declines in initial consumption. The SCC, expressed in 2026 international dollars, is defined in equation (11). It depends on the temperature response to a marginal CO₂ pulse and is therefore independent of the specific warming scenario considered here. All counterfactuals are computed using the fully nonlinear model. Log-linearization is used only for estimation.

These exercises assess the consequences of sustained multi-degree warming using damage functions estimated from smaller temperature shocks. While households optimally adjust consumption and investment in anticipation of future warming, the framework abstracts from additional behavioral and institutional adaptations beyond those embedded in the neoclassical structure.

6.2 Aggregate Mortality Impacts

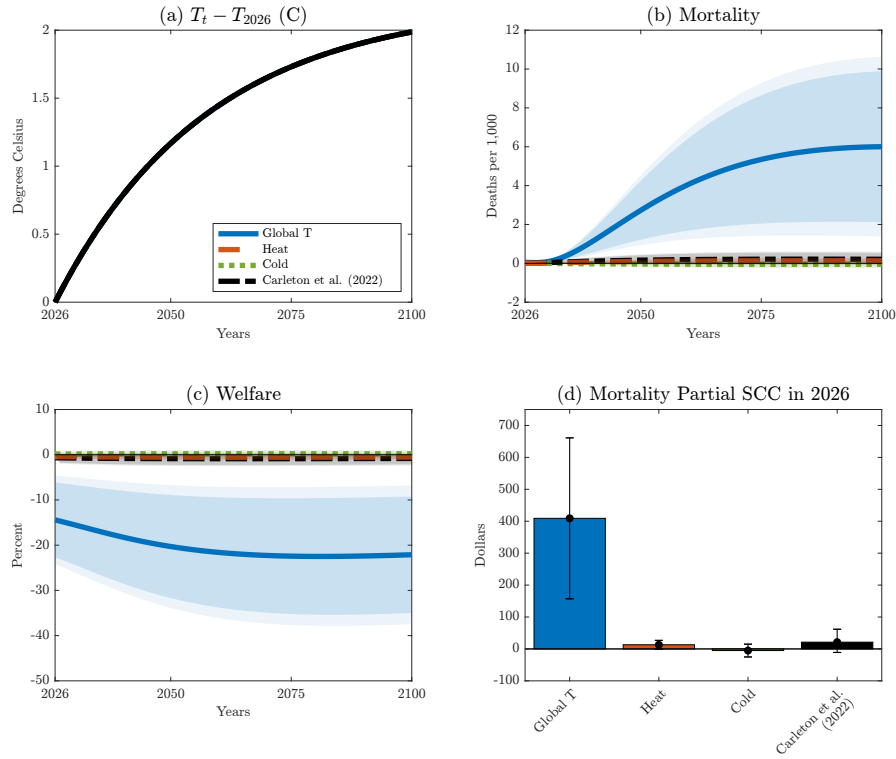
We start by focusing on mortality. Importantly, our counterfactuals include indirect mortality effects through productivity-induced output losses, which are key drivers of mortality through the health production function (7). We isolate direct and indirect effects of climate change on mortality in Section 6.4. Figure 10 presents the results, with Panel (a) showing the temperature path under our warming scenario.

Panel (b) displays implied excess mortality per 1,000 individuals. Under damages based on global temperature, mortality rises steadily throughout the century, reaching around 6 excess deaths per 1,000 by 2100. The increase reflects both the continued rise in temperature and the persistence embedded in the estimated mortality damage function. Even though annual temperature increases are gradual, their cumulative effects generate substantial additional mortality by the end of the century.

In contrast, when damages are constructed using local heat and cold estimates, the mortality effects are considerably smaller and statistically indistinguishable from zero. Appendix Figure F.6 focuses on these impacts for better visibility. This difference mirrors our reduced-form estimates: global temperature shocks generate large and persistent mortality responses, whereas local heat and cold shocks imply smaller and largely transitory effects. Reassuringly, our mortality impacts under local heat and cold are broadly consistent with existing estimates (see e.g. Carleton et al., 2022).

Panel (c) reports welfare losses as consumption-equivalent declines associated with the mortality impacts. Under global temperature damages, the welfare impact of climate

Figure 10: Transitional Dynamics Under Climate Change: Mortality Damages



Notes: Transitional dynamics of the estimated model under the scenario in Panel (a) allowing for mortality damages only. Solid blue lines: model estimated under global temperature. Other lines: model estimated under alternative damage functions: local heat (red dashed), local cold (green dotted), Carleton et al. (2022) damage function (black dash-dotted). The heat and cold damage functions are estimated from the impact effect of local heat and cold exposure only. Shaded areas: 90 and 95% confidence intervals based on the Delta method and accounting for the variance-covariance matrix between empirical GDP and mortality impulse response functions. Panel (d): bars, with 95% confidence intervals.

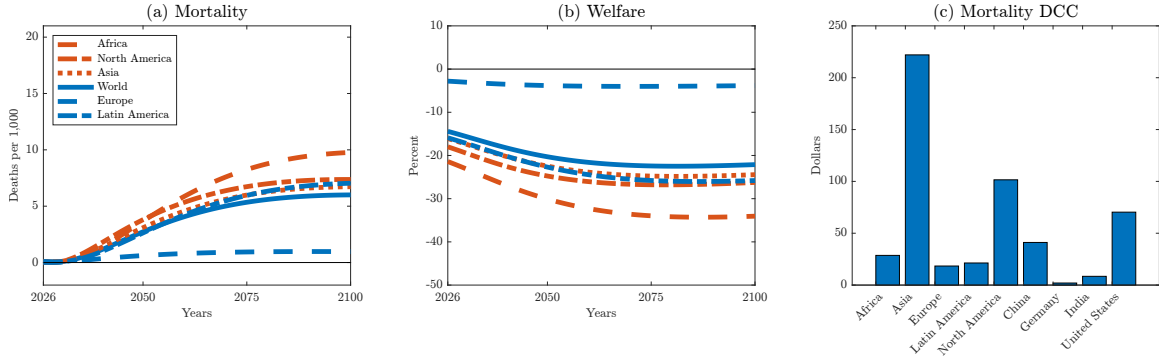
change amounts to a 14% loss in consumption equivalent in 2026, which continues to increase over the century as mortality damages accumulate. In contrast, welfare losses under local heat and cold damages are much more modest.

Panel (d) reports the mortality component of the social cost of carbon in 2026. The SCC implied by global temperature damages is \$400 per ton of CO_2 . This value is over ten times larger than the implied social cost under conventional damage function specifications based on local heat and direct losses only, which is near \$10–\$20 per ton.

6.3 Regional Mortality Impacts

We next examine how the mortality consequences of future warming differ across regions. To do so, we combine the regional reduced-form mortality responses estimated in Appendix D.2.2 with the global temperature path described above. We use these regional impulse responses to re-estimate our model region by region, construct region-specific

Figure 11: Mortality Impacts Across Regions



Notes: Transitional dynamics and DCCs given the warming scenario in Figure 10, panel (a). Model is re-estimated for each region based on regional reduced-form mortality responses estimated in Appendix D.2.2. Panel (a): transitional dynamics of excess deaths per 1,000 individuals. Panel (b): mortality-related welfare losses. Panel (c): mortality-related DCCs by region and country.

mortality damage functions and evaluate mortality, welfare, and the cost of carbon across regions.¹³

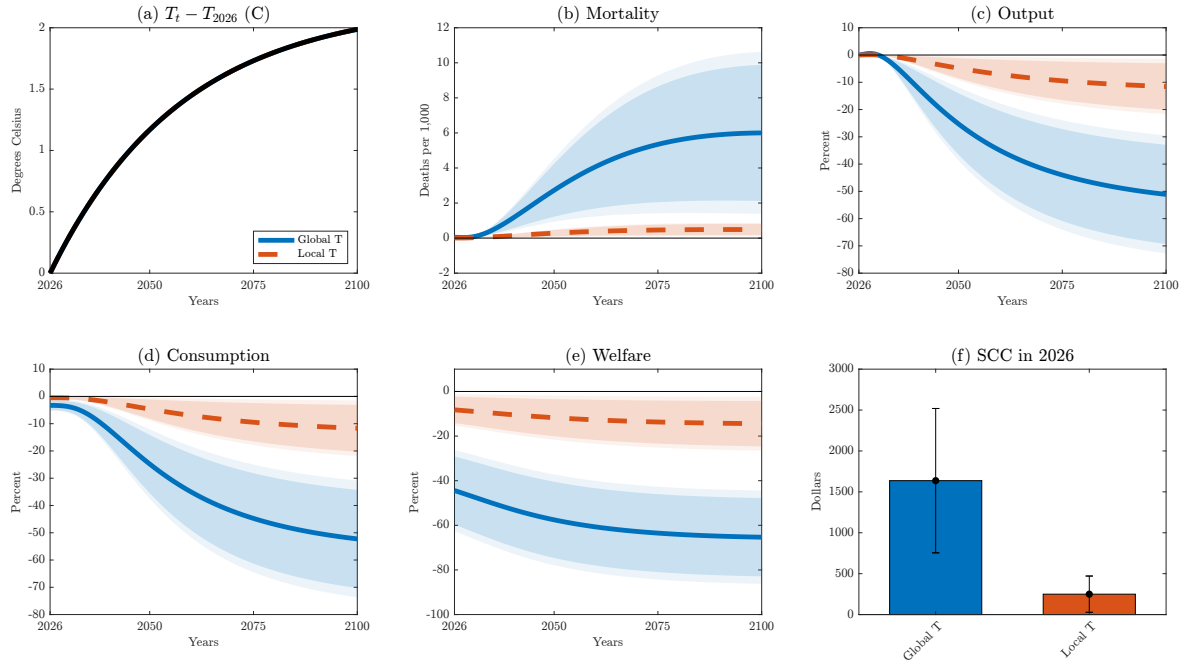
Figure 11 presents the results. Panel (a) displays excess mortality per 1,000 individuals across regions. Mortality effects are largest in Africa—reaching approximately 10 excess deaths per 1,000—followed by North America, Latin America and Asia, and substantially smaller in Europe. These differences reflect heterogeneity in the estimated regional temperature sensitivities embedded in the damage functions.

Panel (b) reports the corresponding welfare losses from mortality in consumption-equivalent terms. Regions with larger mortality responses experience correspondingly larger welfare declines, with over 21% welfare losses from climate-related mortality in Africa, and just under 18% in North America.

Panel (c) reports Domestic Costs of Carbon (DCCs) due to mortality impacts alone, which we term mortality DCCs. The mortality DCC encapsulates only damages circumscribed within a particular region, so that the sum of all regional DCCs is the SCC. The mortality DCC is particularly high in populous regions with large projected mortality effects. In Asia, the mortality DCC exceeds \$200 per ton. In China alone it is just over \$40 per ton. The mortality DCC is also particularly high in less populous regions with high incomes. In the United States, the mortality DCC is just over \$70 per ton.

¹³In that case, the region's aggregate output replaces world output in the health production function. This property is consistent with Bilal, Känzig, and Lisiecki (2026), who find that growth spillovers are regional rather than global.

Figure 12: Transitional Dynamics Under Climate Change: Total Damages



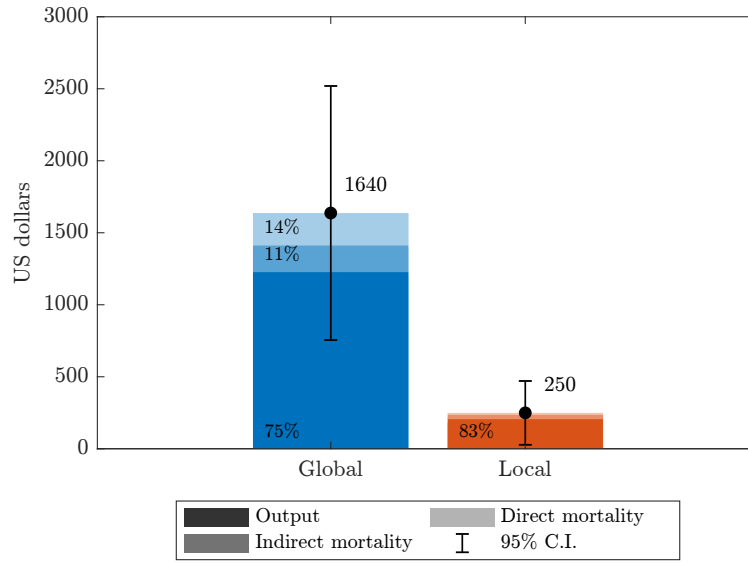
Notes: Transitional dynamics of the estimated model under the scenario in Panel (a). Solid blue lines: model estimated under global temperature. Dashed red lines: model estimated under local temperature and the impact effect of local extreme heat. Shaded areas: 90 and 95% confidence intervals based on the Delta method and accounting for the variance-covariance matrix between empirical GDP and mortality impulse response functions. Panel (f): bars, with 95% confidence intervals.

6.4 Output and Mortality Impacts

We now consider both mortality and output losses from climate change and evaluate the total welfare consequences of future climate change. Figure 12 presents the results. Panels (c) and (d) show the relative path of output and consumption. Under the damage function based on global temperature, output and consumption decline by over 50% over the 21st century. These losses are markedly larger than those implied by conventional damage function specifications based on local temperature, which generate comparatively modest macroeconomic effects of just over 10%.

Panel (e) reports total welfare losses in consumption-equivalent terms. Factoring in mortality and consumption losses, present-value welfare losses exceed 40% in 2026 and rise to over 60% by the end of the century. Importantly, because we report counterfactuals relative to our reference scenario, these figures do not imply that consumption, output and welfare in 2100 are below their 2026 levels. Rather, they indicate that would be over 50% lower than in the reference scenario without climate change, with welfare losses exceeding 60% in consumption-equivalent terms. Even under warming, the economy

Figure 13: Social Cost of Carbon: Mortality and the Economy



Notes: Social Cost of Carbon and its components. Dark area: consumption losses. Intermediate shading area: indirect mortality losses due to output contractions and the health production function (7). Light shaded area: direct mortality losses due to the damage function ζ_s . Left blue bar: damage functions estimated using global temperature. Right red bar: damage function estimates using local temperature and the impact effect of local extreme heat.

continues to grow, albeit at a slower rate (see also Bilal, Ingrand, and Känzig, 2026).

We report the total SCC in 2026, inclusive of both consumption and mortality losses, in Panel (f) and in Figure 13. We obtain a total SCC in excess of \$1,600 per ton when damages are estimated using global temperature shocks. Given the partial mortality SCC of \$400 per ton, mortality accounts for one quarter of the SCC.

Why does mortality not represent a larger share of the total SCC? Consumption and mortality both change by the same order of magnitude relative to their baseline values due to climate change. On the one hand, the VSL per life-year of approximately \$175,000 represents about 3 times annual consumption in the United States. On the other hand, the baseline mortality rate is 0.014, so that absolute changes in mortality risk—which are the welfare-relevant object—are small enough to more than offset a large VSL.

We further split mortality losses into direct and indirect losses. Direct losses are directly driven by the mortality damage function ζ_s . Indirect mortality losses operate through the damage function ζ_s and the resulting declines in consumption and output affecting the health production function (7). We find that indirect losses due to income changes account for just under half of mortality losses. This decomposition differs slightly from Figure 7(b) because our estimation accounts for equilibrium adjustments.

By contrast, when we re-estimate damage functions using local temperature and extreme heat shocks, we obtain a total SCC of \$250 per ton, the majority of which is accounted for by consumption losses. Mortality accounts for 17% of this total, mostly due to indirect effects rather than direct effects.

6.5 Adaptation

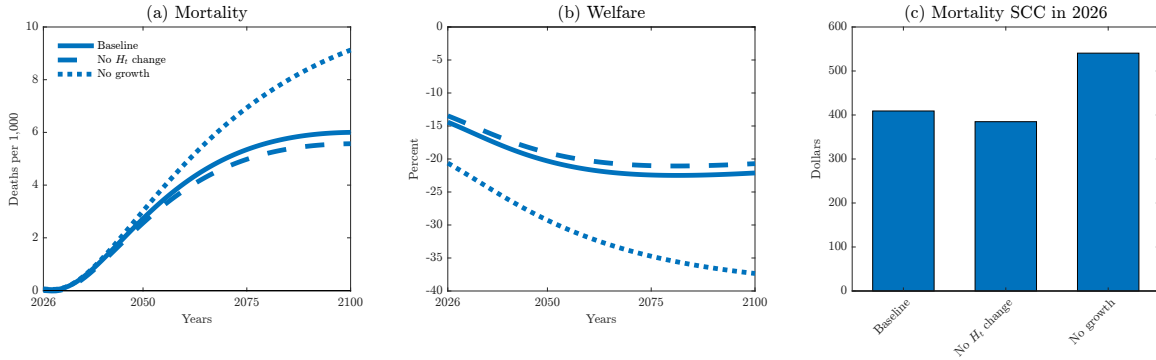
So far we have allowed households to fully adapt to climate change by adjusting their health spending and by letting the economy grow absent climate change. How important is adaptation for our results? We answer this question with two exercises. In the first exercise, we gauge the role of private adaptation. We hold private health consumption H_t fixed at the path that it would take in the reference scenario without climate change, and re-evaluate the effect of climate change. In the second exercise, we evaluate the role of adaptation through economic growth. We shut down background economic growth and re-assess the effect of climate change. Figure 14 displays the results.

Panel (a) shows that shutting down private adaptation does not affect mortality substantially. This result reflects that the elasticity μ of mortality to private health spending is moderate. Perhaps surprisingly, however, mortality is slightly *lower* when we shut down private adaptation to climate change. To understand this result, we exploit the identity: $\frac{M_t}{M_t^{\text{ref}}} = \frac{X_t}{X_t^{\text{ref}}} \left(\frac{H_t}{H_t^{\text{ref}}} \right)^{-\mu} \left(\frac{Y_t}{Y_t^{\text{ref}}} \right)^{-\nu}$. This identity highlights that two opposing forces determine whether private adaptation leads to more or less mortality. In partial equilibrium, households spend more on health, $H_t > H_t^{\text{ref}}$, which lowers mortality relative to the reference scenario without climate change. This effect is moderate because it enters with the elasticity $\mu < \nu$. By contrast, in general equilibrium, preventing households from increasing their health spending leads to more investment and therefore higher output, which ultimately lowers mortality through uninternalized global spillovers. These two effects largely cancel each other, with the latter dominating quantitatively.

By contrast, we find that global economic growth is a strong force that attenuates the mortality effects of climate change. Without global economic growth, climate change would cause 9 excess deaths per 1,000, against 6 excess deaths per 1,000 in the baseline. Economic growth affects the mortality consequences more strongly than private adaptation because the global spillover elasticity ν is substantially larger than μ .

Panels (b) and (c) show that these patterns also affect welfare and the SCC. Without adaptation through global economic growth, initial welfare losses are approximately 21%

Figure 14: Adaptation



Notes: Mortality, mortality-related welfare and mortality partial SCC. Solid line: baseline with background economic growth and private adaptation through private health spending. Dashed line: no private adaptation; H_t held fixed at reference value computed without climate change. Dotted line: no background economic growth. Panel (c): bars, in the same order.

before growing to over 37%, and the SCC exceeds \$500 per ton.

6.6 Age and Income Adjustments

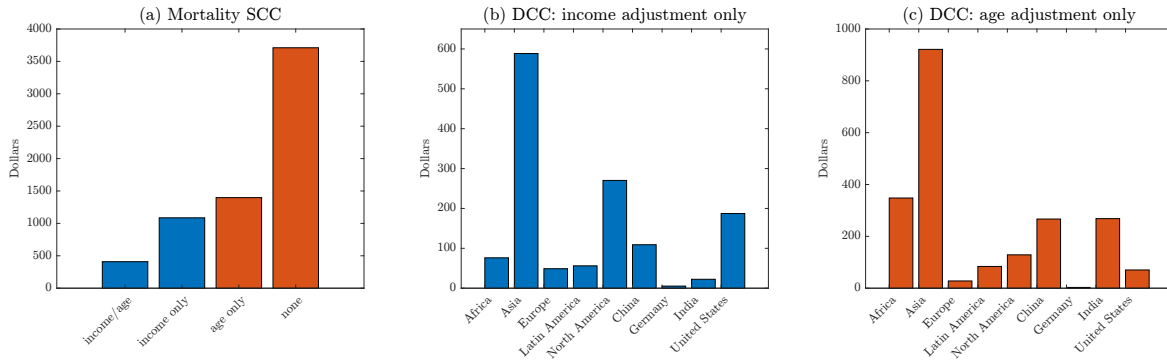
Our next exercises assess the role of age and income adjustments to the VSL. Figure 15 presents the results. Panel (a) shows the mortality partial SCC and DCCs when we remove these adjustments one at a time. When we do not adjust the VSL by income, the mortality SCC triples and is just under \$1,400 per ton. When we assume that mortality impacts of climate change are borne equally by all age groups, the mortality SCC is just under \$1,100 per ton. Without any of these adjustments, the mortality SCC exceeds \$3,700 per ton.

Panel (b) shows regional DCCs without the age adjustment. They are shifted up relative to adjusted DCCs in Figure 11 but retain the same ranking. Panel (c) shows that removing the income adjustment not only increases many DCCs, but also reshuffles their ordering. North America becomes comparatively less important, while Africa exhibits a DCC in excess of \$300 per ton. China and India each have DCCs near \$300 per ton.

6.7 Efficiency

Our framework is inefficient when $\nu \neq 0$ because of global health spillovers (see Appendix E.2). Our final exercise examines whether this inefficiency affects our main counterfactuals. Importantly, while the inefficiency affects the level of economic activity and mortality (Bilal, Känzig, and Lisiecki, 2026), it may not necessarily affect counterfactuals in which the climate changes, relative to a reference that is already inefficient. Appendix

Figure 15: Age and Income Adjustments



Notes: Mortality partial SCC and DCCs. Panel (a): SCC under combinations of age and income adjustments. Leftmost bar: baseline. Panel (b): DCCs without the age adjustment but with the income adjustment. Panel (c): DCCs without the income adjustment but with the age adjustment.

Figure F.7 shows that counterfactuals under the socially efficient allocation are nearly identical to those in the decentralized equilibrium.

7 Conclusion

This paper shows that the mortality consequences of climate change are fundamentally global and macroeconomic in nature. Leveraging plausibly exogenous fluctuations in global mean temperature, we estimate that a permanent 1°C increase raises annual mortality by 2.5 deaths per 1,000 people—an order of magnitude larger than estimates based on local heat alone. Our evidence suggests that these effects are largely mediated by income losses.

Incorporating these findings into a structural model of climate damages fundamentally reshapes welfare assessments. Mortality contributes roughly one quarter of the total Social Cost of Carbon, amounting to \$400 per ton of CO₂. In a world that warms by an additional 2°C by the end of the century, climate change may be responsible for 6 excess deaths per 1,000 individuals. While private adaptation appears to be weak, adaptation through background economic growth and global health spillovers offsets about 3 out of the 9 excess deaths per 1,000 that would occur without growth.

Our results suggest two broader implications. First, they strengthen the case for unilateral climate mitigation by uncovering a large and previously underappreciated mortality externality operating through global economic linkages. In particular, because mortality losses scale with population and income exposure, unilateral decarbonization may

be cost-effective even for relatively lower-income, but populous, economies. Second, our findings suggest an important role for policies that sustain income growth and stabilize economic activity in the face of climate change. By preserving the economy, such policies reduce not only economic losses but also the mortality consequences of climate change.

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Online Appendix

Global Temperature and Global Mortality

Adrien Bilal

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Contents

A. Data	55
A.1. Mortality Data	55
A.2. Climate Data	57
A.3. Economic Data	58
B. Additional Results for Baseline Estimates	59
B.1. Age-Specific Mortality	59
B.2. Transitory Temperature Shock	60
B.3. Nonlinearities and Asymmetries	61
B.4. Estimation Uncertainty and One-Step Local Projections	62
C. Additional Robustness for Global Temperature Shocks	63
C.1. Jackknife Exercise	63
C.2. Alternative Detrending Methods	63
C.3. Lag Orders and Trends	64
D. Additional Results for Channels	65
D.1. Extreme Events	65
D.1.1. Sensitivity to Global Controls	69
D.2. The Income Channel	70
D.2.1. Cause of Death	70
D.2.2. Regional Heterogeneity	71
D.2.3. Heterogeneity by Income and Climate	72
D.2.4. Global GDP and Global Temperature	73
D.2.5. Mortality and GDP	74

E. Model	76
E.1. Decentralized equilibrium	76
E.2. Planner problem	77
E.3. Steady state	77
E.3.1. Decentralized equilibrium	77
E.3.2. Planner	78
E.4. Log-linearized Transitional Dynamics	78
E.4.1. Decentralized Equilibrium: Counterfactuals	78
E.4.2. Model Inversion: Proof of Proposition 1	80
E.4.3. Planner	84
E.5. Nonlinear transitional dynamics	85
F. Additional Model Results	88
F.1. Estimation	88
F.2. Damage Functions	90
F.3. Counterfactuals Using Local Heat	91
F.4. Efficient Allocation	91

A Data

A.1 Mortality Data

We construct our mortality variables by combining data from several sources, with the Human Mortality Database (HMD, 2024) serving as the primary input. The HMD is widely regarded as the gold standard for mortality data. It is based on high-quality vital registration data, complemented by census counts, birth records, and population estimates, and applies a harmonized methodology across countries. Crucially for our purposes, it relies on observed annual death counts rather than model-based imputation across years, making it well suited to high-frequency analysis that exploits year-to-year variation. The HMD covers most of Western and Northern Europe, North America, Australia, and a number of Eastern European and Asian countries. We download mortality counts `deaths.zip` and population and live-birth data, `population.zip` and `births.zip`, from www.mortality.org/Data/ZippedDataFiles. For each age group—total, ages 0–4, ages 5–64, and ages 65 and above—we divide death counts by population and multiply by 1,000 to obtain mortality rates per 1,000 individuals. This normalization is applied consistently across all data sources to ensure comparability.

Because HMD coverage is largely confined to high-income countries, we supplement the HMD series with data from the World Health Organization (WHO, 2024) Mortality Database to expand coverage and fill gaps. The WHO database compiles death counts reported by national statistical authorities from civil registration systems, with data disaggregated by age, sex, and cause of death and subject to varying degrees of completeness across countries and over time. Relative to the HMD, it provides broader geographic coverage, extending to Latin America and the Caribbean, the Middle East, parts of Asia, Eastern Europe, and Africa. We download mortality counts and population data from www.who.int/data/data-collection-tools/who-mortality-database. For countries where the entire HMD series is missing, we use the corresponding WHO mortality counts to construct our measures. For countries with partial gaps, we preserve the level of the HMD series and use the growth rates implied by the WHO data to interpolate and extrapolate missing years.

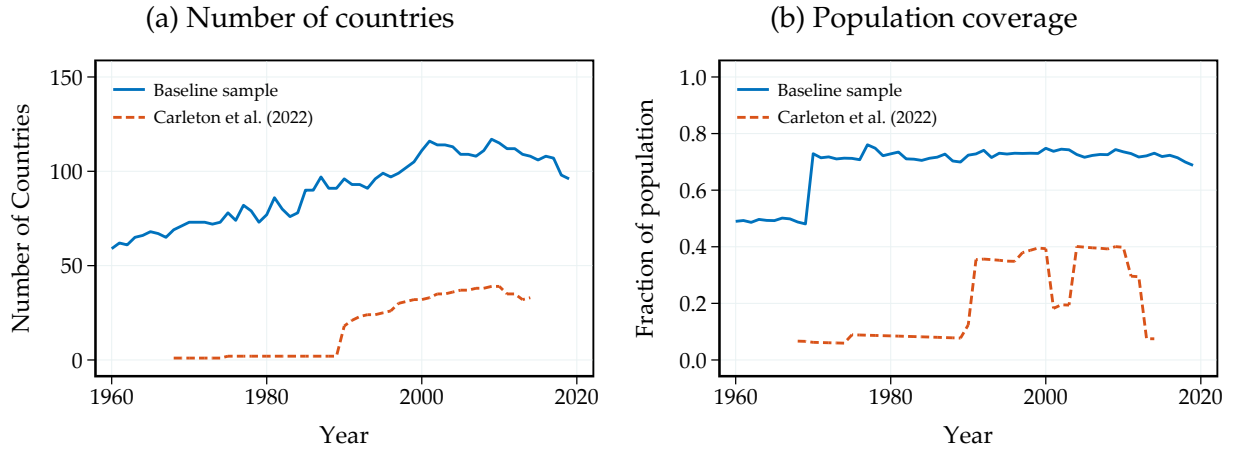
Despite the expanded coverage, two crucial countries remain missing from the baseline sources: India and China. For these countries, we therefore draw on additional sources. We obtain mortality rates for China directly from the National Bureau of Statis-

tics of China (<https://data.stats.gov.cn/english/easyquery.htm?cn=C01>), which reports annual crude death rates based on official statistics. For India, we obtain mortality counts from Vital Statistics India (VSI) and combine them with population data from the United Nations World Population Prospects. Starting from 2009, VSI reports are available at <https://censusindia.gov.in/census.website/data/VSREPORT>. For earlier years we digitize these data from university library sources. Finally, we add mortality rates for Liechtenstein from Carleton et al. (2022). Together, these sources yield a mortality panel with coverage of around 60 countries in 1960, rising to over 100 countries in the second half of our sample period.

Despite this broad coverage, reliable annual mortality data remain unavailable for many countries, particularly in Africa, where civil registration systems are often incomplete or absent. For these countries, we rely on estimates from the United Nations World Population Prospects (UNWPP, 2024), accessed from population.un.org/wpp/. Unlike the HMD and WHO databases, which are based on directly observed deaths from civil registration systems, the UNWPP constructs internally consistent demographic series by combining a wide range of available sources—including population censuses, household surveys, partial registration systems, and international datasets—and applying demographic models to reconcile discrepancies and interpolate missing observations. In particular, the estimates are derived using a cohort-component framework that ensures consistency between population levels and the underlying components of change, namely fertility, mortality, and migration. While this approach provides comprehensive global coverage and is well suited for capturing long-run trends and cross-country differences, it necessarily smooths high-frequency variation and is therefore less appropriate for our empirical design, which exploits annual fluctuations in mortality. Still, the UNWPP data provide the best available source for achieving broad global coverage. We therefore use them to construct an extended mortality panel for robustness exercises. Specifically, for any country where the HMD-WHO-based series contains fewer than 20 observations, we replace it with a mortality rate constructed from UNWPP death counts and population figures. This extended data set covers up to 210 countries and is used in sensitivity checks and for the regional analysis in Africa.

To decompose the mortality response by cause of death, we use the 2021 release of Global Burden of Disease (GBD) data from The Institute for Health Metrics and Evaluation (Naghavi et al., 2024). The GBD combines vital registration, verbal autopsy, sur-

Figure A.1: Mortality Data Coverage



Notes: Panel (a): number of countries with mortality data over time. Panel (b): fraction of world population covered by mortality data. Mortality data for Europe, North America, Asia, Latin America, and Oceania based on Human Mortality Database (HMD), supplemented by WHO when HMD is missing, with direct sources for India and China.

veys, and surveillance data with statistical models to produce internally consistent cause-specific death counts for 204 countries and territories. We download total annual death counts for twelve cause groups from <https://vizhub.healthdata.org/gbd-results/> for 1988-2021. As with the other sources, we divide counts by population and multiply by 1,000 to obtain death rates per 1,000 individuals. We report cardiovascular diseases, chronic respiratory diseases, HIV/AIDS, neglected tropical diseases and malaria, and environmental heat and cold exposure separately, and group the remaining seven categories into a residual category.

A.2 Climate Data

Our climate dataset builds on Bilal and Känzig (2026) and combines data on global and country-level temperature as well as information on extreme climatic events.

We use annual global temperature anomalies (land and ocean surface) from NOAA National Centers for Environmental Information (2023) and Berkeley Earth (Rohde and Hausfather, 2020), which we convert to temperature levels using their respective climatological baselines.

For spatially disaggregated analysis, we rely on gridded temperature data from Berkeley Earth (Rohde and Hausfather, 2020), available at a $1^\circ \times 1^\circ$ resolution over a long historical period. We aggregate these data to the country level using population weights, although results are robust to area weighting. As a validation exercise, aggregating the

gridded data to the global level closely reproduces the official NOAA temperature series.

To capture broader climatic conditions, we complement temperature data with gridded information on temperature, precipitation, and wind from ISIMIP (Lange et al., 2023), available at a $0.5^\circ \times 0.5^\circ$ spatial resolution. To characterize extreme climatic conditions, we construct indicators based on daily gridded data and percentile thresholds of the historical distribution. Specifically, we define extreme events relative to the distribution over a baseline period (1950–1980), distinguishing between country-specific (relative) thresholds and global (absolute) thresholds. Our baseline uses relative thresholds, capturing deviations from local climatic norms. We compute, for each country and year, the fraction of days in which temperature, precipitation, or wind exceed the relevant threshold, and consider both country-level and grid-level aggregation approaches. The results are robust to alternative threshold definitions, baseline periods, and aggregation methods.

A.3 Economic Data

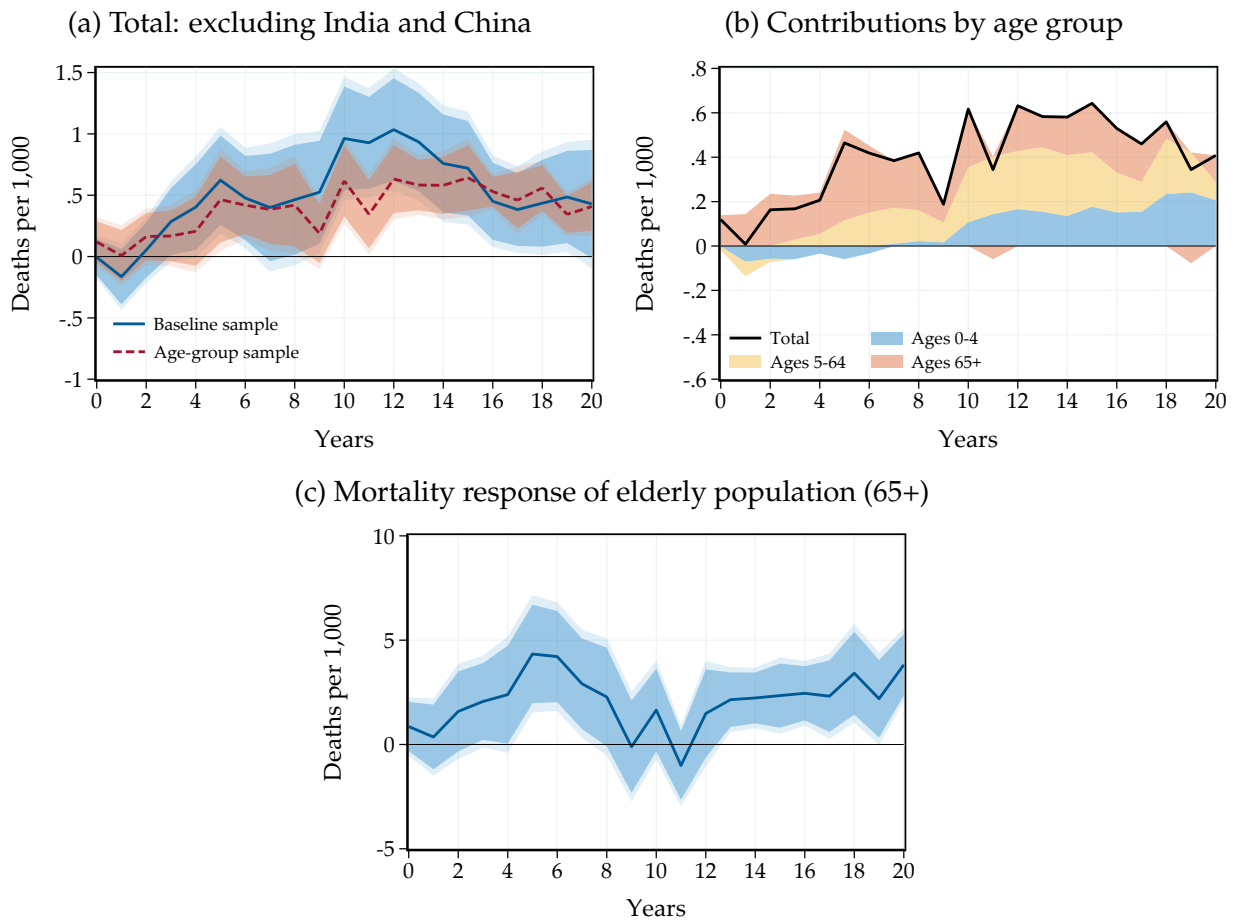
We obtain GDP and population data from the Penn World Tables (PWT; Feenstra, Inklaar, and Timmer, 2015), which provides comprehensive coverage for our sample countries from 1960 to 2019. Our main output measure is real GDP per capita from the national accounts (rgdpna/pop). Based on these data, we construct a measure of world real GDP by chain-weighting country-level growth rates. We obtain a global commodity price index from Bloomberg. World recession dates are taken from the World Bank (Kose, Sugawara, and Terrones, 2020). Data on wars are drawn from the Uppsala Conflict Data Program. Information on major epidemics and pandemics is compiled from publicly available historical sources.

B Additional Results for Baseline Estimates

B.1 Age-Specific Mortality

We study how mortality responses vary across age groups and which groups account for the mortality response to global temperature shocks. To this end, we decompose total mortality into the contributions of individuals aged 0–4, 5–64, and 65 and above. We multiply each age-specific mortality rate by the corresponding population share, so that all contributions are expressed in deaths per 1,000 individuals in the total population.

Figure B.1: Age Decomposition of the Mortality Response



Notes: Impulse responses to a 1°C global temperature shock. Panel (a) compares the baseline mortality response with the response estimated on the subsample for which mortality rates are available for all three age groups (ages 0–4, 5–64, and 65+). Both responses are expressed in deaths per 1,000 individuals in the total population. The age-group sample excludes India and China. Panel (b) decomposes the mortality response on this common sample into population-weighted contributions from the three age groups, after proportionally rescaling age-specific mortality rates to match reported total mortality. Colored areas show the age-group contributions. Panel (c) reports the mortality response for the population aged 65 and above. All regressions use the same baseline controls. Shaded areas in Panels (a) and (c) denote 90 and 95% confidence bands based on Driscoll-Kraay standard errors.

We estimate the local projections for all three contributions on the same sample and with the same controls, including lags of total mortality changes. The estimated responses therefore add up to the total response on this sample. Because of data limitations, the sample for this decomposition excludes India and China. However, Figure B.1(a) shows that the mortality response in the age-group sample is comparable to our baseline sample.

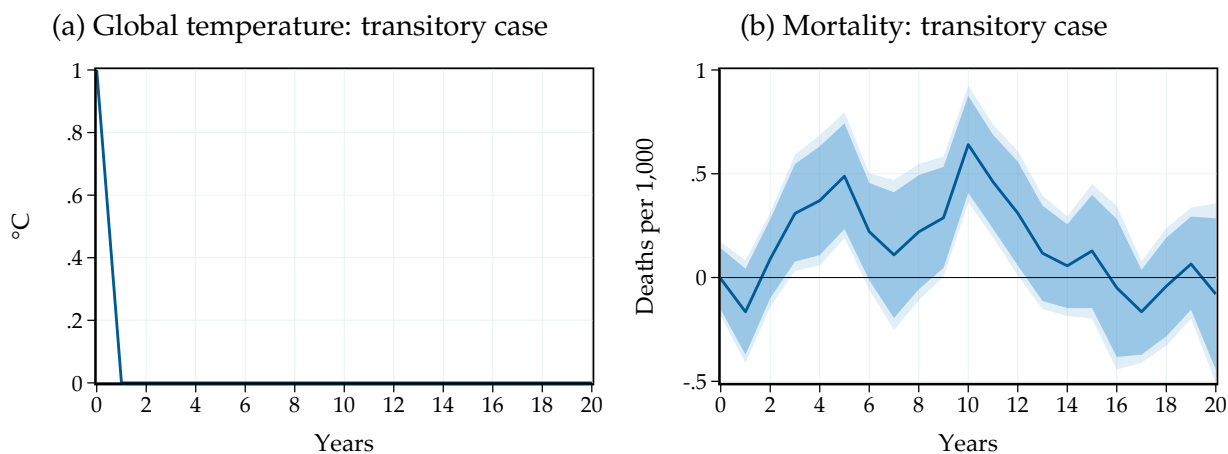
Figure B.1(b) shows the total response in the restricted sample together with the contributions of the three different age groups. Mortality among individuals aged 65 and above accounts for much of the response in the first decade. The contributions of younger individuals become more important at longer horizons.

Finally, Figure B.1(c) shows the mortality response for the elderly population, expressed relative to the population aged 65 and above. The peak effect is around five times larger among this age group.

B.2 Transitory Temperature Shock

Figure 3 in the main text shows that mortality effects build up over time as global temperature remains persistently elevated following a shock. To isolate the role of temperature persistence, we construct a counterfactual mortality response to a purely transitory temperature shock using the method of Sims (1986).

Figure B.2: Mortality Response to a Transitory Temperature Shock



Notes: Counterfactual impulse responses to a purely transitory 1°C temperature increase, constructed using the method of Sims (1986). A sequence of temperature shocks offsets the persistence of the initial shock, so that temperature returns to baseline after impact. The mortality response includes all transmission channels under this counterfactual temperature path.

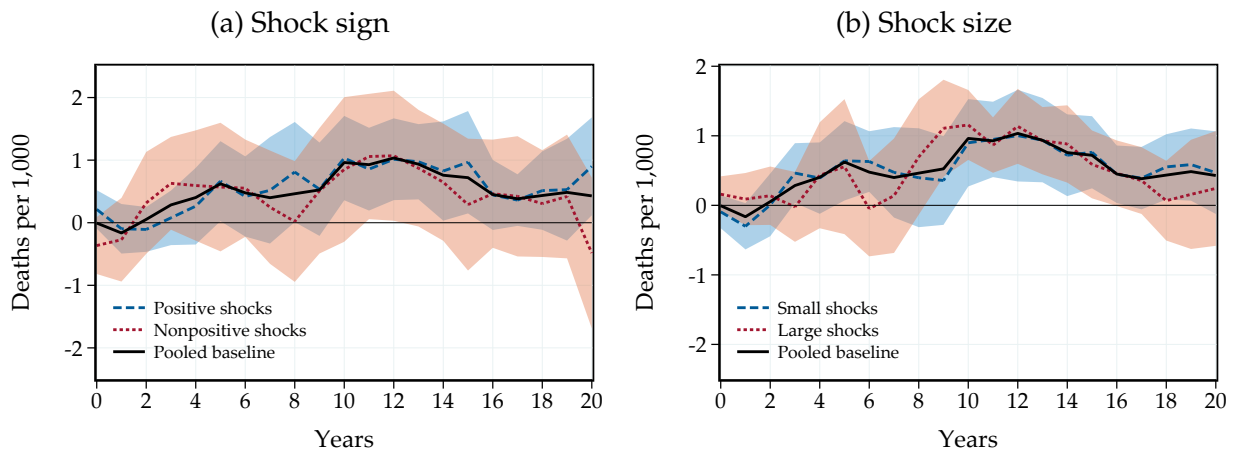
Figure B.2 shows the results. Panel (a) displays the transitory temperature path: by construction, temperature rises by 1°C on impact and then immediately returns to base-

line. Panel (b) shows the corresponding mortality response. Under this transitory shock, the peak mortality effect is approximately 0.6 deaths per 1,000 at year 10, compared to about 1 death per 1,000 in year 12 in the baseline. Removing temperature persistence thus reduces the peak mortality response by roughly 40% and leads to a less persistent mortality response.

B.3 Nonlinearities and Asymmetries

Our baseline specification imposes a linear relationship between global temperature shocks and mortality. We assess this assumption by allowing the coefficient on the contemporaneous shock to vary with its sign or size, while retaining the baseline controls. For the sign specification, we include positive and nonpositive shocks as separate regressors. For the size specification, we distinguish shocks with an absolute value above 0.2°C from smaller shocks.

Figure B.3: Nonlinearities and Asymmetries in Mortality Responses



Notes: Mortality responses per 1°C global temperature shock, in deaths per 1,000. Panel (a) estimates separate coefficients on positive and nonpositive shocks. Coefficients for nonpositive shocks are reported with the same sign convention as those for positive shocks: a positive coefficient implies that a negative temperature shock lowers mortality. Panel (b) estimates separate coefficients for shocks with absolute value at most 0.2°C and above 0.2°C . Each regression includes both shock components and the baseline controls. Black line: pooled baseline. Colored lines and shaded areas: group-specific estimates and 95% confidence bands based on Driscoll-Kraay standard errors.

Figure B.3 shows the results. Positive and nonpositive shocks imply similar mortality effects per degree over the medium term, as do small and large shocks. In particular, the large, delayed increase in mortality is also present when we only use the variation in positive shocks. The group-specific estimates are less precise than the pooled baseline, and their confidence bands overlap substantially. These results support using the pooled

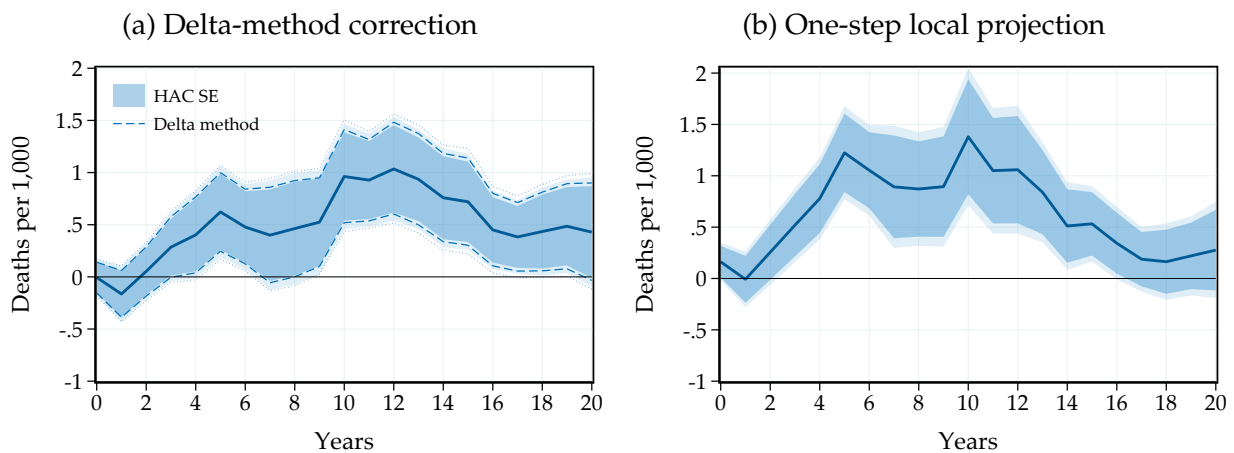
linear specification within the range of temperature shocks observed in our sample, but do not rule out nonlinearities under larger future temperature changes.

B.4 Estimation Uncertainty and One-Step Local Projections

Our baseline specification first constructs global temperature shocks using the Hamilton filter and then estimates their effects on mortality. The baseline standard errors treat the estimated shocks as given. To assess the importance of this first-stage estimation uncertainty, we apply a Delta-method correction to the covariance matrix of the local projection estimates. The correction propagates uncertainty in the fitted Hamilton coefficients to the mortality responses and retains covariance between the two estimation stages.

Figure B.4(a) compares the corrected confidence bands with our baseline bands. The two are very close, suggesting that this correction for shock estimation uncertainty has little effect on our inference.

Figure B.4: Estimation Uncertainty and One-Step Estimation



Notes: Mortality responses to a 1°C increase in global temperature, in deaths per 1,000. Panel (a): baseline point estimates and confidence bands treating the temperature shock as given, together with bands incorporating a Delta-method correction for uncertainty in the Hamilton regression. Panel (b): one-step local projection using current global temperature and six of its lags, with the baseline mortality and economic controls. Dark and light shaded areas: 90 and 95% confidence bands based on Driscoll-Kraay standard errors. Dashed and dotted lines in Panel (a): corrected 90 and 95% confidence bands.

We also consider a one-step local projection that projects cumulative changes in mortality directly on global temperature, controlling for six lags of temperature and the baseline mortality and economic controls. This specification estimates the mortality response without first constructing a temperature shock. Panel (b) shows a somewhat larger response in the first decade, followed by a decline, with precision comparable to our baseline estimates. The delayed and persistent mortality increase therefore also appears under

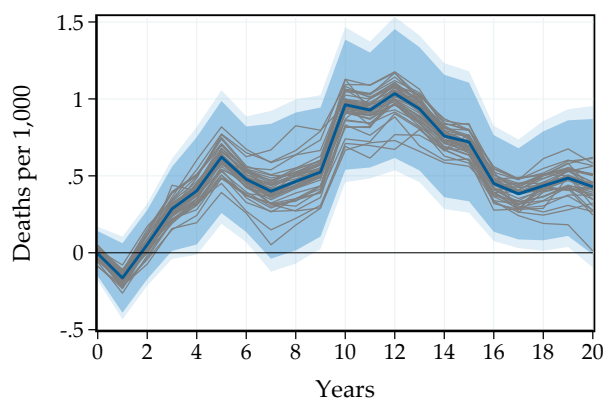
this alternative estimator.

C Additional Robustness for Global Temperature Shocks

C.1 Jackknife Exercise

In this appendix, we assess whether individual temperature shocks drive our results. For each year, we set the corresponding temperature shock to zero and re-estimate the local projections. Figure C.1 shows the resulting responses. The delayed mortality increase remains present across these estimates, suggesting that it is not driven by any single shock.

Figure C.1: Jackknife Analysis: Leave-One-Out Temperature Shocks

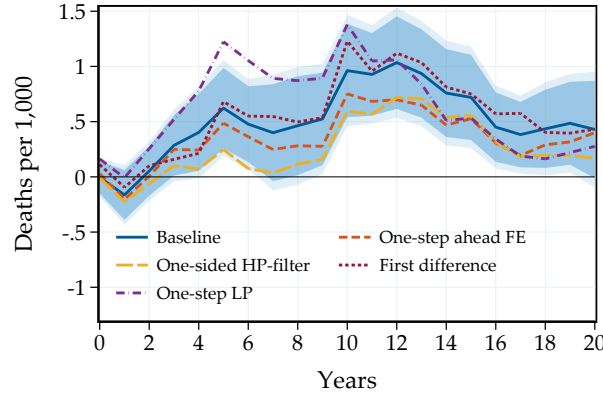


Notes: Impulse responses of mortality to a 1°C global temperature shock. Thin gray lines: estimates setting one year's temperature shock to zero at a time. Solid blue line: baseline estimate. Dark and light shaded areas: baseline 90 and 95% confidence bands based on Driscoll-Kraay standard errors.

C.2 Alternative Detrending Methods

Our baseline uses two-year-ahead forecast errors from the Hamilton filter to construct temperature shocks. Figure C.2 compares these estimates with those based on one-year-ahead forecast errors, a one-sided HP filter with smoothing parameter 100, and first differences of temperature. For reference, we also show the one-step local projection from Appendix B.4. The estimated responses are very similar across specifications. The one-year-ahead forecast-error and HP-filter estimates are somewhat smaller, while the first-difference and one-step estimates closely track the baseline over much of the horizon. All specifications imply a delayed increase in mortality followed by a gradual decline.

Figure C.2: Alternative Temperature Shock Construction Methods



Notes: Mortality responses using the baseline Hamilton filter ($h = 2, p = 2$), one-year-ahead forecast errors with two lags, a one-sided HP filter with smoothing parameter 100, first differences of temperature, and a one-step local projection with six temperature lags. Responses are in deaths per 1,000 per 1°C temperature shock. Lines: point estimates. Dark and light shaded areas: baseline 90 and 95% confidence bands based on Driscoll-Kraay standard errors.

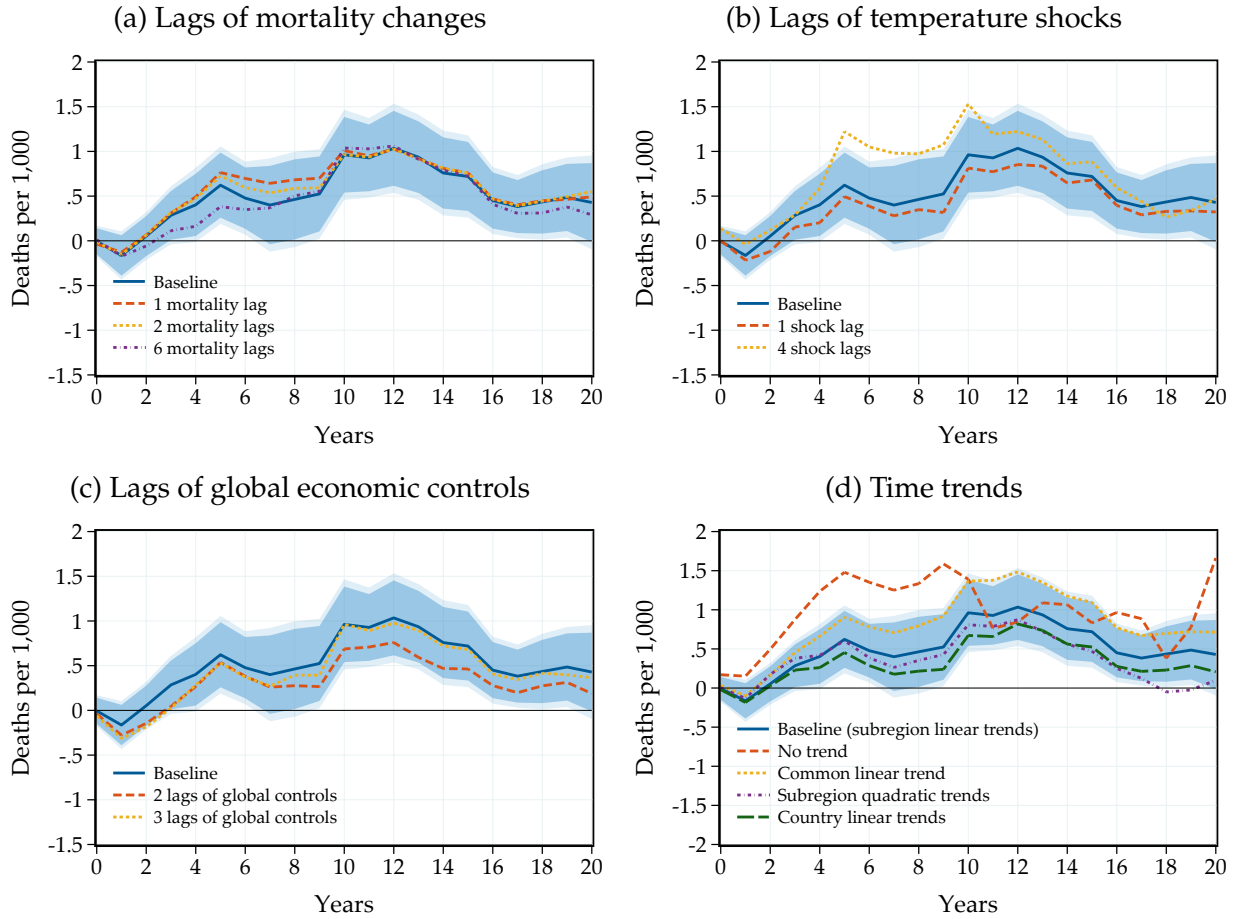
C.3 Lag Orders and Trends

We also assess the sensitivity of our mortality estimates to the dynamic controls and time trends in the local projections. Figure C.3 collects the results.

Number of lags. Panels (a), (b), and (c) vary the lag orders for mortality changes, global temperature shocks, and global economic controls, respectively. We vary one set of lags at a time, keeping the remaining specification at its baseline value. The large, delayed increase in mortality is present throughout. Changing the number of mortality lags has little effect on the response. The magnitude is somewhat more sensitive to the lag orders of the temperature shock and global controls, but the medium-term response remains large and positive.

Time trends. Our baseline controls for subregion-specific linear trends. Panel (d) alternatively omits trends, includes a common linear trend, adds subregion-specific quadratic trends, or uses country-specific linear trends. The estimated response is somewhat larger when trends are omitted or when only a common trend is included, and somewhat smaller at some horizons under the more flexible trend specifications. Overall, however, the mortality response is not sensitive to the treatment of trends.

Figure C.3: Sensitivity to Lag Orders and Time Trends



Notes: Impulse responses of mortality to a 1°C global temperature shock, in deaths per 1,000. Panels (a)–(c) vary the lag order for the indicated regressors, keeping other lag orders fixed. In Panel (c), the country-level GDP growth controls remain unchanged and the contemporaneous recession indicator is retained. Panel (d) varies the time trends. All specifications retain country fixed effects and lagged population weights. Solid blue line: baseline estimate. Dark and light shaded areas: baseline 90 and 95% confidence bands based on Driscoll-Kraay standard errors. Other lines: alternative point estimates.

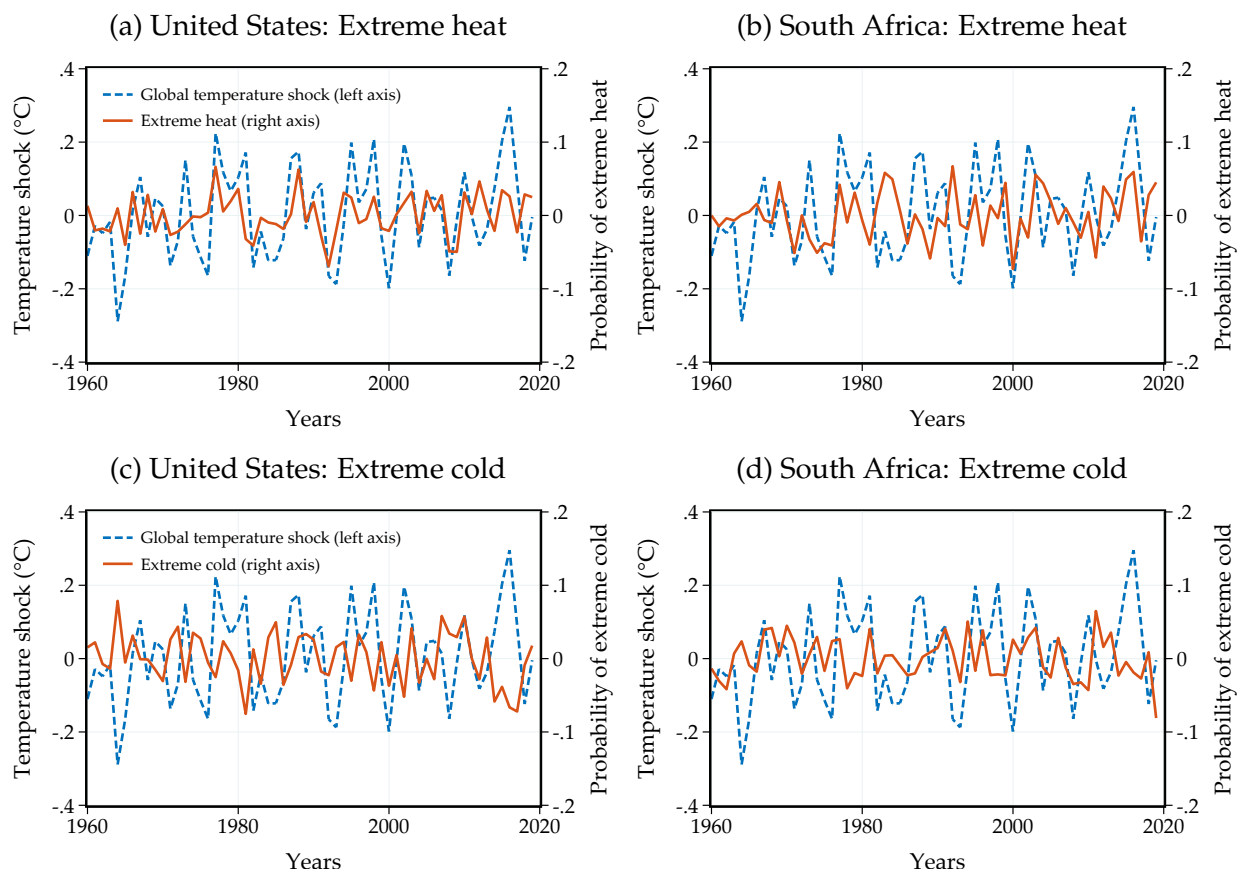
D Additional Results for Channels

D.1 Extreme Events

Figure D.1 illustrates the identifying variation in local extreme temperature. Specifically we show the exposure to extreme heat and cold in two example countries: the United States and South Africa. While local temperature extremes partly co-move with fluctuations in global mean temperature, they also display substantial country-specific variation. As a result, countries can experience unusually high exposure to extreme heat or cold even in years when global temperature movements are relatively modest. This variation reflects the fact that local extremes are shaped not only by global temperature changes,

but also by regional weather dynamics.

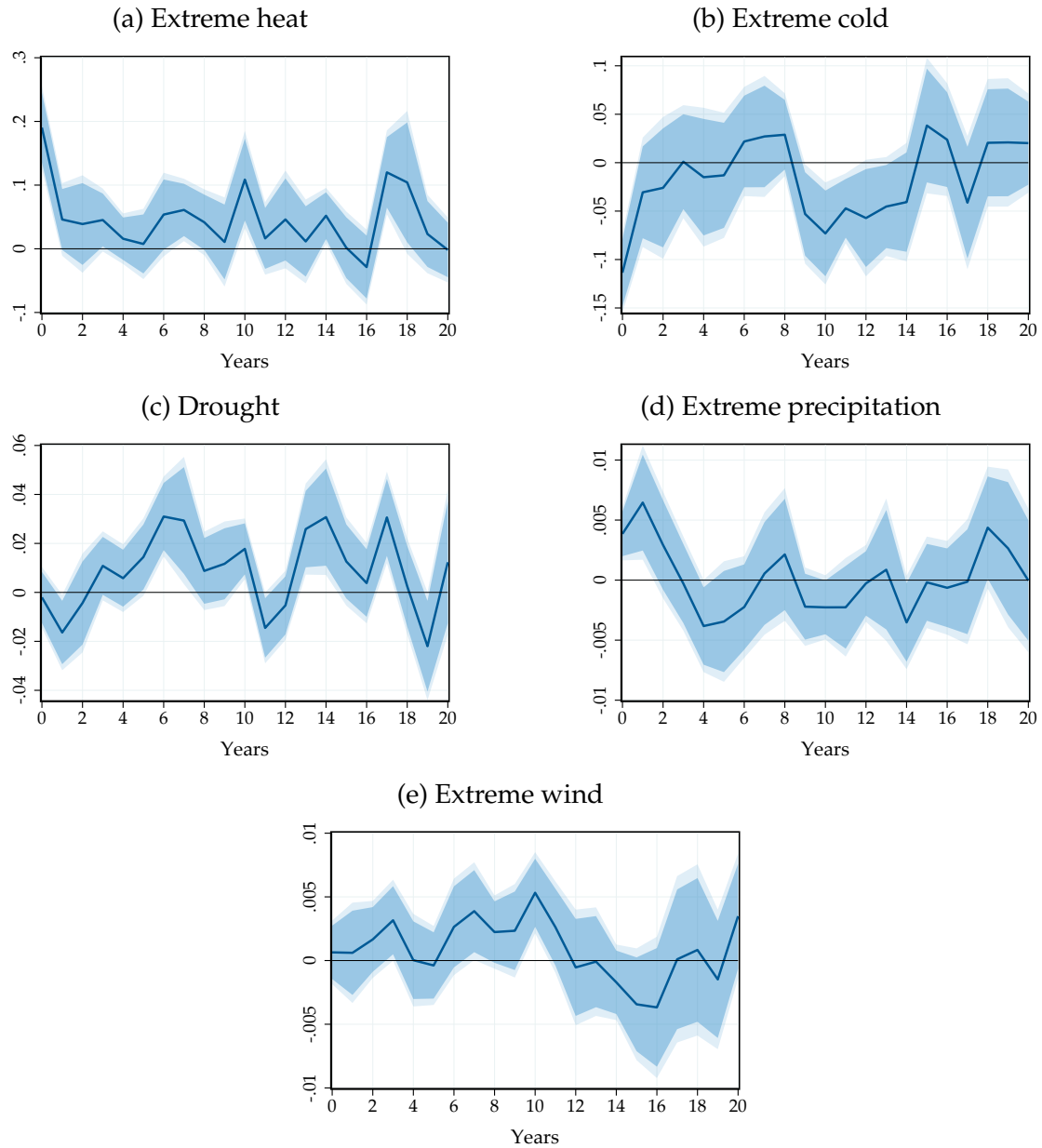
Figure D.1: Local Extreme Heat and Cold Shocks



Notes: Local extreme heat and cold shocks for the United States and South Africa. Extreme heat exposure is measured as the fraction of days with temperature above the 95th percentile of the 1950–1980 country-specific temperature distribution, while extreme cold exposure is measured as the fraction of days with temperature below the 25th percentile. The shock series are constructed by purging each exposure measure of its own lags. The top panels show extreme heat shocks and the bottom panels extreme cold shocks; the left panels correspond to the United States and the right panels to South Africa.

Figure D.2 reports the responses of local extreme weather exposure to global temperature shocks. Extreme heat exposure rises persistently, while extreme cold exposure declines, and drought becomes more prevalent. Extreme precipitation and wind exposure also respond meaningfully, although their effects vary more over the horizon. Taken together, the results show that global temperature shocks systematically shift the incidence of a broad range of local weather extremes. These patterns are broadly consistent with Bilal and Känzig (2026), with quantitative differences reflecting the slightly different sample of countries.

Figure D.2: Extreme Weather Events and Global Temperature Shocks

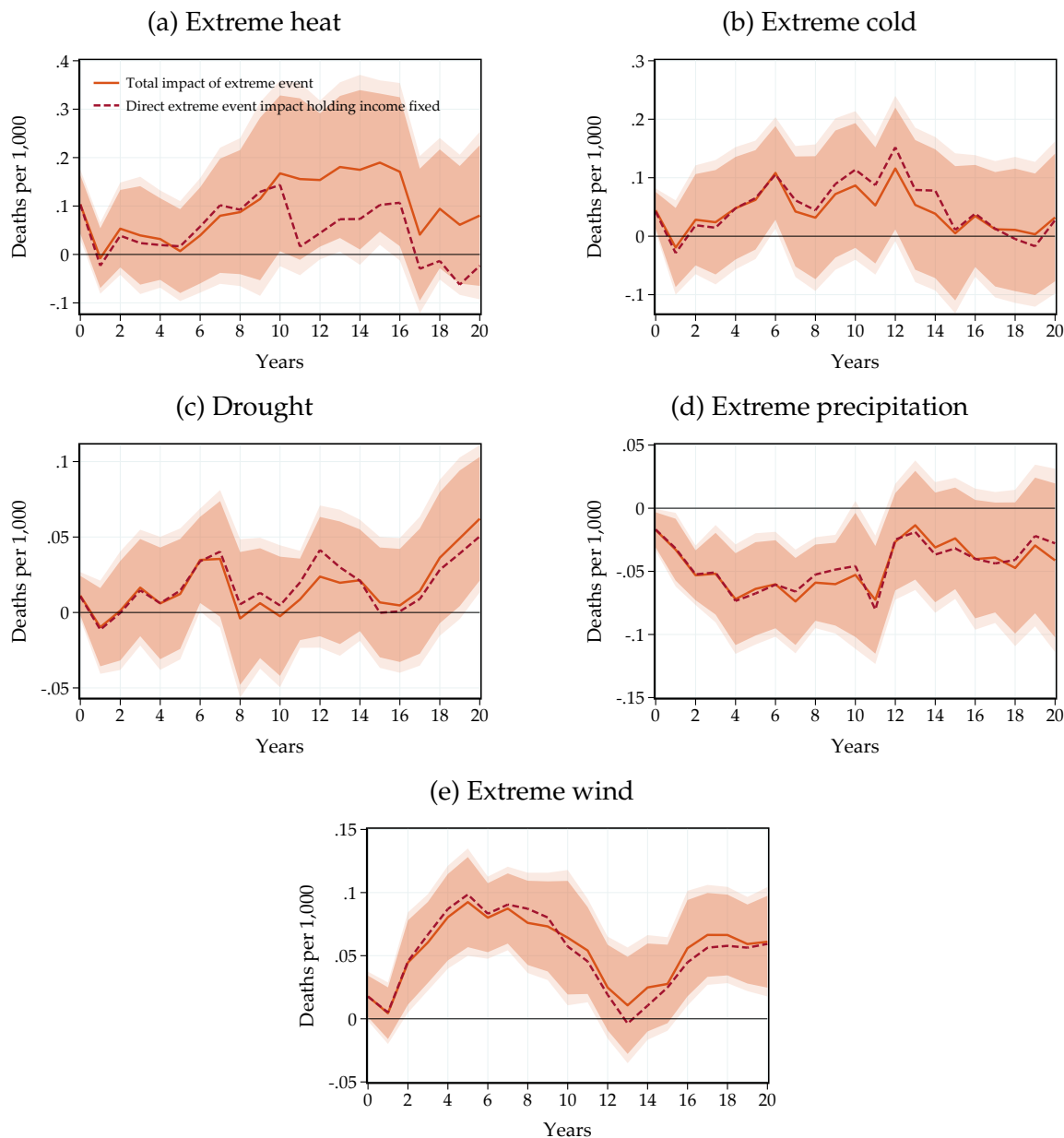


Notes: Impulse responses of local extreme weather exposure indices to a 1°C global temperature shock. Indices use country-specific thresholds from the 1950–1980 daily weather distribution: the 95th percentile for heat, the 25th percentile for cold and low precipitation, and the 99th percentile for high precipitation and wind. Precipitation and wind indices are two-year moving averages of annual exposure. Units: changes in the fraction of days. Dark and light shaded areas: 90 and 95% confidence bands based on Driscoll-Kraay standard errors.

Figure D.3 then estimates the mortality effects of these exposures, including the five weather measures jointly. For comparison across events, we scale the mortality responses by the magnitude of the largest change in the corresponding annual exposure associated with a 1°C global temperature shock. The cold panel in Figure D.3 reports the effect of an

increase in cold exposure; the robustness exercise below instead uses the decline in cold exposure associated with warming.

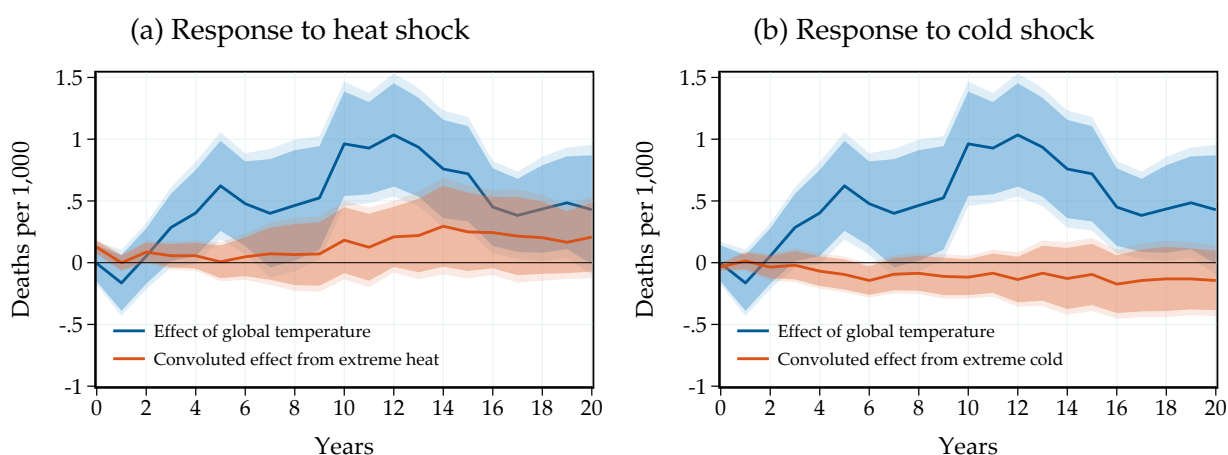
Figure D.3: Mortality and Extreme Weather Events



Notes: Mortality responses to increases in local extreme weather exposure, in deaths per 1,000. Each regression includes the contemporaneous values and six lags of all five weather measures, in addition to the baseline mortality and economic controls. Solid lines: total responses. Dashed lines: responses additionally controlling for the path of global GDP through the projection horizon. Responses are scaled by the peak increase in the corresponding annual exposure following a 1°C global temperature shock; for cold, we use the absolute value of its largest decline. Thus, the cold panel shows the effect of increased cold exposure. Thresholds and moving averages are as in Figure D.2. Dark and light shaded areas: 90 and 95% confidence bands for the total responses, based on Driscoll-Kraay standard errors.

Figure D.4 accounts for differences in treatment dynamics by convolving the mortality responses to local extreme heat and cold shocks with the corresponding exposure responses to a global temperature shock, following Sims (1986). This imposes the magnitude and persistence of local extreme-temperature exposure induced by global warming. The adjustment increases the implied mortality effect of local extremes, particularly for heat, with much of the additional effect emerging at longer horizons. Nevertheless, the resulting responses remain substantially smaller than the mortality response to global temperature, and are estimated relatively imprecisely.

Figure D.4: Mortality Impacts of Persistent Local Heat and Cold



Notes: Impulse responses of mortality to a local extreme heat shock (Panel (a)) and to a local extreme cold shock (Panel (b)), convolved with the response of the corresponding exposure to a 1°C global temperature shock using the method of Sims (1986), so that exposure follows the same path as after a global temperature shock. Solid blue line: effect of the global temperature shock. Solid red line: convoluted effect of the local extreme. Shaded areas: 90% and 95% confidence bands based on Driscoll-Kraay standard errors.

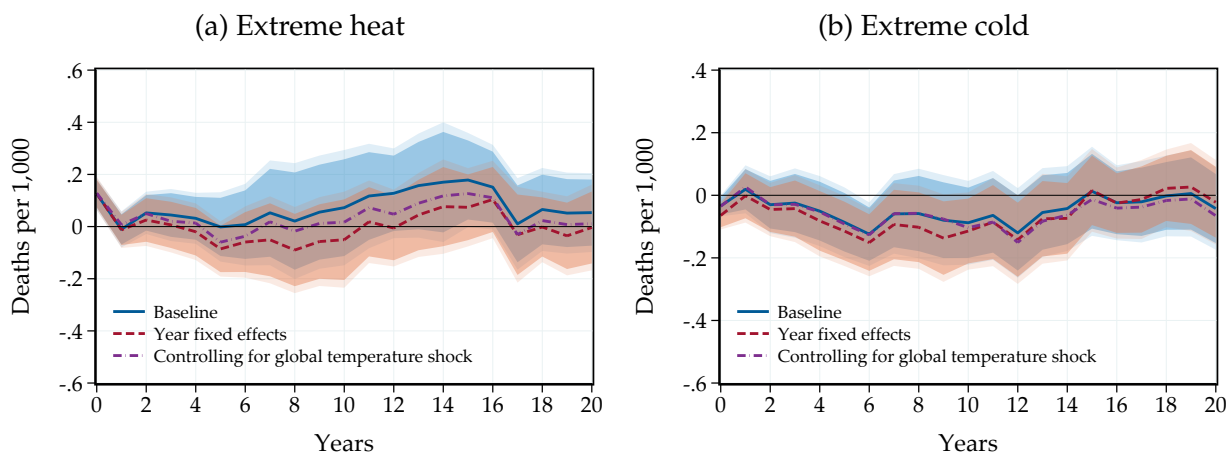
D.1.1 Sensitivity to Global Controls

Our comparison of global temperature and local heat uses observable global controls in both specifications. Local heat and cold exposure also vary across countries, which allows us to absorb common global developments more flexibly. We therefore assess whether the local mortality responses change when we replace the global economic controls with year fixed effects. In a separate specification, we retain the baseline controls and add the contemporaneous global temperature shock and two of its lags. In each exercise, we retain country fixed effects, subregion-specific linear trends, four lags of mortality changes and country-level GDP growth, and six lags of the local exposure measure.

Figure D.5 shows the results. Panel (a) reports mortality responses to an increase in heat exposure. Controlling for the global temperature shock reduces the persistent com-

ponent of the response, and year fixed effects reduce it further. The impact response changes little. Panel (b) reports the response to a reduction in cold exposure, using the decline associated with a global temperature increase for normalization. The resulting mortality reduction is similar across the three specifications. Thus, allowing for more flexible global controls does not increase the mortality effects attributed to local heat and cold or close the gap with the global temperature response.

Figure D.5: Sensitivity of Heat and Cold Responses to Global Controls



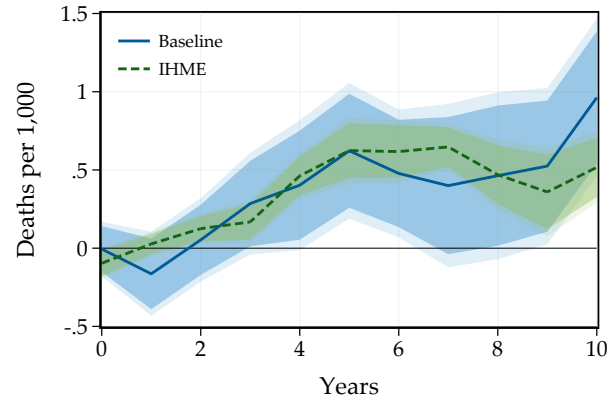
Notes: Impulse responses of mortality to local heat and cold exposure, in deaths per 1,000. Heat is the fraction of days above the country-specific 95th percentile of daily temperature in 1950–1980; cold is the fraction below the 25th percentile. Heat responses are scaled to the peak increase in heat exposure following a 1°C global temperature shock. Cold responses are scaled to the largest decline in cold exposure following the same shock, so that a negative mortality response represents a benefit from reduced cold. Solid blue lines: baseline global controls. Dashed red lines: year fixed effects. Dash-dotted purple lines: baseline controls plus the contemporaneous global temperature shock and two lags. Shaded areas: 90 and 95% confidence bands for the baseline and year fixed effects specifications, based on Driscoll-Kraay standard errors.

D.2 The Income Channel

D.2.1 Cause of Death

Our cause-of-death analysis relies on IHME data and therefore covers a smaller sample of countries due to data availability. Figure D.6 shows that the total mortality response in this restricted sample closely matches our baseline response, supporting the representativeness of the sample used for the cause-of-death analysis.

Figure D.6: Comparison of Baseline and IHME Mortality Data

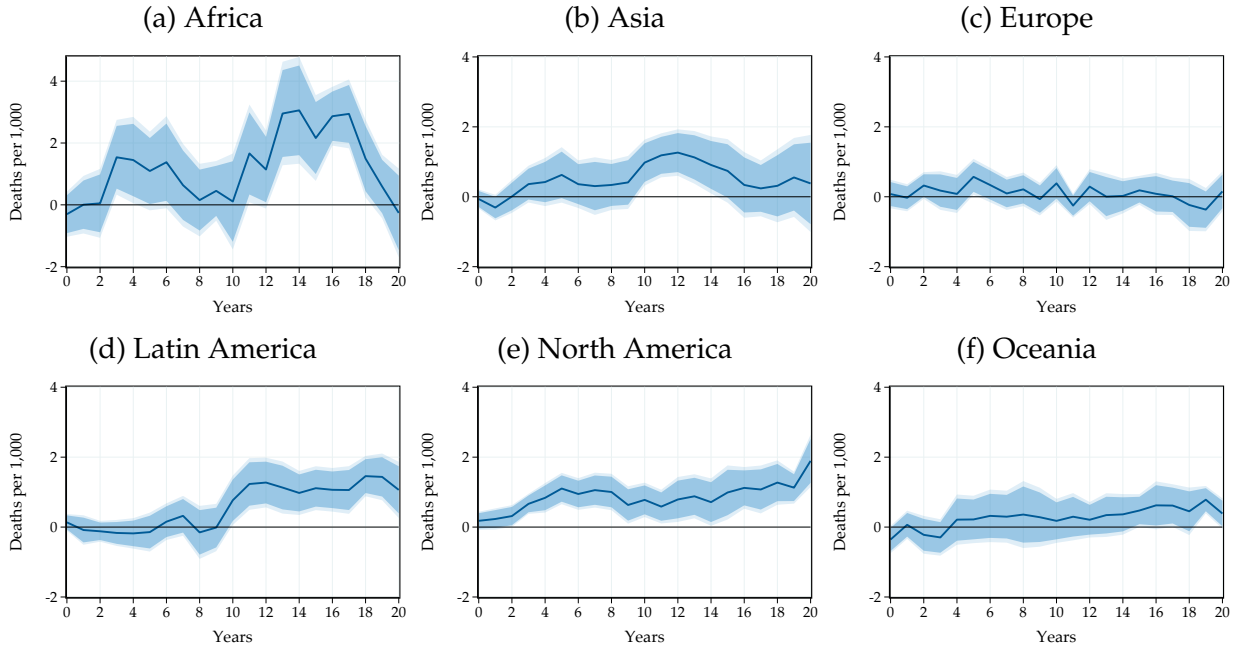


Notes: Impulse responses to a 1°C global temperature shock, in deaths per 1,000. Blue line: baseline mortality response. Green dashed line: response of mortality summed across the twelve causes available in the IHME data. The IHME specification uses three lags of changes in this total mortality rate and economic controls, two lags of the temperature shock, country fixed effects, subregion-specific linear trends, and lagged population weights. Dark and light shaded areas: 90 and 95% confidence bands based on Driscoll-Kraay standard errors.

D.2.2 Regional Heterogeneity

Figure D.7 shows mortality responses estimated separately by region. The estimates suggest larger mortality increases in Africa and Asia, and smaller responses in Europe and Oceania. The African estimates use the broader mortality series, supplementing countries with fewer than 20 baseline mortality observations with UN World Population Prospects data. As discussed in Appendix A.1, these data smooth annual variation, so the African response should be interpreted as only suggestive. Regional differences may reflect several correlated characteristics, including income, climate, and healthcare provision. We examine income and climate more directly below.

Figure D.7: Regional Heterogeneity in Mortality Responses



Notes: Impulse responses of mortality to a 1°C global temperature shock, estimated separately by region using country fixed effects and lagged population weights. The African panel supplements countries with fewer than 20 baseline mortality observations with UN World Population Prospects data. Units: deaths per 1,000. Solid lines: point estimates. Dark and light shaded areas: 90 and 95% confidence bands based on Driscoll-Kraay standard errors.

D.2.3 Heterogeneity by Income and Climate

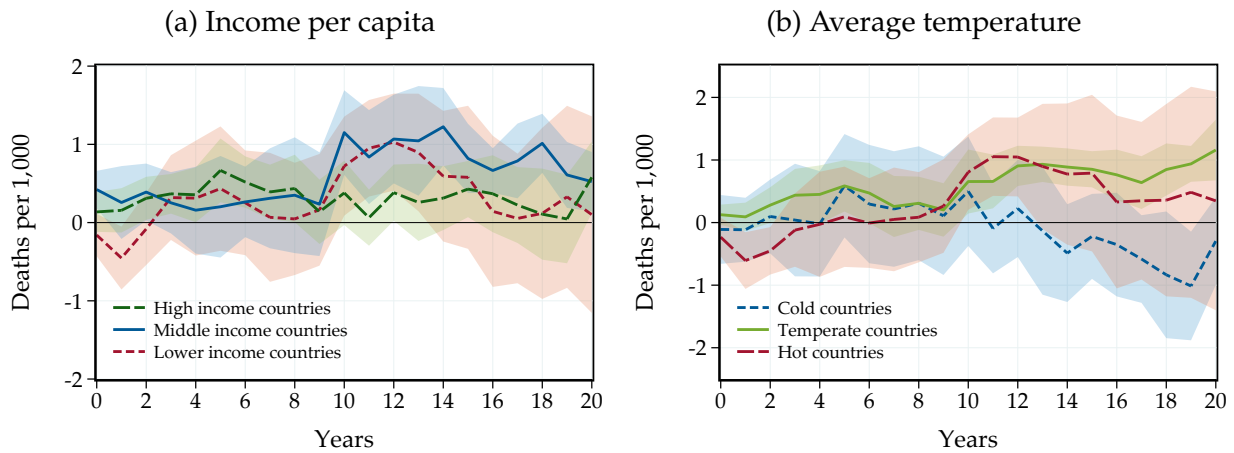
We further study how mortality responses vary with countries' initial income and climate. Following Bilal and Känzig (2026), we classify countries using their average characteristics in 1957–1959, before our estimation sample begins. We then estimate the baseline local projection separately for each group, using lagged population weights within groups.

Income per capita. Figure D.8(a) groups countries by expenditure-side real GDP per capita in purchasing power parity terms: at most \$4,000, between \$4,000 and \$8,000, and above \$8,000. Lower- and middle-income countries exhibit a pronounced increase in mortality around ten years after the shock. The response in high-income countries is smaller at these horizons, although it is positive earlier in the projection window. The pattern is therefore not a uniform income gradient at every horizon. It is consistent with greater medium-term vulnerability outside the high-income group, but the overlapping confidence bands warrant caution in interpreting differences between groups.

Average temperature. Panel (b) groups countries by population-weighted mean temperature: at most 10°C, between 10°C and 20°C, and above 20°C. Hot and temperate coun-

tries display positive medium-term mortality responses. Cold countries show a smaller response that turns even negative at longer horizons. The confidence bands are wide, particularly for hot countries, but the point estimates suggest that the global mortality increase is concentrated in countries with warmer baseline climates. Since income, climate, and healthcare systems are correlated across countries, these comparisons do not separately identify their contributions to vulnerability.

Figure D.8: Mortality Responses by Initial Income and Climate

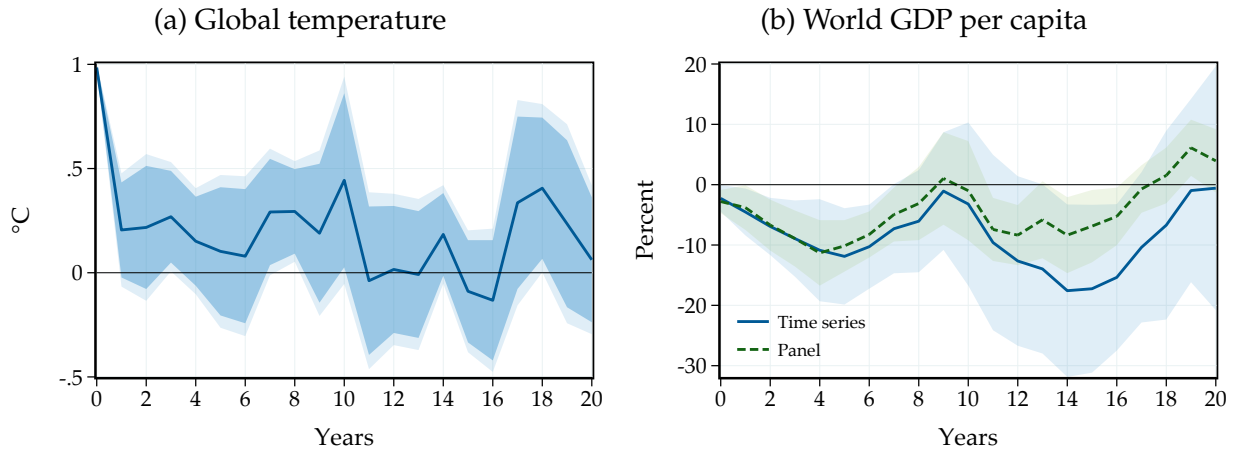


Notes: Impulse responses of mortality to a 1°C global temperature shock, estimated separately by country group using the baseline controls, country fixed effects, and lagged population weights. Groups are fixed using averages over available observations in 1957–1959. Panel (a): expenditure-side real GDP per capita from the PWT in purchasing power parity terms, with cutoffs at \$4,000 and \$8,000. Panel (b): population-weighted country temperature from Berkeley Earth, with cutoffs at 10°C and 20°C. Countries without the corresponding presample measure are excluded. Lines: point estimates. Shaded areas: 95% confidence bands based on Driscoll-Kraay standard errors. Units: deaths per 1,000.

D.2.4 Global GDP and Global Temperature

We report here the temperature and GDP responses to a global temperature shock used in the indirect aggregation exercise for the income channel. Figure D.9 reproduces the results from Bilal and Känzig (2026) extended to a 20-year horizon. Panel (a) shows the impulse response of global temperature to a temperature shock, while Panel (b) shows the corresponding GDP response. We present both, the response estimated based on the world real GDP time series, and the average GDP response from our panel, conditioning on our country sample.

Figure D.9: Global Temperature and World GDP

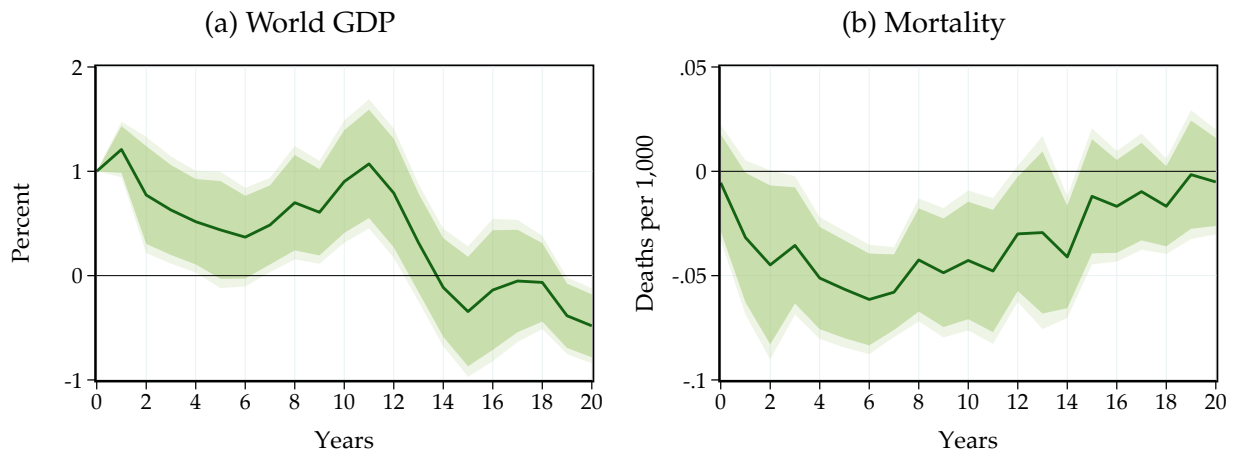


Notes: Impulse responses to a 1°C global temperature shock, based on Bilal and Känzig (2026) and extended to a 20-year horizon. Panel (a) reports the response of global mean temperature. Shaded areas denote 90% and 95% confidence bands. Panel (b) reports two estimates of the GDP response. The solid blue line shows the time-series response of world real GDP per capita, while the dashed green line shows the corresponding panel response estimated using the set of countries covered in our mortality analysis. Shaded areas denote 95% confidence bands.

D.2.5 Mortality and GDP

We report here the GDP and mortality responses to a GDP shock used in the indirect aggregation exercise for the income channel. Figure D.10 applies the empirical approach of Bilal, Känzig, and Lisiecki (2026) to our sample and extends the response horizon to 20 years. The panels report simple local projection estimates based on global GDP shocks extracted using the Hamilton (2018) filter. Panel (a) shows the dynamic response of GDP per capita to a 1% global GDP per capita shock, while Panel (b) shows the corresponding mortality response.

Figure D.10: GDP and Mortality



Notes: Impulse responses to a 1% global GDP per capita shock, following Bilal, Känzig, and Lisiecki (2026) and extended to a 20-year horizon. Global GDP shocks are constructed using the Hamilton (2018) filter. Panel (a) reports the response of global real GDP per capita, while Panel (b) reports the corresponding mortality response. Solid lines denote point estimates. Shaded areas denote 90% and 95% confidence bands.

The estimated responses are closely aligned with Bilal, Känzig, and Lisiecki (2026): global GDP rises persistently following the shock, while mortality declines gradually, reaching its largest reduction after about five to six years before fading at longer horizons.

E Model

E.1 Decentralized equilibrium

Firm optimality and market clearing. We obtain: $\alpha Z_t K_t^{\alpha-1} = r_t + \delta$ and $r_t K_t + w_t = Y_t - \delta K_t = Z_t K_t^\alpha - \delta K_t$.

Household optimality. The Bellman equation of the household is:

$$\rho V_t(K) = \max_{C,H} U(C_t) + (r_t K + w_t - C - H) V_{tK}(K) - \bar{M} X_t H^{-\mu} Y_t^{-\nu} \bar{V} + \frac{\partial V_t(K)}{\partial t}.$$

The household's first-order condition with respect to consumption C_t is: $U'(C_t) = V_{tK}(K_t)$. The envelope condition is:

$$\rho V_{tK} = r_t V_{tK} + \frac{dK_t}{dt} V_{tKK} + \frac{dV_{tK}}{dt},$$

where crucially the household does not internalize the effect of savings on mortality improvements through aggregate output. Together with the consumption optimality condition, the envelope condition leads to the standard Euler equation:

$$\frac{\dot{C}_t}{C_t} = \frac{r_t - \rho}{\gamma} = \frac{\alpha Z_t K_t^{\alpha-1} - \rho - \delta}{\gamma}.$$

Health demand. The household's first-order condition with respect to health consumption H_t is $V_{tK} = \mu \frac{M_t}{H_t} \bar{V}$, which implies $\frac{H_t}{C_t^\gamma} = \mu M_t \bar{V}$. Substituting in the expression for M_t , we obtain:

$$H_t = \bar{H} X_t^{\frac{1}{1+\mu}} C_t^{\frac{\gamma}{1+\mu}} Y_t^{-\frac{\nu}{1+\mu}},$$

where $\bar{H} \equiv (\mu \bar{M} \bar{V})^{\frac{1}{1+\mu}}$. Substituting into mortality, we obtain:

$$M_t = \tilde{M} C_t^{-\frac{\mu\gamma}{1+\mu}} Y_t^{-\frac{\nu}{1+\mu}} X_t^{\frac{1}{1+\mu}},$$

where $\tilde{M} = \bar{M}^{\frac{1}{1+\mu}} (\mu \bar{V})^{-\frac{\mu}{1+\mu}}$. Finally, capital accumulates according to: $\dot{K}_t = Z_t K_t^\alpha - \delta K_t - C_t - H_t$.

E.2 Planner problem

Consumption and Euler equation. The only inefficient decision is savings, encoded by the Euler equation (envelope condition). The planner internalizes the effects of higher capital on output. The envelope condition with the shadow interest rate is:

$$\rho V_{tK} = r_t V_{tK} + \frac{dK_t}{dt} V_{tKK} + v \frac{M_t}{Y_t} \frac{\partial Y}{\partial K} \bar{V} + \frac{dV_{tK}}{dt}.$$

Substituting in the consumption optimality condition:

$$(\rho - r_t) u'(C_t) = u''(C_t) \frac{dC_t}{dt} + v \frac{M_t}{Y_t} (r_t + \delta) \bar{V}.$$

Re-arranging: $\frac{\dot{C}_t}{C_t} = \frac{r_t - \rho}{\gamma} + v \frac{M_t C_t^\gamma}{\gamma Y_t} (r_t + \delta) \bar{V}$. Substituting in the MPK:

$$\frac{\dot{C}_t}{C_t} = \frac{\alpha Z_t K_t^{\alpha-1} - \rho - \delta}{\gamma} + \frac{\alpha v}{\gamma} \frac{C_t^\gamma}{K_t} M_t \bar{V}.$$

Consumption grows faster for the planner: consumption today is lower because the planner invests more to lower mortality going forward.

Other equations. All the other equations are the same as in the decentralized equilibrium. Only the Euler equation changes.

E.3 Steady state

We start by analyzing the steady state of this economy with constant TFP $Z_t = \bar{Z}$ and mortality shifter $X_t = \bar{X}$. We normalize $X_0 = 1$ and consider a steady state without warming, so that $\bar{X} = X_0$.

E.3.1 Decentralized equilibrium

The Euler equation implies: $\rho + \delta = \alpha \bar{Z} K_{ss}^{\alpha-1}$, which pins down the steady-state capital stock: $K_{ss} = \left(\frac{\alpha \bar{Z}}{\rho + \delta} \right)^{\frac{1}{1-\alpha}}$ and steady-state output $Y_{ss} = \bar{Z} K_{ss}^\alpha$. The capital accumulation equation pins down steady-state consumption given the steady-state capital stock and given steady-state output: $C_{ss} + \bar{H} Y_{ss}^{-\frac{\nu}{1+\mu}} C_{ss}^{\frac{\gamma}{1+\mu}} = \bar{Z} K_{ss}^\alpha - \delta K_{ss}$. From there, we obtain: $H_{ss} = \bar{H} C_{ss}^{\frac{\gamma}{1+\mu}} Y_{ss}^{-\frac{\nu}{1+\mu}}$, $M_{ss} = \tilde{M} C_{ss}^{\frac{-\mu\gamma}{1+\mu}} Y_{ss}^{-\frac{\nu}{1+\mu}}$, and $I_{ss} = \delta K_{ss}$.

E.3.2 Planner

We temporarily omit subscripts for simplicity. The only difference is in the Euler equation: $\rho + \delta = \alpha Z K^{\alpha-1} + \alpha \nu \frac{C^\gamma}{K} M \bar{V} = \alpha Z K^{\alpha-1} + \eta Z^{-\frac{\nu}{1+\mu}} C^{\frac{\gamma}{1+\mu}} K^{-\left(1+\frac{\alpha\nu}{1+\mu}\right)}$, where $\eta \equiv \alpha \nu \bar{V} \tilde{M}$. We have to solve jointly for consumption and capital. The resource constraint delivers, as in the laissez-faire: $C + \bar{H} Z^{-\frac{\nu}{1+\mu}} K^{-\frac{\alpha\nu}{1+\mu}} C^{\frac{\gamma}{1+\mu}} = \bar{Z} K^\alpha - \delta K$. The rest of the allocation is the same as in the laissez-faire, conditional on C_{ss}, K_{ss} .

E.4 Log-linearized Transitional Dynamics

We now denote steady-state variables without time subscripts. We denote deviations from steady state with hats.

E.4.1 Decentralized Equilibrium: Counterfactuals

We start by noting that:

$$\frac{\hat{H}_t}{H_t} = \frac{\gamma}{1+\mu} \frac{\hat{C}_t}{C_t} - \frac{\nu}{1+\mu} \frac{\hat{Y}_t}{Y_t} + \frac{1}{1+\mu} \frac{\hat{X}_t}{X_t} = \frac{\gamma}{1+\mu} \frac{\hat{C}_t}{C_t} - \frac{\alpha\nu}{1+\mu} \frac{\hat{K}_t}{K_t} - \frac{\nu}{1+\mu} \frac{\hat{Z}_t}{Z_t} + \frac{1}{1+\mu} \frac{\hat{X}_t}{X_t}.$$

Denote $\hat{x}_t = \log \frac{X_t}{X_0}$. To first order:

$$\hat{H}_t = \frac{\gamma}{1+\mu} \frac{H}{C} \hat{C}_t - \frac{\alpha\nu}{1+\mu} \frac{H}{K} \hat{K}_t - \frac{\nu}{1+\mu} H \hat{z}_t + \frac{H}{1+\mu} \hat{x}_t.$$

We also note that:

$$\frac{\hat{M}_t}{M} = -\frac{\mu\gamma}{1+\mu} \frac{\hat{C}_t}{C} - \frac{\nu}{1+\mu} \frac{\hat{Y}_t}{Y} + \frac{1}{1+\mu} \hat{x}_t.$$

We then linearize the resource constraint:

$$\begin{aligned} \frac{d\hat{K}_t}{dt} &= (\alpha Z K^{\alpha-1} - \delta) \hat{K}_t - \hat{C}_t - \hat{H}_t + \hat{Z}_t K^\alpha = \rho \hat{K}_t - \hat{C}_t - \hat{H}_t + Y \hat{z}_t \\ &= \phi_K \hat{K}_t - \phi_C \hat{C}_t + \left(Y + \frac{\nu}{1+\mu} H\right) \hat{z}_t - \frac{H}{1+\mu} \hat{x}_t \end{aligned}$$

where we denoted $\hat{z}_t = \hat{Z}_t/Z$ and defined: $\phi_K \equiv \rho + \frac{\alpha\nu}{1+\mu} \frac{H}{K}$ and $\phi_C \equiv 1 + \frac{\gamma}{1+\mu} \frac{H}{C}$. For the planner, both this resource constraint and the Euler equation change. We derive the

corresponding system below. Next, we linearize the Euler equation:

$$\frac{d\hat{C}_t}{dt} = \frac{C}{\gamma} \left(-\alpha(1-\alpha)ZK^{\alpha-2}\hat{K}_t + \alpha K^{\alpha-1}\hat{Z}_t \right) = \frac{C}{\gamma} \left(-\frac{(1-\alpha)(r+\delta)}{K}\hat{K}_t + (r+\delta)\hat{Z}_t \right).$$

We define:

$$P_t = \begin{pmatrix} \hat{K}_t \\ \hat{C}_t \end{pmatrix}.$$

We can summarize the linearized resource constraint and Euler equation as:

$$\dot{P}_t = AP_t + S_t,$$

where:

$$A = \begin{pmatrix} \phi_K & -\phi_C \\ -\frac{(1-\alpha)(r+\delta)C}{\gamma K} & 0 \end{pmatrix}, \quad S_t = \hat{z}_t B_z + \hat{x}_t B_m, \quad B_z = \begin{pmatrix} Y + \frac{\nu}{1+\mu}H \\ \frac{(r+\delta)C}{\gamma} \end{pmatrix}, \quad B_m = \begin{pmatrix} -\frac{H}{1+\mu} \\ 0 \end{pmatrix}.$$

We have an initial condition $\hat{K}_0$, and a terminal condition $\hat{C}_t \rightarrow 0$. We now apply standard Blanchard-Kahn arguments. Let $A = M^{-1}DM$, with D diagonal. For determinacy we require that parameters are such that D has a positive eigenvalue in the top left position, and a negative eigenvalue in the bottom right position. We denote by $\mathcal{P}_t = MP_t$, so that

$$\dot{\mathcal{P}}_t = D\mathcal{P}_t + MS_t.$$

We then solve explicitly for $\mathcal{P}_t$:

$$\mathcal{P}_t = e^{tD} \left[\mathcal{P}_0 + \int_0^t e^{-sD} (MS_s) ds \right].$$

Hence, long-run stability requires the top entry of the bracket to be zero as time grows. That is:

$$0 = \mathcal{P}_{0,1} + \int_0^\infty e^{-sD_1} (MS_s)_1 ds.$$

Therefore,

$$M_{1\bullet}P_0 = - \int_0^\infty e^{-sD_1} M_{1\bullet} S_s ds.$$

We can thus solve for initial consumption:

$$\widehat{C}_0 = -\frac{1}{M_{12}} \left[M_{11}\widehat{K}_0 + \int_0^\infty e^{-sD_1} M_{1\bullet} S_s ds \right].$$

We denote $\varepsilon_K = -\frac{M_{11}}{M_{12}}$, $\varepsilon_S = -\frac{1}{M_{12}} M_{1\bullet}$ and $\varepsilon_{S,s} = e^{-sD_1} \varepsilon_S$. We can write more compactly:

$$\widehat{C}_0 = \varepsilon_K \widehat{K}_0 + \int_0^\infty \varepsilon_{S,s} S_s ds.$$

Of course, this condition must hold at all times:

$$\widehat{C}_t = \varepsilon_K \widehat{K}_t + \int_0^\infty \varepsilon_{S,s} S_{t+s} ds.$$

To compute consumption somewhat more efficiently, we exploit that:

$$\begin{aligned} \widehat{C}_t &= \varepsilon_K \widehat{K}_t + \varepsilon_S B_z \int_0^\infty e^{-sD_1} \widehat{z}_{t+s} ds + \varepsilon_S B_m \int_0^\infty e^{-sD_1} \widehat{x}_{t+s} ds \\ &= \varepsilon_K \widehat{K}_t + e^{tD_1} \varepsilon_S B_z \int_t^\infty e^{-sD_1} \widehat{z}_s ds + e^{tD_1} \varepsilon_S B_m \int_t^\infty e^{-sD_1} \widehat{x}_s ds, \end{aligned}$$

after changing variables in the integral.

E.4.2 Model Inversion: Proof of Proposition 1

We substitute the solution for linearized consumption into the law of motion of capital:

$$\frac{d\widehat{K}_t}{dt} = A_{11}\widehat{K}_t + A_{12}\widehat{C}_t + S_{1t} = (A_{11} + A_{12}\varepsilon_K)\widehat{K}_t + S_{1t} + A_{12} \int_0^\infty \varepsilon_{S,s} S_{t+s} ds.$$

Denote $\kappa = -(A_{11} + A_{12}\varepsilon_K)$ and $\mathcal{S}_t = S_{1t} + A_{12} \int_0^\infty \varepsilon_{S,s} S_{t+s} ds$ so that:

$$\frac{d\widehat{K}_t}{dt} = -\kappa \widehat{K}_t + \mathcal{S}_t.$$

We obtain:

$$\widehat{K}_t = e^{-\kappa t} \widehat{K}_0 + e^{-\kappa t} \int_0^t e^{\kappa s} \mathcal{S}_s ds.$$

In small log deviations:

$$\widehat{k}_t = e^{-\kappa t} \widehat{k}_0 + \frac{e^{-\kappa t}}{K} \int_0^t e^{\kappa s} \mathcal{S}_s ds.$$

Since (in small log deviations): $\widehat{y}_t = \widehat{z}_t + \alpha \widehat{k}_t$, we are left with calculating $\widehat{k}_t$ as a function of the sequence $\widehat{z}$. We express:

$$\begin{aligned} \int_0^t e^{\kappa s} \mathcal{S}_s ds &= \int_0^t e^{\kappa s} \left(S_{1s} + A_{12} \int_0^\infty \varepsilon_{S,r} S_{s+r} dr \right) ds \\ &= \int_0^t e^{\kappa s} S_{1s} ds + A_{12} \int_0^t \int_0^\infty \mathbb{1}[s \leq t] \varepsilon_{S,r} S_{s+r} e^{\kappa s} ds dr \\ &= \int_0^t e^{\kappa s} S_{1s} ds + A_{12} \int_0^t \int_0^\infty \mathbb{1}[s \leq t] \varepsilon_S S_{s+r} e^{\kappa s - D_1 r} ds dr. \end{aligned}$$

Changing variables to $\tau = s + r$ over r , we obtain

$$\begin{aligned} \int_0^t e^{\kappa s} \mathcal{S}_s ds &= \int_0^t e^{\kappa s} S_{1s} ds + A_{12} \varepsilon_S \int_0^\infty \int_0^\infty \mathbb{1}[s \leq t, s \leq \tau] S_\tau e^{\kappa s - D_1(\tau-s)} ds d\tau \\ &= \int_0^t e^{\kappa s} S_{1s} ds + A_{12} \varepsilon_S \int_{\tau=0}^\infty e^{-D_1 \tau} S_\tau \int_{s=0}^{\min\{t, \tau\}} e^{(D_1 + \kappa)s} ds d\tau \\ &\equiv \int_0^t e^{\kappa s} S_{1s} ds + A_{12} \varepsilon_S \int_{\tau=0}^\infty J_{t,\tau} S_\tau d\tau, \end{aligned}$$

where we defined:

$$J_{t,\tau} = e^{-D_1 \tau} \int_{s=0}^{\min\{t, \tau\}} e^{(D_1 + \kappa)s} ds = e^{-D_1 \tau} \frac{e^{(D_1 + \kappa) \min\{t, \tau\}} - 1}{D_1 + \kappa}.$$

Thus,

$$\begin{aligned} \widehat{k}_t &= e^{-\kappa t} \widehat{k}_0 + \frac{e^{-\kappa t}}{K} \int_0^\infty \left\{ B_{z,1} \mathbb{1}[s \leq t] e^{\kappa s} + A_{12} (\varepsilon_S B_z) J_{t,s} \right\} \widehat{z}_s ds \\ &\quad + \frac{e^{-\kappa t}}{K} \int_0^\infty \left\{ B_{m,1} \mathbb{1}[s \leq t] e^{\kappa s} + A_{12} (\varepsilon_S B_m) J_{t,s} \right\} \widehat{x}_s ds \\ &\equiv e^{-\kappa t} \widehat{k}_0 + \int_0^\infty \mathcal{K}_{t,s}^z \widehat{z}_s ds + \int_0^\infty \mathcal{K}_{t,s}^m \widehat{x}_s ds, \end{aligned}$$

where we defined:

$$\mathcal{K}_{t,s}^z = \frac{e^{-\kappa t}}{K} \left\{ B_{z,1} \mathbb{1}[s \leq t] e^{\kappa s} + A_{12}(\varepsilon_S B_z) J_{t,s} \right\}.$$

$$\mathcal{K}_{t,s}^m = \frac{e^{-\kappa t}}{K} \left\{ B_{m,1} \mathbb{1}[s \leq t] e^{\kappa s} + A_{12}(\varepsilon_S B_m) J_{t,s} \right\}.$$

Hence, we have obtained:

$$\hat{y}_t = \hat{z}_t + \alpha \left\{ e^{-\kappa t} \hat{k}_0 + \int_0^\infty \mathcal{K}_{t,s}^z \hat{z}_s ds + \int_0^\infty \mathcal{K}_{t,s}^m \hat{x}_s ds \right\}.$$

Recall that

$$\hat{C}_t = \varepsilon_K \hat{K}_t + e^{tD_1} \varepsilon_S B_z \int_t^\infty e^{-sD_1} \hat{z}_s ds + e^{tD_1} \varepsilon_S B_m \int_t^\infty e^{-sD_1} \hat{x}_s ds,$$

So in log deviations

$$\hat{c}_t = \varepsilon_K \hat{k}_t \frac{K}{C} + \frac{1}{C} e^{tD_1} \varepsilon_S B_z \int_t^\infty e^{-sD_1} \hat{z}_s ds + \frac{1}{C} e^{tD_1} \varepsilon_S B_m \int_t^\infty e^{-sD_1} \hat{x}_s ds,$$

Plugging in the formula for $\hat{k}_t$ we get

$$\begin{aligned} \hat{c}_t &= \varepsilon_K \frac{K}{C} \left\{ e^{-\kappa t} \hat{k}_0 + \int_0^\infty \mathcal{K}_{t,s}^z \hat{z}_s ds + \int_0^\infty \mathcal{K}_{t,s}^m \hat{x}_s ds \right\} \\ &\quad + \frac{e^{tD_1}}{C} \left[\varepsilon_S B_z \int_t^\infty e^{-sD_1} \hat{z}_s ds + \varepsilon_S B_m \int_t^\infty e^{-sD_1} \hat{x}_s ds \right]. \end{aligned}$$

Defining:

$$\begin{aligned} \mathcal{C}_{t,s}^z &= \varepsilon_K \frac{K}{C} \mathcal{K}_{t,s}^z + \frac{e^{tD_1}}{C} (\varepsilon_S B_z) e^{-sD_1} \mathbb{1}[s \geq t], \\ \mathcal{C}_{t,s}^m &= \varepsilon_K \frac{K}{C} \mathcal{K}_{t,s}^m + \frac{e^{tD_1}}{C} (\varepsilon_S B_m) e^{-sD_1} \mathbb{1}[s \geq t], \end{aligned}$$

we obtain:

$$\hat{c}_t = \varepsilon_K \frac{K}{C} e^{-\kappa t} \hat{k}_0 + \int_0^\infty C_{t,s}^z \hat{z}_s ds + \int_0^\infty C_{t,s}^m \hat{x}_s ds.$$

Also recall the log linear expression for mortality:

$$\hat{m}_t = -\frac{\mu\gamma}{1+\mu} \hat{c}_t - \frac{\nu}{1+\mu} \hat{y}_t + \frac{1}{1+\mu} \hat{x}_t.$$

Plugging in expressions for $\hat{c}_t$ and $\hat{y}_t$, this becomes

$$\begin{aligned} \hat{m}_t &= -\frac{\mu\gamma}{1+\mu} \left\{ \varepsilon_K \frac{K}{C} e^{-\kappa t} \hat{k}_0 + \int_0^\infty C_{t,s}^z \hat{z}_s ds + \int_0^\infty C_{t,s}^m \hat{x}_s ds \right\} \\ &\quad - \frac{\nu}{1+\mu} \left\{ \hat{z}_t + \alpha e^{-\kappa t} \hat{k}_0 + \alpha \int_0^\infty \mathcal{K}_{t,s}^z \hat{z}_s ds + \alpha \int_0^\infty \mathcal{K}_{t,s}^m \hat{x}_s ds \right\} \\ &\quad + \frac{1}{1+\mu} \hat{x}_t. \end{aligned}$$

Collecting the $\hat{k}_0$ coefficient:

$$\begin{aligned} \hat{m}_t &= - \left[\frac{\mu\gamma}{1+\mu} \varepsilon_K \frac{K}{C} + \frac{\nu\alpha}{1+\mu} \right] e^{-\kappa t} \hat{k}_0 \\ &\quad - \frac{\nu}{1+\mu} \hat{z}_t + \frac{1}{1+\mu} \hat{x}_t \\ &\quad - \int_0^\infty \left[\frac{\mu\gamma}{1+\mu} C_{t,s}^z + \frac{\nu\alpha}{1+\mu} \mathcal{K}_{t,s}^z \right] \hat{z}_s ds \\ &\quad - \int_0^\infty \left[\frac{\mu\gamma}{1+\mu} C_{t,s}^m + \frac{\nu\alpha}{1+\mu} \mathcal{K}_{t,s}^m \right] \hat{x}_s ds. \end{aligned}$$

Defining:

$$\begin{aligned} \mathcal{M}_{t,s}^z &= -\frac{\mu\gamma}{1+\mu} C_{t,s}^z - \frac{\nu\alpha}{1+\mu} \mathcal{K}_{t,s}^z, \\ \mathcal{M}_{t,s}^m &= -\frac{\mu\gamma}{1+\mu} C_{t,s}^m - \frac{\nu\alpha}{1+\mu} \mathcal{K}_{t,s}^m, \end{aligned}$$

we obtain the compact form:

$$\hat{m}_t = - \left[\frac{\mu\gamma}{1+\mu} \varepsilon_K \frac{K}{C} + \frac{\nu\alpha}{1+\mu} \right] e^{-\kappa t} \hat{k}_0 - \frac{\nu}{1+\mu} \hat{z}_t + \frac{1}{1+\mu} \hat{x}_t + \int_0^\infty \mathcal{M}_{t,s}^z \hat{z}_s ds + \int_0^\infty \mathcal{M}_{t,s}^m \hat{x}_s ds.$$

The joint linear mapping from $\{\hat{z}_t, \hat{x}_t\}$ to $\{\hat{y}_t, \hat{m}_t\}$ takes the form:

$$\begin{pmatrix} \hat{y} \\ \hat{m} \end{pmatrix} = \begin{pmatrix} \text{Id} + \alpha \mathcal{K}^z & \alpha \mathcal{K}^m \\ -\frac{\nu}{1+\mu} \text{Id} + \mathcal{M}^z & \frac{1}{1+\mu} \text{Id} + \mathcal{M}^m \end{pmatrix} \begin{pmatrix} \hat{z} \\ \hat{x} \end{pmatrix}.$$

Identification requires this matrix to be invertible.

E.4.3 Planner

The planner faces a different Euler equation, but the log-linearization of the resource constraint is the same conditional on C. The Euler equation becomes, to first order:

$$\frac{d\hat{C}_t}{dt} = \left(\alpha Z_t K_t^{\alpha-1} - \rho - \delta + \widehat{\eta Z_t^{-\frac{\nu}{1+\mu}} C_t^{\frac{\gamma}{1+\mu}} K_t^{-(1+\frac{\alpha\nu}{1+\mu})} X_t^{\frac{1}{1+\mu}}} \right) \frac{C}{\gamma} + 0$$

and so:

$$\begin{aligned} \frac{\gamma}{C} \frac{d\hat{C}_t}{dt} &= -\alpha(1-\alpha)ZK^{\alpha-2}\hat{K}_t + \alpha K^{\alpha-1}\hat{Z}_t \\ &\quad + \frac{\alpha\nu\bar{V}C^\gamma M}{K} \left\{ -\frac{\nu}{1+\mu}\hat{z}_t + \frac{\gamma}{1+\mu}\frac{\hat{C}_t}{C} - \left(1 + \frac{\alpha\nu}{1+\mu}\right)\frac{\hat{K}_t}{K} + \frac{1}{1+\mu}\hat{x}_t \right\} \\ &= -\frac{(1-\alpha)(r+\delta)}{K}\hat{K}_t + (r+\delta)\hat{z}_t - \nu\mathcal{M}\hat{z}_t + \gamma\mathcal{M}\frac{\hat{C}_t}{C} - (1+\mu+\alpha\nu)\mathcal{M}\frac{\hat{K}_t}{K} + \mathcal{M}\hat{x}_t \\ &= -\frac{(1+\mu+\alpha\nu)\mathcal{M} + (1-\alpha)(r+\delta)}{K}\hat{K}_t + (r+\delta-\nu\mathcal{M})\hat{z}_t + \gamma\mathcal{M}\frac{\hat{C}_t}{C} + \mathcal{M}\hat{x}_t \end{aligned}$$

where $\mathcal{M} = \frac{\alpha\nu\bar{V}C^\gamma M}{(1+\mu)K}$. The resource constraint is:

$$\begin{aligned} \frac{d\hat{K}_t}{dt} &= (\alpha ZK^{\alpha-1} - \delta)\hat{K}_t - \hat{C}_t - \hat{H}_t + \hat{Z}_t K^\alpha \\ &= (\rho - (1+\mu)\mathcal{M})\hat{K}_t - \hat{C}_t - \hat{H}_t + \gamma\hat{z}_t \\ &= (\phi_K - \psi_{KK})\hat{K}_t - \phi_C\hat{C}_t + (Y + \frac{\nu}{1+\mu}H)\hat{z}_t - \frac{H}{1+\mu}\hat{x}_t \end{aligned}$$

Thus, for the planner, we must change the A and B matrices to:

$$A = \begin{pmatrix} \phi_K - \psi_{KK} & -\phi_C \\ -\frac{(1-\alpha)(r+\delta)+\psi_K}{\gamma}\frac{C}{K} & \psi_C \end{pmatrix}, \quad S_t = \hat{z}_t B_z + \hat{x}_t B_m, \quad B_z = \begin{pmatrix} Y + \frac{\nu}{1+\mu}H \\ \frac{r+\delta-\psi_z}{\gamma}C \end{pmatrix}, \quad B_m = \begin{pmatrix} -\frac{H}{1+\mu} \\ \frac{\psi_C C}{\gamma} \end{pmatrix}.$$

where $\psi_K = (1 + \mu + \alpha\nu)\mathcal{M}$, $\psi_{KK} = (1 + \mu)\mathcal{M}$, $\psi_C = \mathcal{M}$ and $\psi_z = \nu\mathcal{M}$.

E.5 Nonlinear transitional dynamics

We can use a quasi-Newton algorithm that uses the linearized transitional dynamics as the gradient. We collect the equilibrium conditions:

$$\begin{aligned}\dot{C}_t &= \frac{\alpha Z_t K_t^{\alpha-1} + E_t(C_t, K_t) - \rho - \delta}{\gamma} C_t \equiv F_t(K_t, C_t) \\ \dot{K}_t &= Z_t K_t^\alpha - \delta K_t - C_t - \bar{H} C_t^{\frac{\gamma}{1+\mu}} (Z_t K_t^\alpha)^{-\frac{\nu}{1+\mu}} X_t^{\frac{1}{1+\mu}} \equiv G_t(K_t, C_t),\end{aligned}$$

where in the decentralized equilibrium $E_t \equiv 0$. For the planner, we have:

$$E_t(C_t, K_t) = \eta C_t^{\frac{\gamma}{1+\mu}} Z_t^{-\frac{\nu}{1+\mu}} K_t^{-\frac{\alpha\nu+1+\mu}{1+\mu}} X_t^{\frac{1}{1+\mu}}.$$

Discretized, these become:

$$C_{t+1} - C_t = F_t(K_t, C_t) \quad K_{t+1} - K_t = G_t(K_t, C_t),$$

together with initial condition $K_0 = \bar{K}_0$ given, and terminal condition $C_T = \bar{C}_T$ given. We are looking for a path of $\mathbf{X} = \{C_t, K_t\}_t$ that sets all these equations to zero, which we rewrite:

$$0 = \mathcal{F}_t(K_t, C_t, C_{t+1}) \quad 0 = \mathcal{G}_t(K_t, K_{t+1}, C_t),$$

with $\mathcal{F} = F + C_t - C_{t+1}$ and $\mathcal{G} = G + K_t - K_{t+1}$. We collect all these equations under the name $\mathcal{H}$. A Newton updating algorithm can be constructed as follows:

$$\mathbf{X}^{n+1} = \mathbf{X}^n - \mathcal{J}(\mathbf{X}^n)^{-1} \mathcal{H}(\mathbf{X}^n),$$

where $\mathcal{J}$ is the Jacobian of $\mathcal{H}$ w.r.t. $\mathbf{X}$. To calculate this Jacobian consistently, we have to specify the order in which we stack the unknowns and the equations. We posit:

$$\mathbf{X}_t = \begin{pmatrix} K_1 \\ K_2 \\ \vdots \\ K_t \\ \vdots \\ K_{T-1} \\ C_0 \\ C_1 \\ \vdots \\ C_t \\ \vdots \\ C_{T-1} \end{pmatrix} \quad \mathcal{H}(\mathbf{X}) = \begin{pmatrix} \mathcal{G}_0(\bar{K}_0, K_1, C_0) \\ \mathcal{G}_1(K_1, K_2, C_1) \\ \vdots \\ \mathcal{G}_{t-1}(K_{t-1}, K_t, C_{t-1}) \\ \vdots \\ \mathcal{G}_{T-2}(K_{T-2}, K_{T-1}, C_{T-2}) \\ \mathcal{F}_0(\bar{K}_0, C_0, C_1) \\ \mathcal{F}_1(K_1, C_1, C_2) \\ \vdots \\ \mathcal{F}_t(K_t, C_t, C_{t+1}) \\ \vdots \\ \mathcal{F}_{T-1}(K_{T-1}, C_{T-1}, \bar{C}_T) \end{pmatrix}.$$

Note that (1) there is no $\mathcal{G}_{T-1}$ condition, as it would only determine K_T , which enters no other condition, and (2) there is no $\mathcal{F}_T$ condition because we already enforced $C_T = \bar{C}_T$ in the last Euler equation. The Jacobian keeps $\mathcal{H}$ on the rows and $\mathbf{X}$ on the columns; it is

sparse and satisfies:

$$\mathcal{J} = \begin{pmatrix} \frac{\partial \mathcal{G}_0}{\partial K_1} & & & 0 & \frac{\partial \mathcal{G}_0}{\partial C_0} & & 0 \\ \frac{\partial \mathcal{G}_1}{\partial K_1} & \frac{\partial \mathcal{G}_1}{\partial K_2} & & & \frac{\partial \mathcal{G}_1}{\partial C_1} & & \\ & \ddots & \ddots & & & \ddots & \\ & & \frac{\partial \mathcal{G}_{t-1}}{\partial K_{t-1}} & \frac{\partial \mathcal{G}_{t-1}}{\partial K_t} & & \frac{\partial \mathcal{G}_{t-1}}{\partial C_{t-1}} & \\ & & & \ddots & & & \ddots \\ & & & & \frac{\partial \mathcal{G}_{T-2}}{\partial K_{T-2}} & \frac{\partial \mathcal{G}_{T-2}}{\partial K_{T-1}} & 0 & & \frac{\partial \mathcal{G}_{T-2}}{\partial C_{T-2}} & 0 \\ 0 & & & & 0 & \frac{\partial \mathcal{F}_0}{\partial C_0} & \frac{\partial \mathcal{F}_0}{\partial C_1} & & & \\ 0 & \frac{\partial \mathcal{F}_1}{\partial K_1} & 0 & & & \frac{\partial \mathcal{F}_1}{\partial C_1} & \frac{\partial \mathcal{F}_1}{\partial C_2} & & & \\ 0 & & \ddots & \ddots & & & \ddots & \ddots & & \\ & & & \frac{\partial \mathcal{F}_{t-1}}{\partial K_{t-1}} & 0 & & \frac{\partial \mathcal{F}_{t-1}}{\partial C_{t-1}} & \frac{\partial \mathcal{F}_{t-1}}{\partial C_t} & & \\ & & & & \ddots & & & \ddots & & \\ & & & & & \frac{\partial \mathcal{F}_{T-2}}{\partial K_{T-2}} & 0 & & \frac{\partial \mathcal{F}_{T-2}}{\partial C_{T-2}} & \frac{\partial \mathcal{F}_{T-2}}{\partial C_{T-1}} \\ 0 & & & & & \frac{\partial \mathcal{F}_{T-1}}{\partial K_{T-1}} & 0 & & & \frac{\partial \mathcal{F}_{T-1}}{\partial C_{T-1}} \end{pmatrix}.$$

We thus have to calculate:

$$\begin{aligned} \frac{\partial \mathcal{G}_t}{\partial K_t} &= 1 + \frac{\partial G_t}{\partial K_t} = 1 + \alpha Z_t K_t^{\alpha-1} - \delta + \frac{\alpha \nu}{1+\mu} H_t / K_t \\ \frac{\partial \mathcal{G}_t}{\partial K_{t+1}} &= -1 \\ \frac{\partial \mathcal{G}_t}{\partial C_t} &= -1 - \frac{\gamma}{1+\mu} H_t / C_t \\ \frac{\partial \mathcal{F}_t}{\partial C_t} &= 1 + \frac{\alpha Z_t K_t^{\alpha-1} + E_t - \rho - \delta}{\gamma} + \frac{1}{1+\mu} E_t \\ \frac{\partial \mathcal{F}_t}{\partial C_{t+1}} &= -1 \\ \frac{\partial \mathcal{F}_t}{\partial K_t} &= \frac{\alpha(\alpha-1) Z_t K_t^{\alpha-2} - \frac{\alpha \nu + 1 + \mu}{1+\mu} E_t / K_t}{\gamma} C_t \end{aligned}$$

with $H_t = \bar{H} C_t^{\frac{\gamma}{1+\mu}} Z_t^{-\frac{\nu}{1+\mu}} K_t^{-\frac{\alpha \nu}{1+\mu}} X_t^{\frac{1}{1+\mu}}$.

For welfare, we use steady-state welfare in the decentralized equilibrium as the base-

line (for both the DE and SP cases). We have:

$$V_{ss}^{DE} + \frac{u((1 + \omega_t)C_{ss}^{DE}) - u(C_{ss}^{DE})}{\rho} = V_t,$$

so that:

$$u((1 + \omega_t)C_{ss}^{DE}) - u(C_{ss}^{DE}) = \rho(V_t - V_{ss}^{DE}),$$

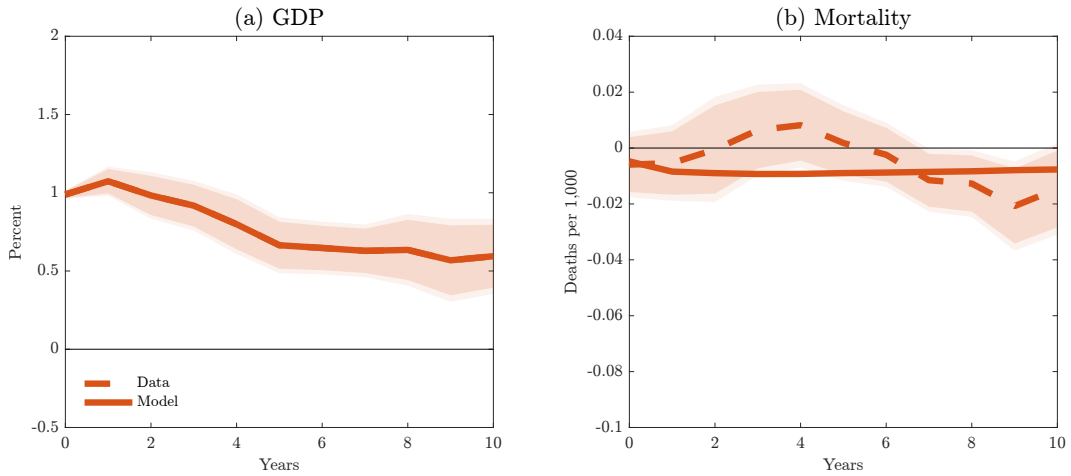
and hence:

$$\omega_t = \frac{1}{C_{ss}^{DE}} u^{-1} \left[u(C_{ss}^{DE}) + \rho(V_t - V_{ss}^{DE}) \right] - 1$$

F Additional Model Results

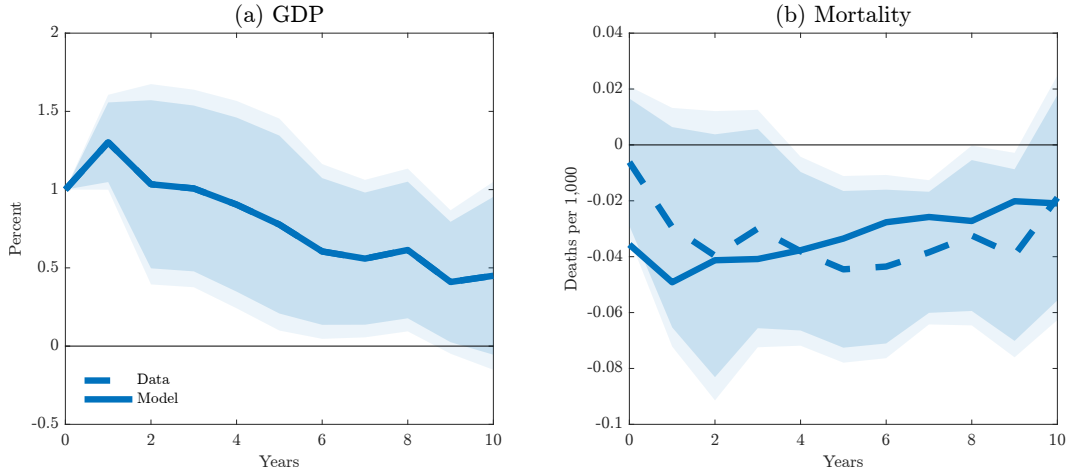
F.1 Estimation

Figure F.1: Estimation of μ



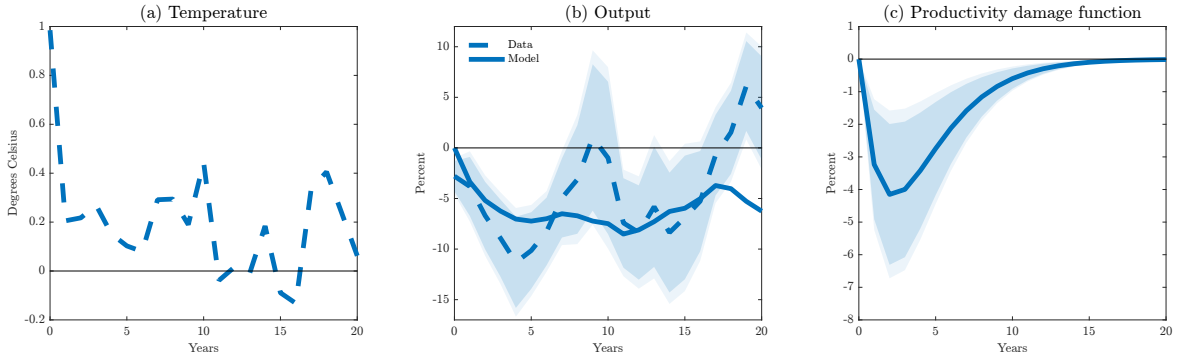
Notes: Panel (a): empirical (dashed) and model (solid) impulse response function of GDP to a local GDP shock. Panel (b): empirical (dashed) and model (solid) impulse response function of mortality to a local GDP shock. Model responses evaluated at point estimate of μ . GDP shock scaled to 1% in horizon 0. Empirical impulse response functions from Bilal, Känzig, and Lisiecki (2026).

Figure F.2: Estimation of ν



Notes: Panel (a): empirical (dashed) and model (solid) impulse response function of GDP to a global GDP shock. Panel (b): empirical (dashed) and model (solid) impulse response function of mortality to a global GDP shock. Model responses evaluated at point estimate of ν . GDP shock scaled to 1% in horizon 0. Empirical impulse response functions from Bilal, Känzig, and Lisiecki (2026).

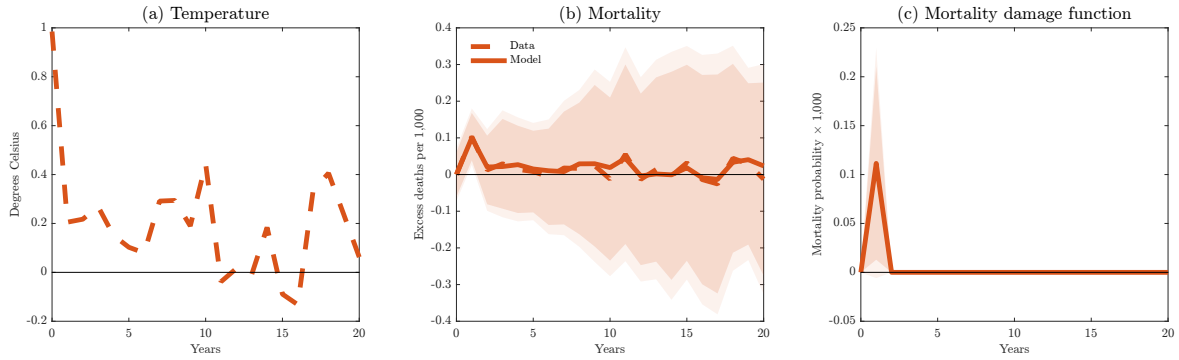
Figure F.3: Productivity Damage Function Estimates



Notes: Estimated productivity damage functions obtained by matching model-implied impulse responses to the empirical responses to global temperature shocks. Dashed lines: empirical responses. Solid lines: model fit. Dark and light shaded areas: 90 and 95% confidence bands, computed using the Delta method and accounting for the variance-covariance matrix between empirical GDP and mortality impulse response functions.

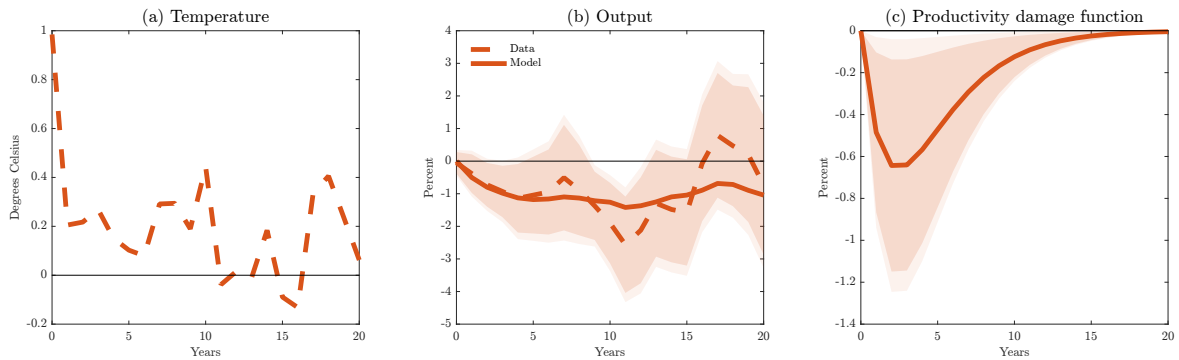
F.2 Damage Functions

Figure F.4: Mortality Damage Functions From Local Heat



Notes: Estimated mortality damage functions obtained by matching model-implied impulse responses to the empirical responses to local heat shocks, using the impact effect of local heat only. Dashed lines denote empirical responses; solid lines denote model fits. Dark and light shaded areas represent 90 and 95% confidence bands, computed using the Delta method.

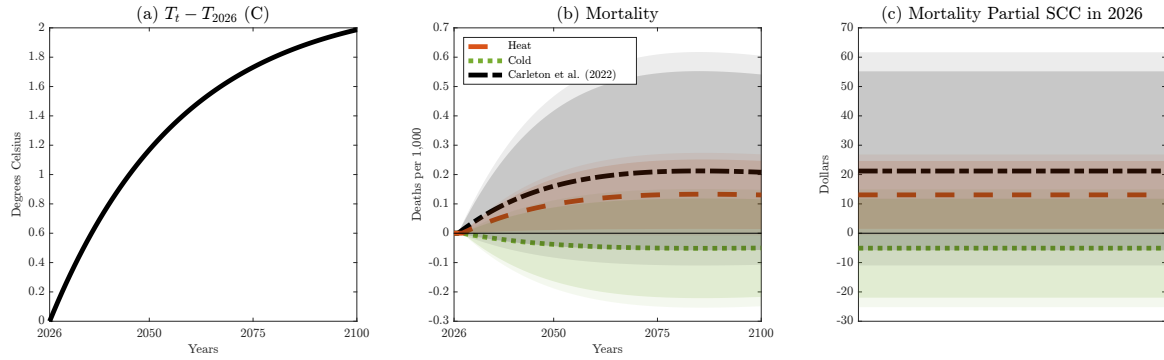
Figure F.5: Productivity Damage Functions From Local Temperature



Notes: Estimated productivity damage functions obtained by matching model-implied impulse responses to the empirical responses to local temperature. Dashed lines denote empirical responses; solid lines denote model fits. Dark and light shaded areas represent 90 and 95% confidence bands, computed using the Delta method.

F.3 Counterfactuals Using Local Heat

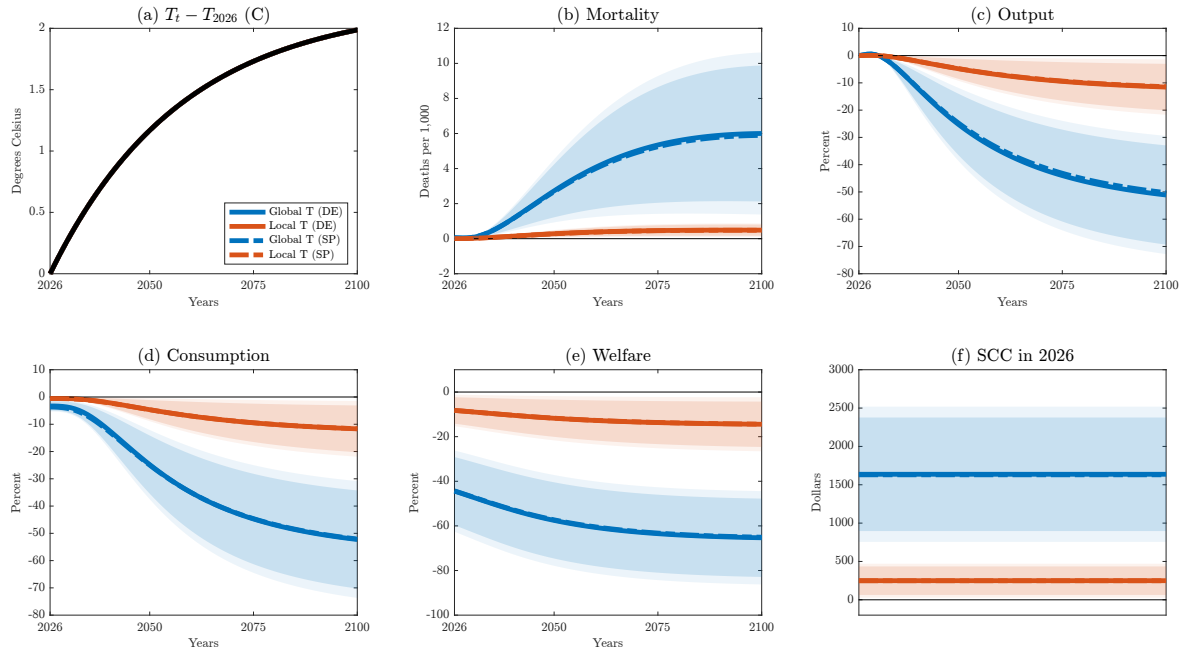
Figure F.6: Transitional Dynamics: Local Heat



Notes: Transitional dynamics of the model given the warming scenario in Panel (a) based on alternative damage function specifications: local heat (red dashed), local cold (green dotted), Carleton et al. (2022) damage function (black dash-dotted). The heat and cold damage functions are estimated from the impact effect of local heat and cold exposure only. Shaded areas: 90 and 95% confidence intervals based on the Delta method.

F.4 Efficient Allocation

Figure F.7: Counterfactuals in the Efficient Allocation.



Notes: Transitional dynamics of the model given the warming scenario in Panel (a) based on alternative specifications. Solid lines: decentralized equilibrium allocations (DE). Dash-dotted lines: efficient allocations (SP). Reference and counterfactual both calculated in the same specification.

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